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(54) **RECOMBINANT CHIMERIC ADENOVIRAL VECTOR SUBSTITUTED BY KNOB GENE OF CHIMPANZEE ADENOVIRUS SEROTYPE 6, AND APPLICATION THEREOF**

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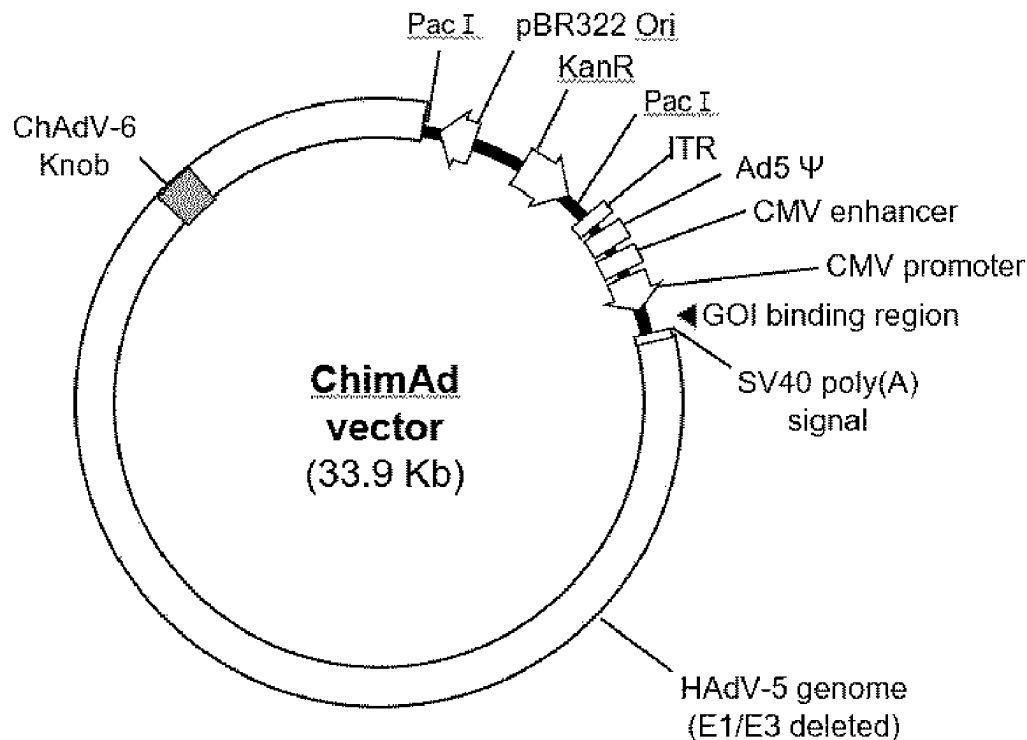
Mar. 21, 2022 (KR) 10-2022-0034885

ABSTRACT

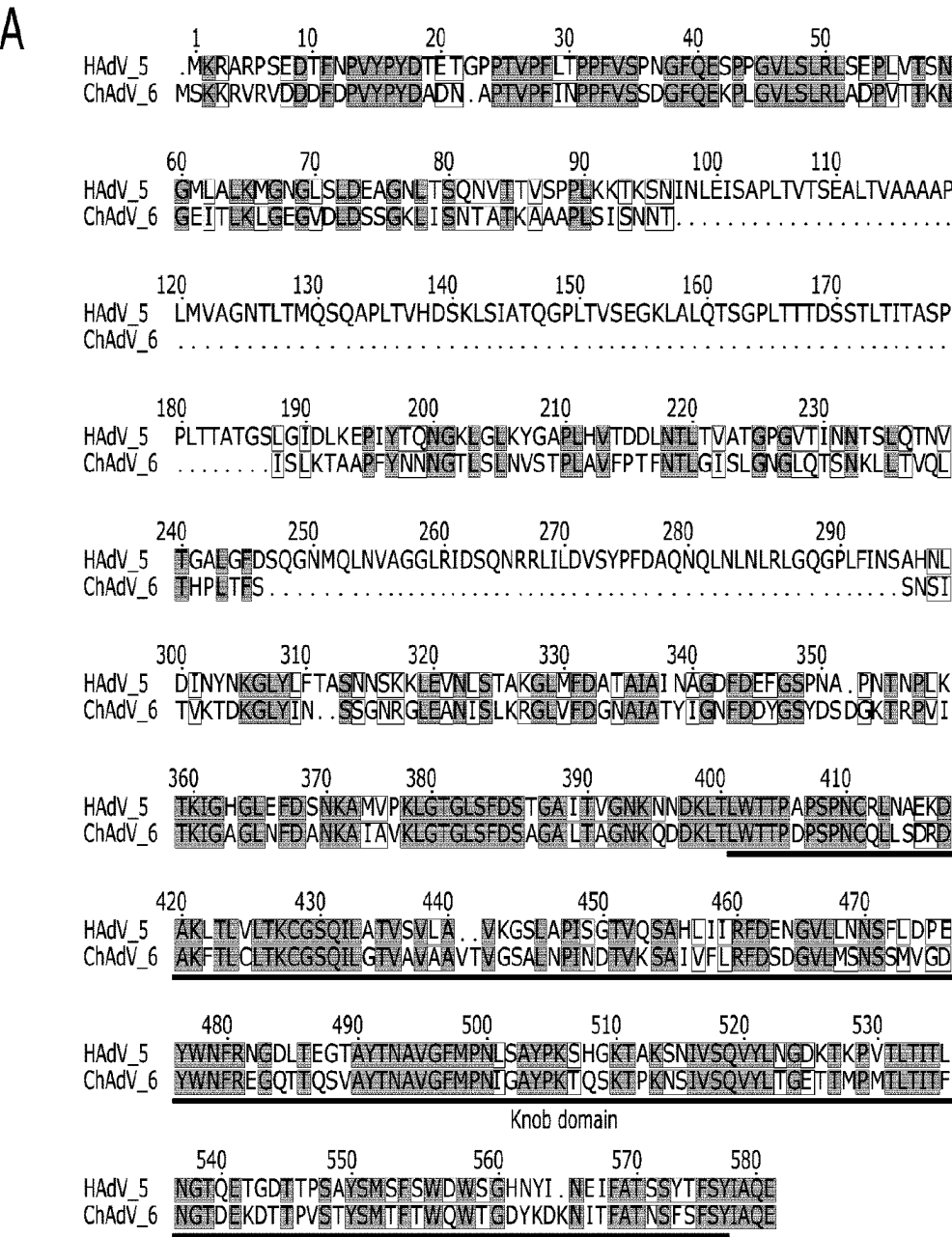
The present invention is a chimeric adenovirus vector in which the knob domain of the end of the fiber protein of human adenovirus type 5 is replaced with the knob gene of chimpanzee adenovirus serotype 6 and/or in addition the hexon protein of human adenovirus type 5 is replaced with hypervariable regions 1-7 of human adenovirus serotype 28. The present invention not only provides the optimal adenovirus vector in the development of treatments or vaccines for various diseases, but also when the chimeric adenovirus vector produced in the present invention is infected with a host cell for production, it can contribute to improved productivity by exhibiting superior cell infection ability compared to the recombinant HAdV-5 vector-based vaccine.

Specification includes a Sequence Listing.

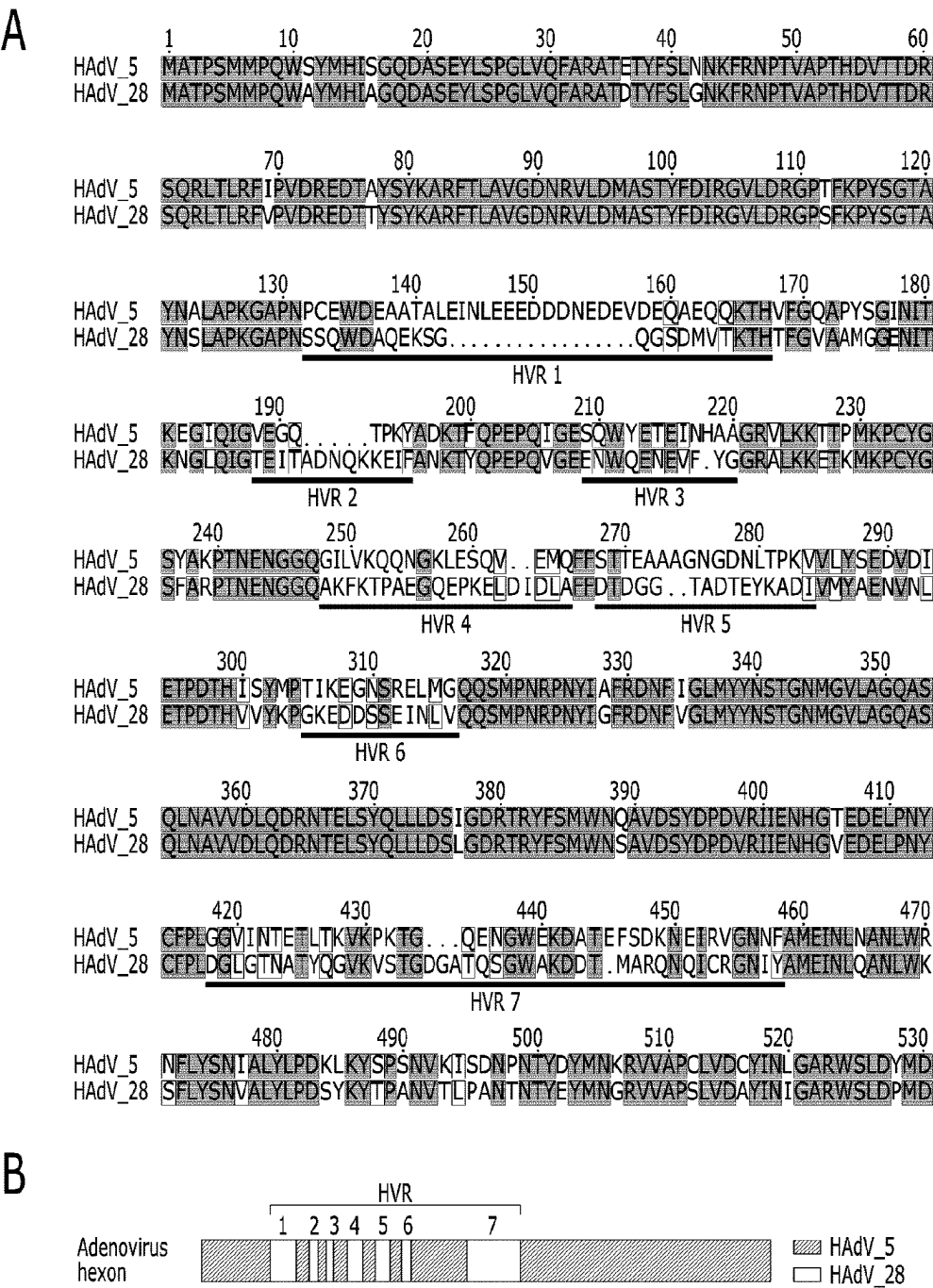
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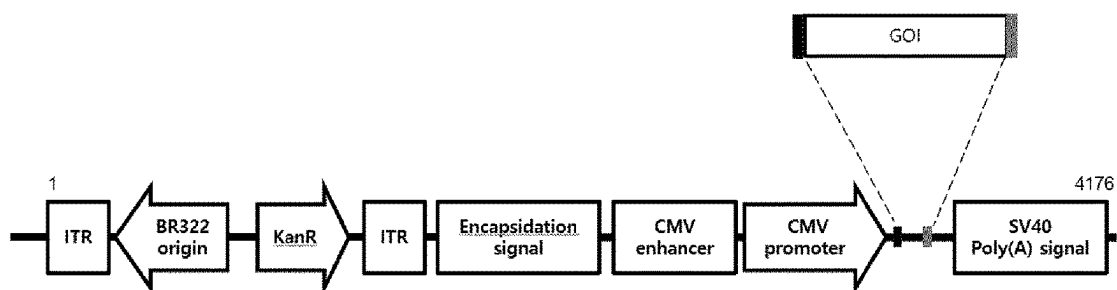
【Fig. 1】



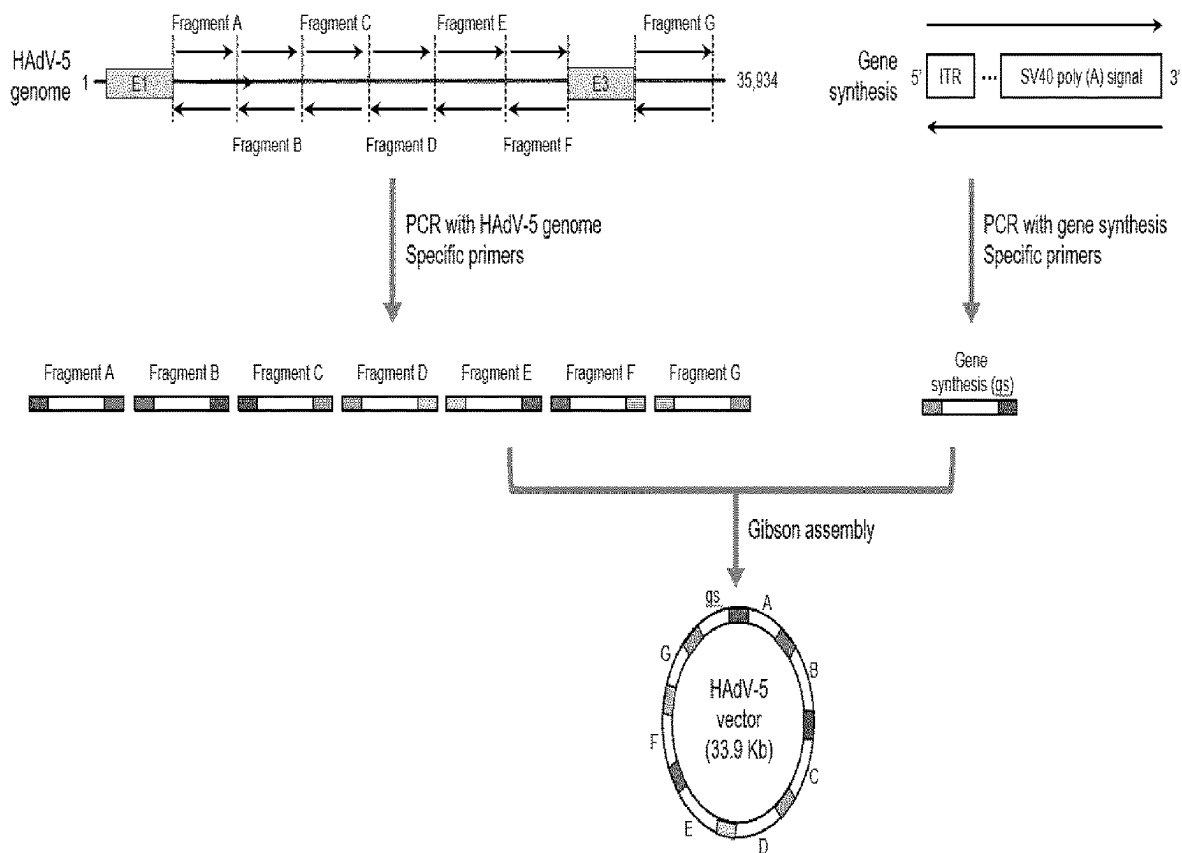
【Fig. 2】



【Fig. 3】

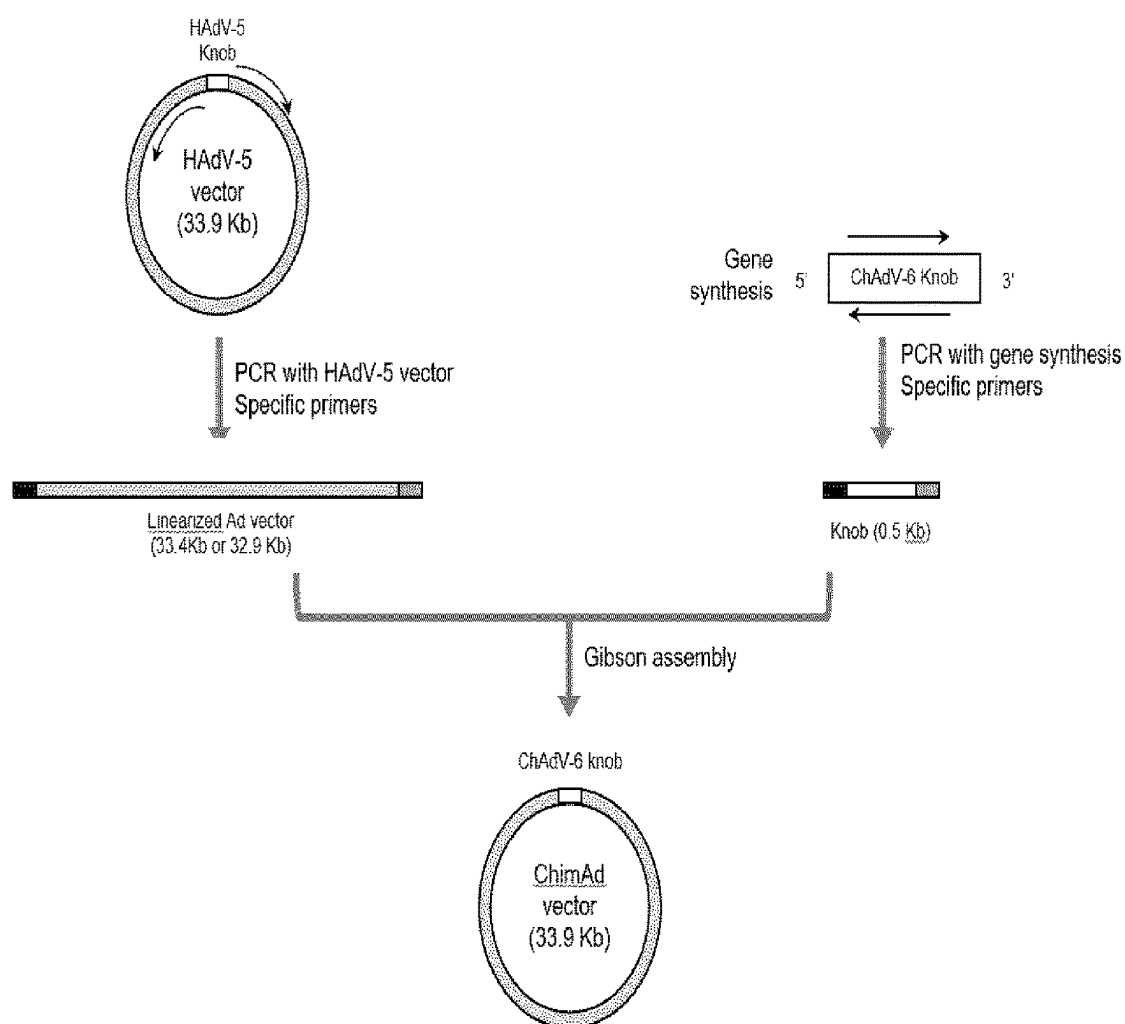


【Fig. 4】



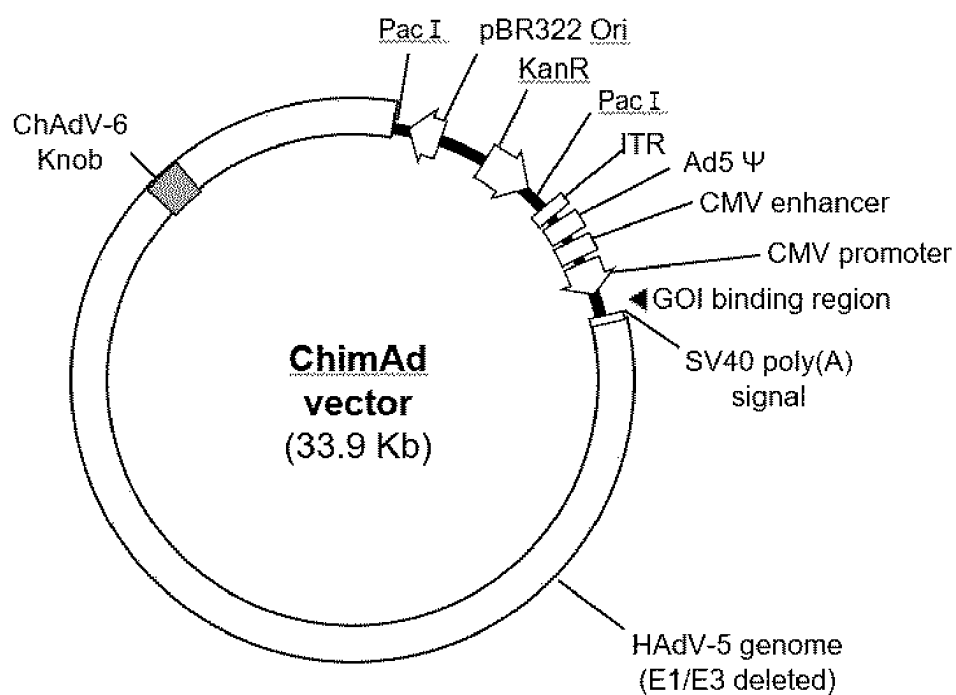
【Fig. 5】

A

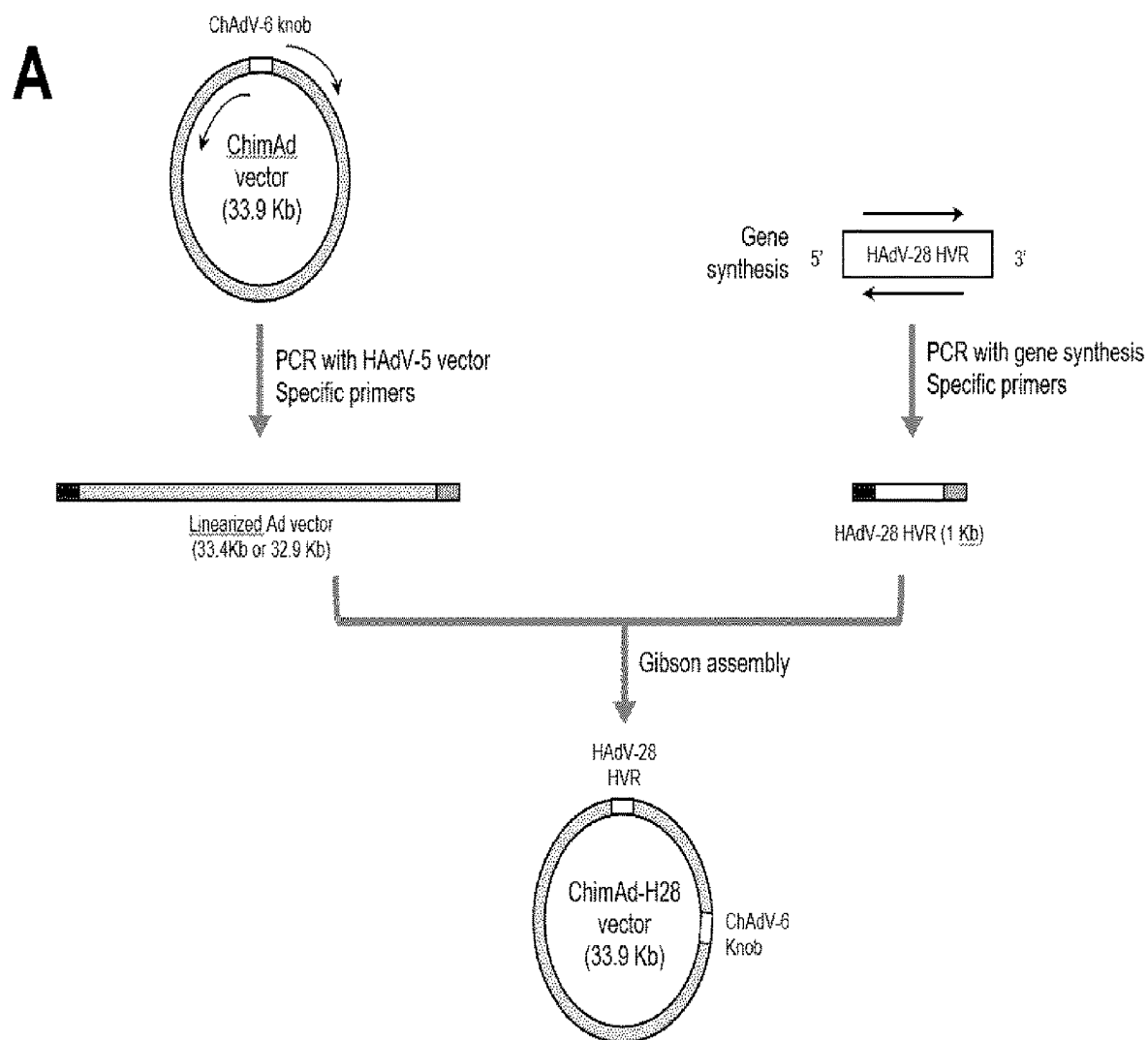


【Fig. 6】

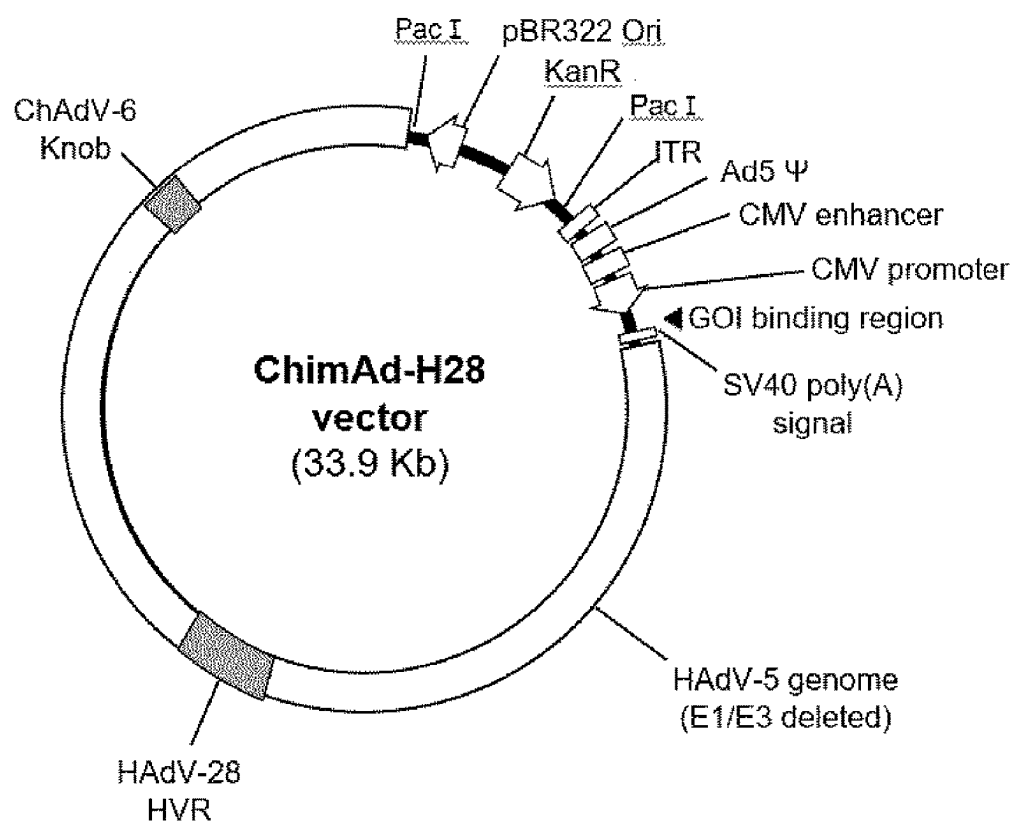
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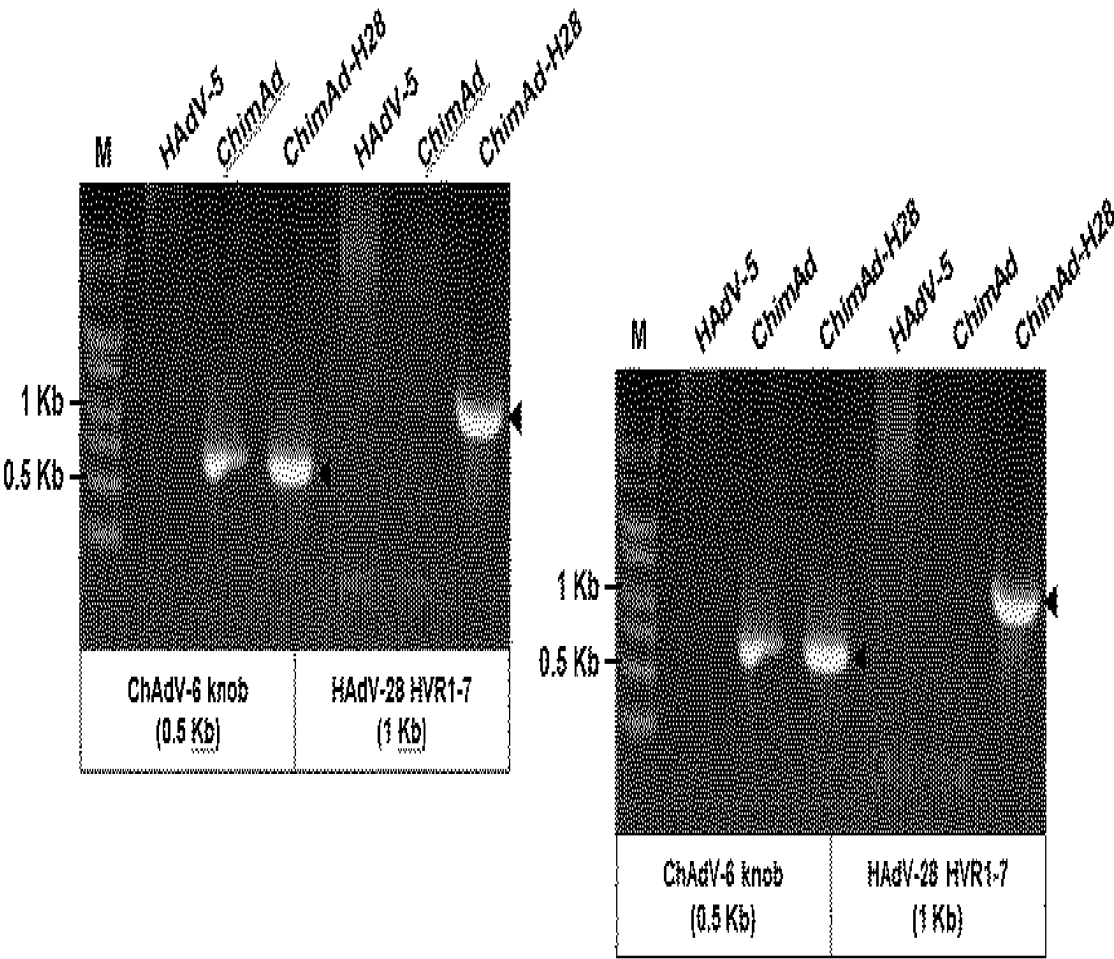
【Fig. 7】



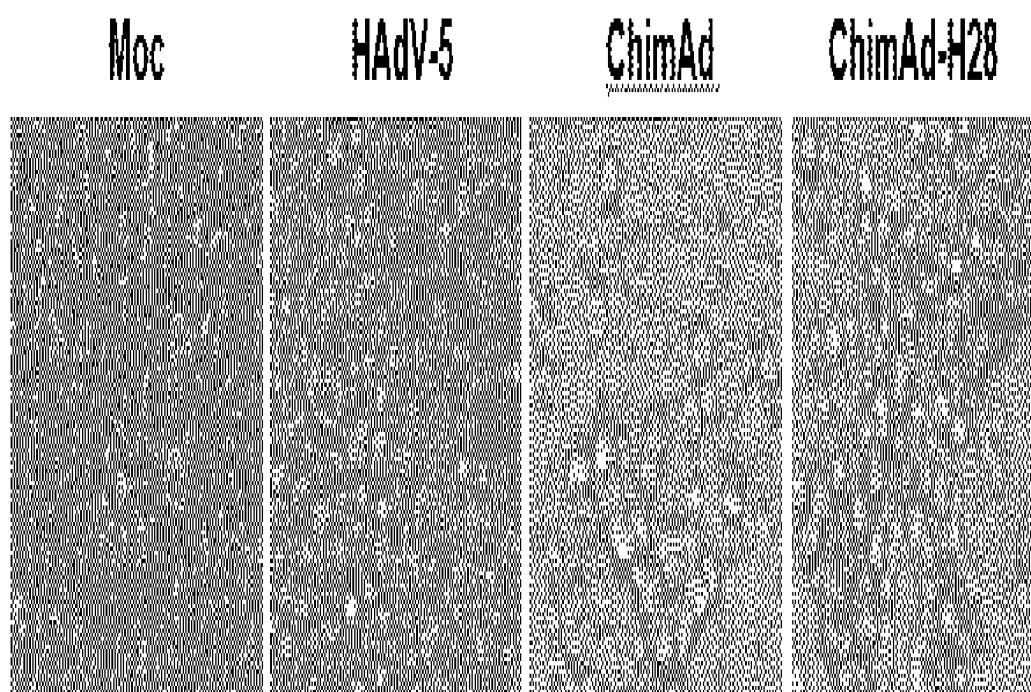
【Fig. 8】

B

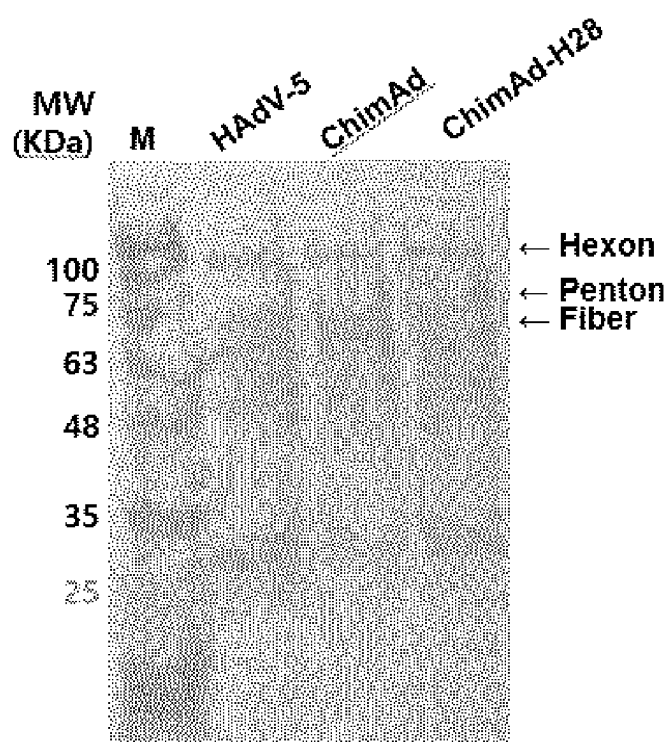
【Fig. 9】



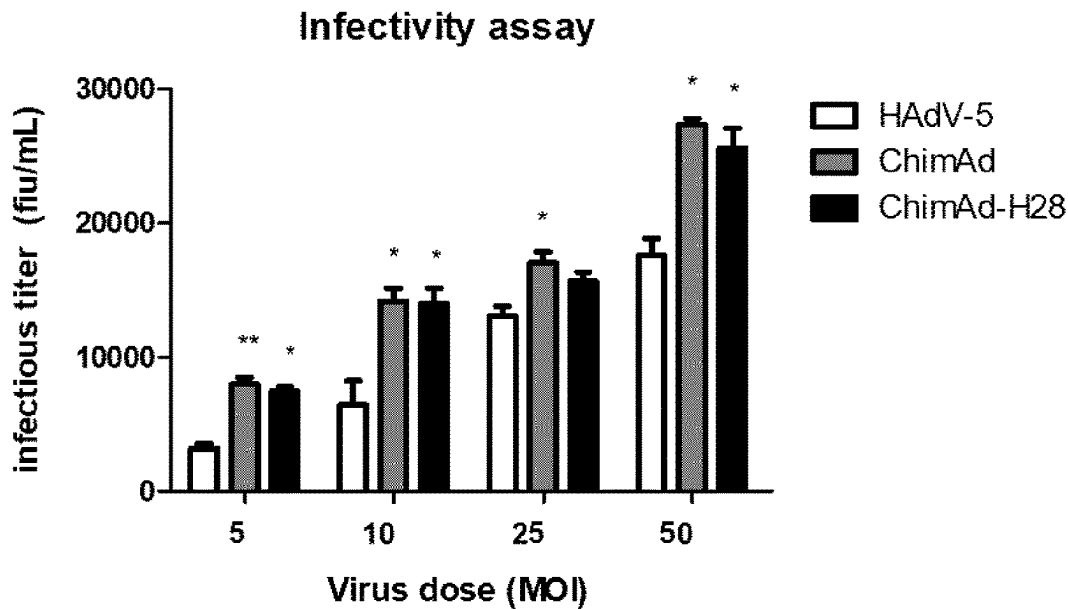
【Fig. 10】



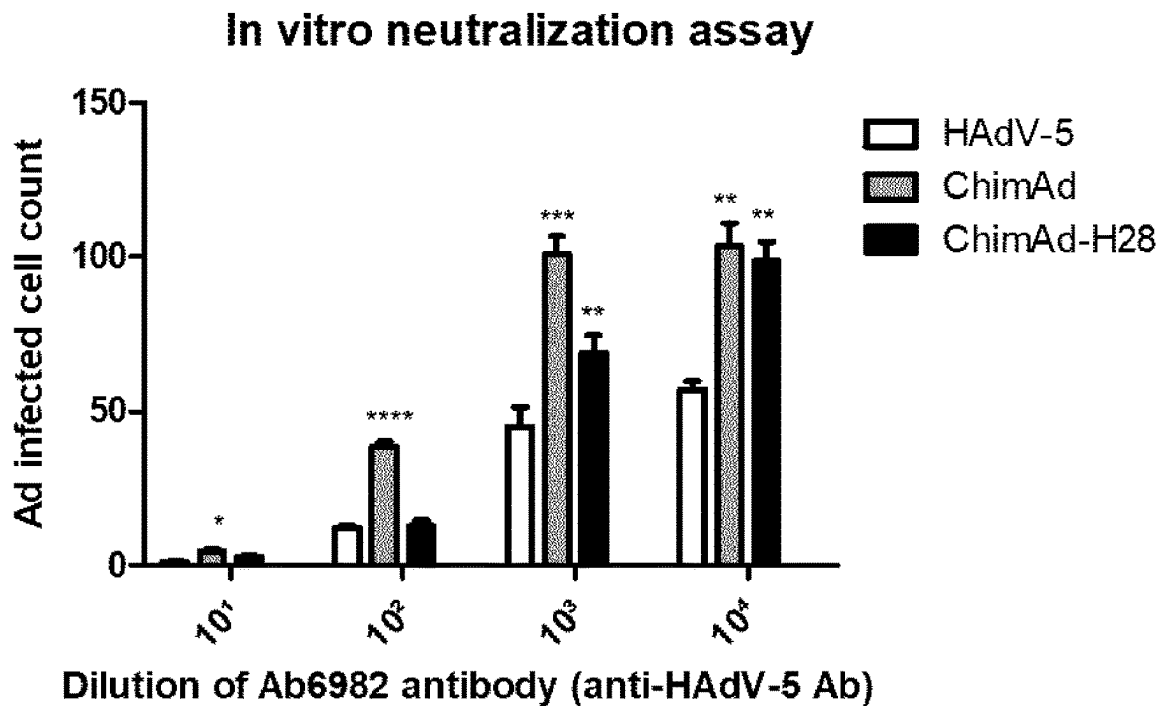
【Fig. 11】



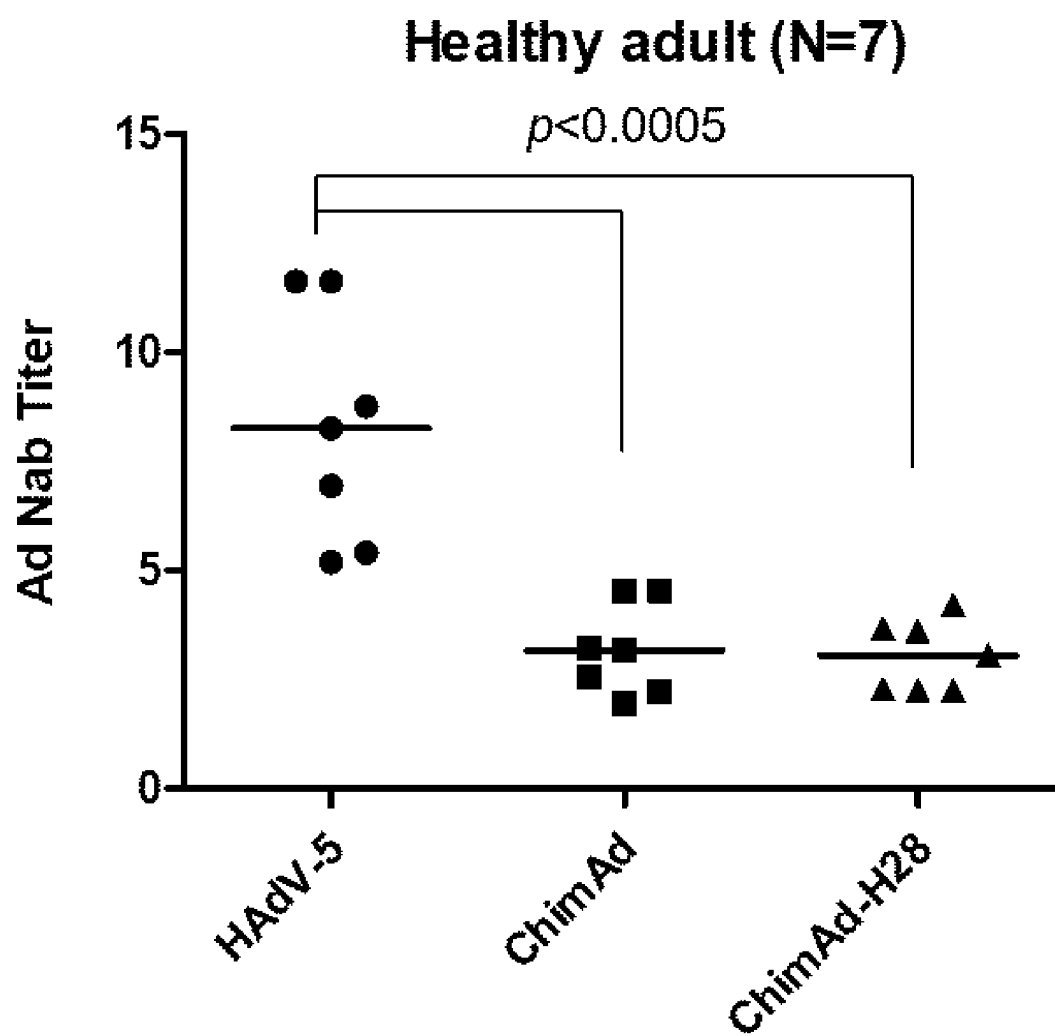
【Fig. 12】



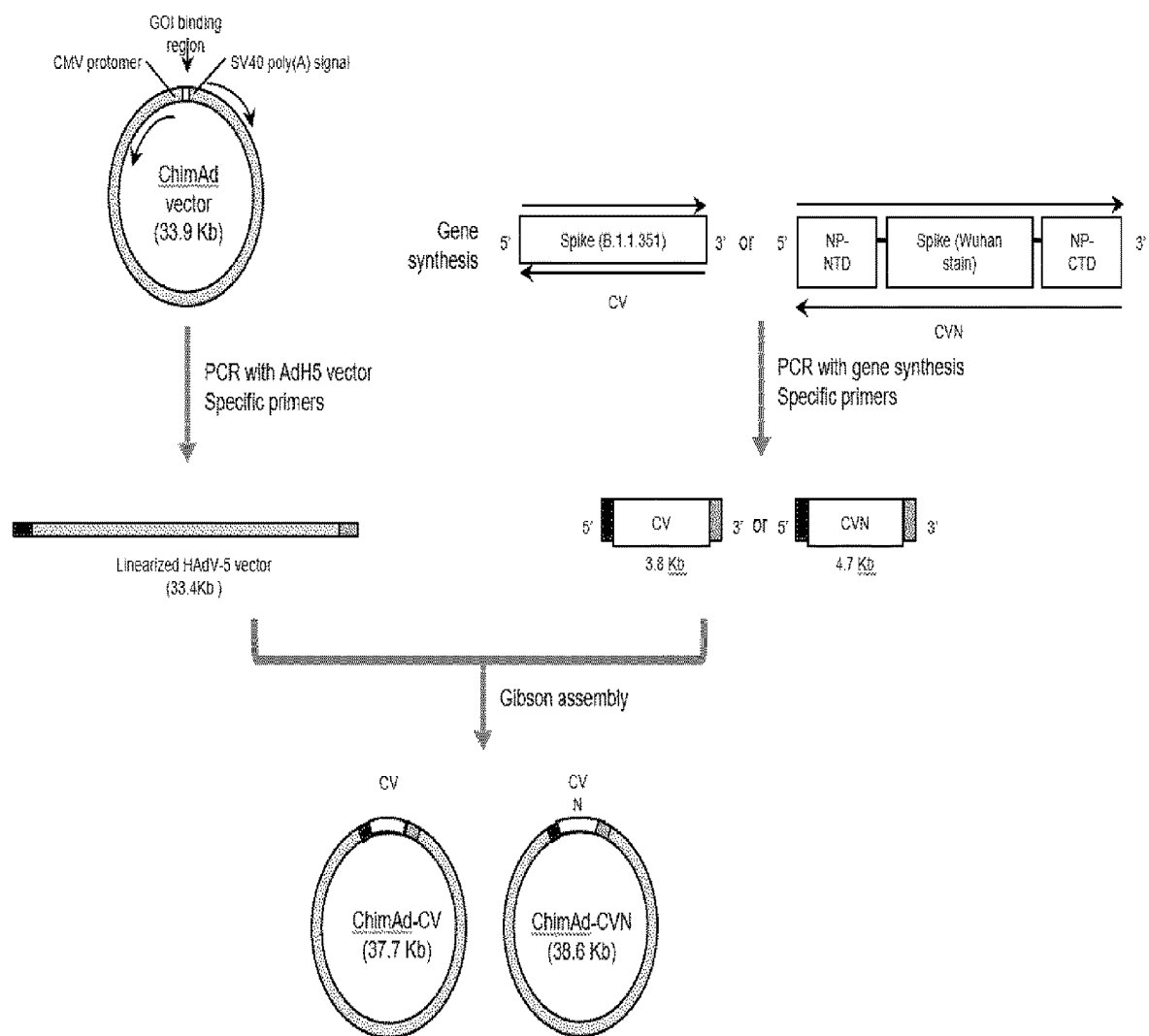
【Fig. 13】



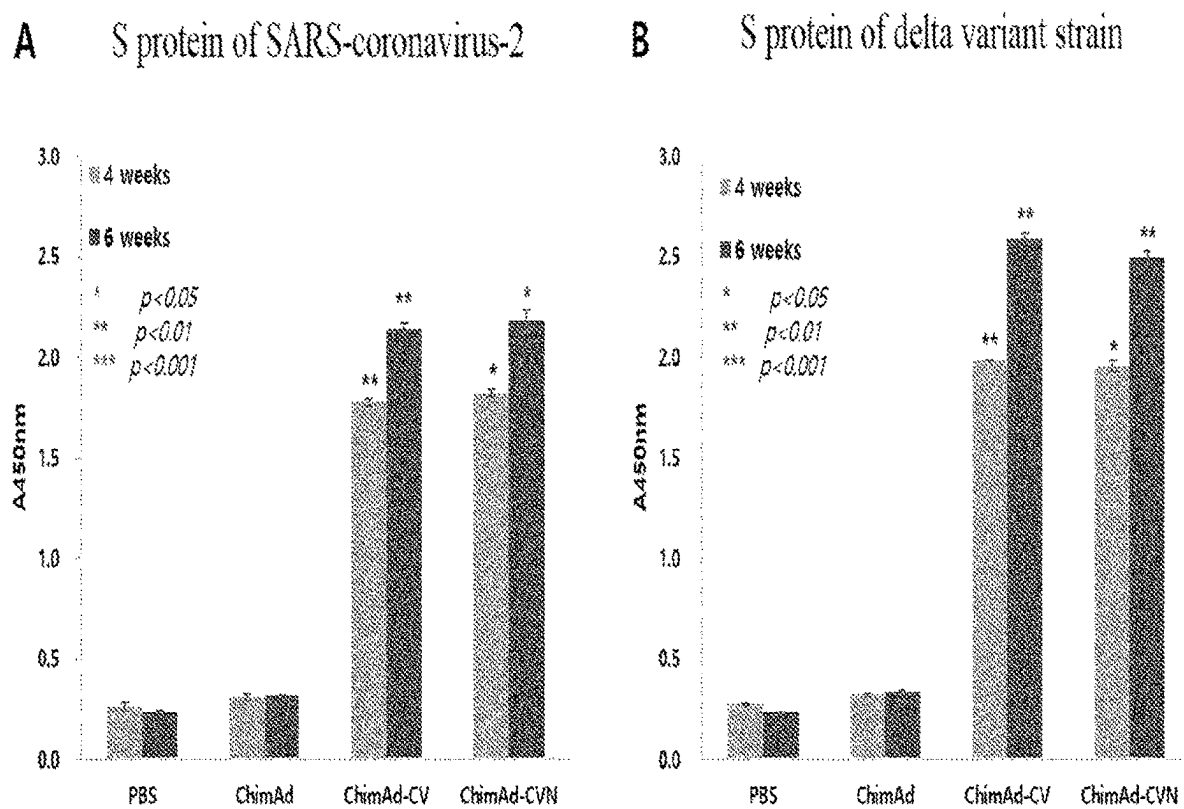
【Fig. 14】



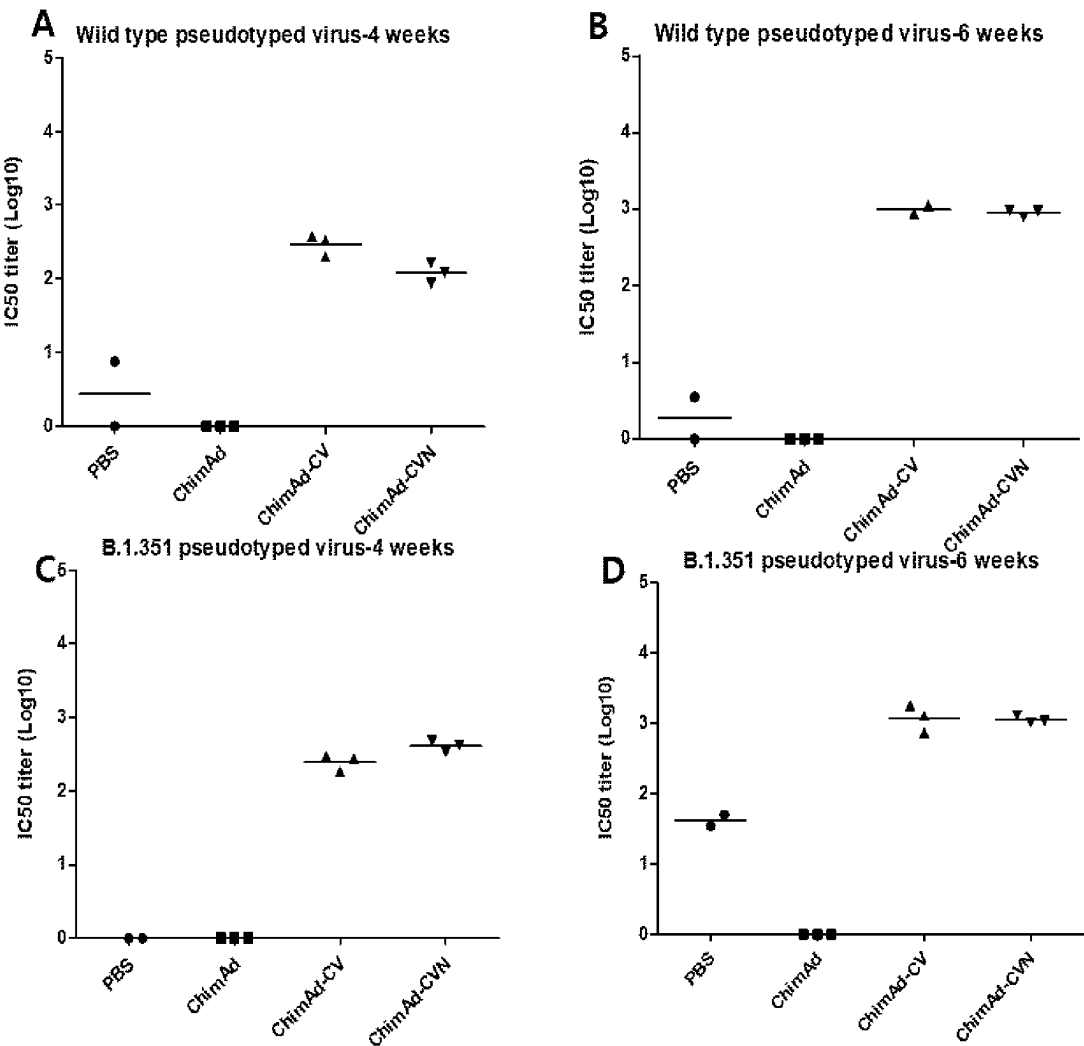
【Fig. 15】



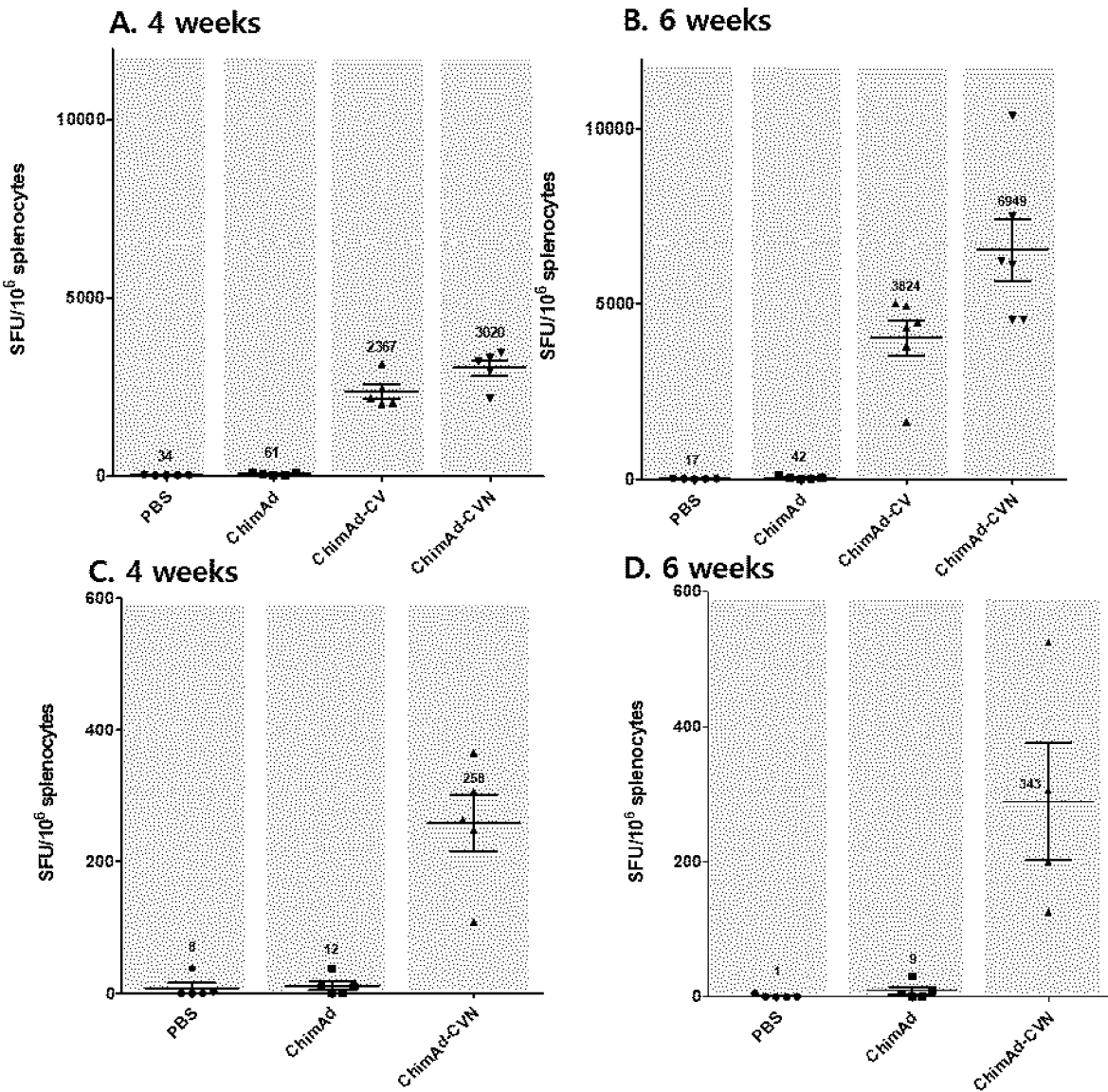
【Fig. 16】



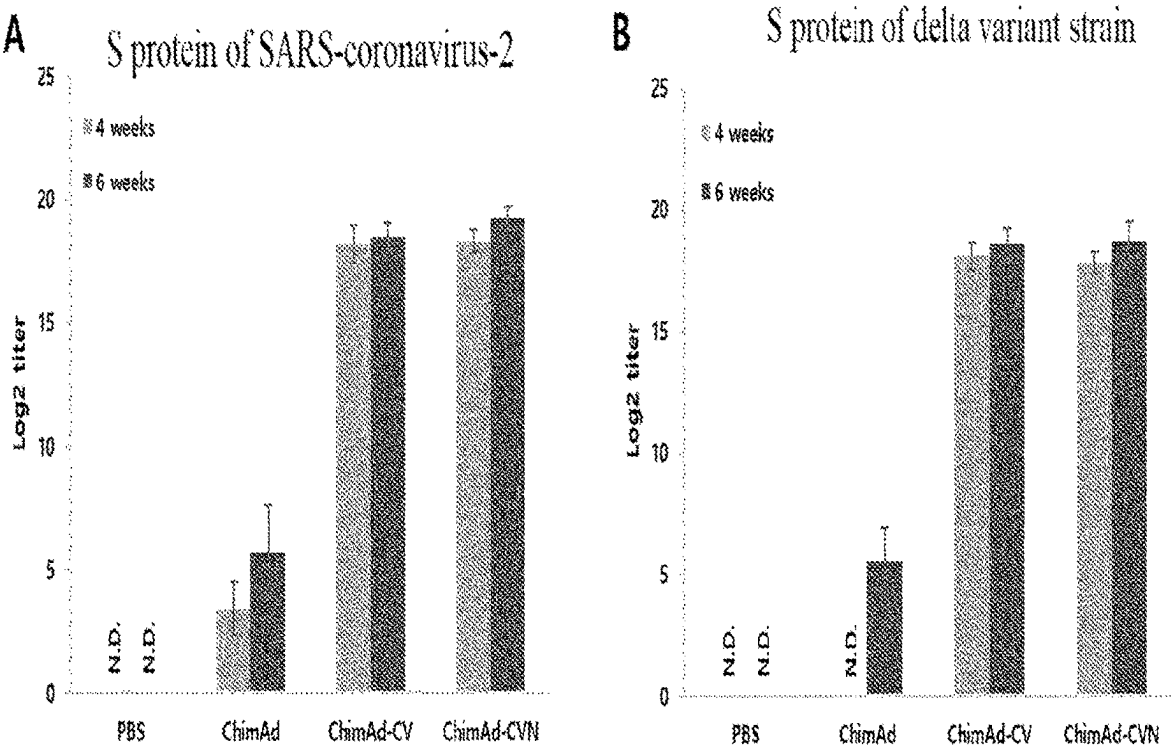
【Fig. 17】



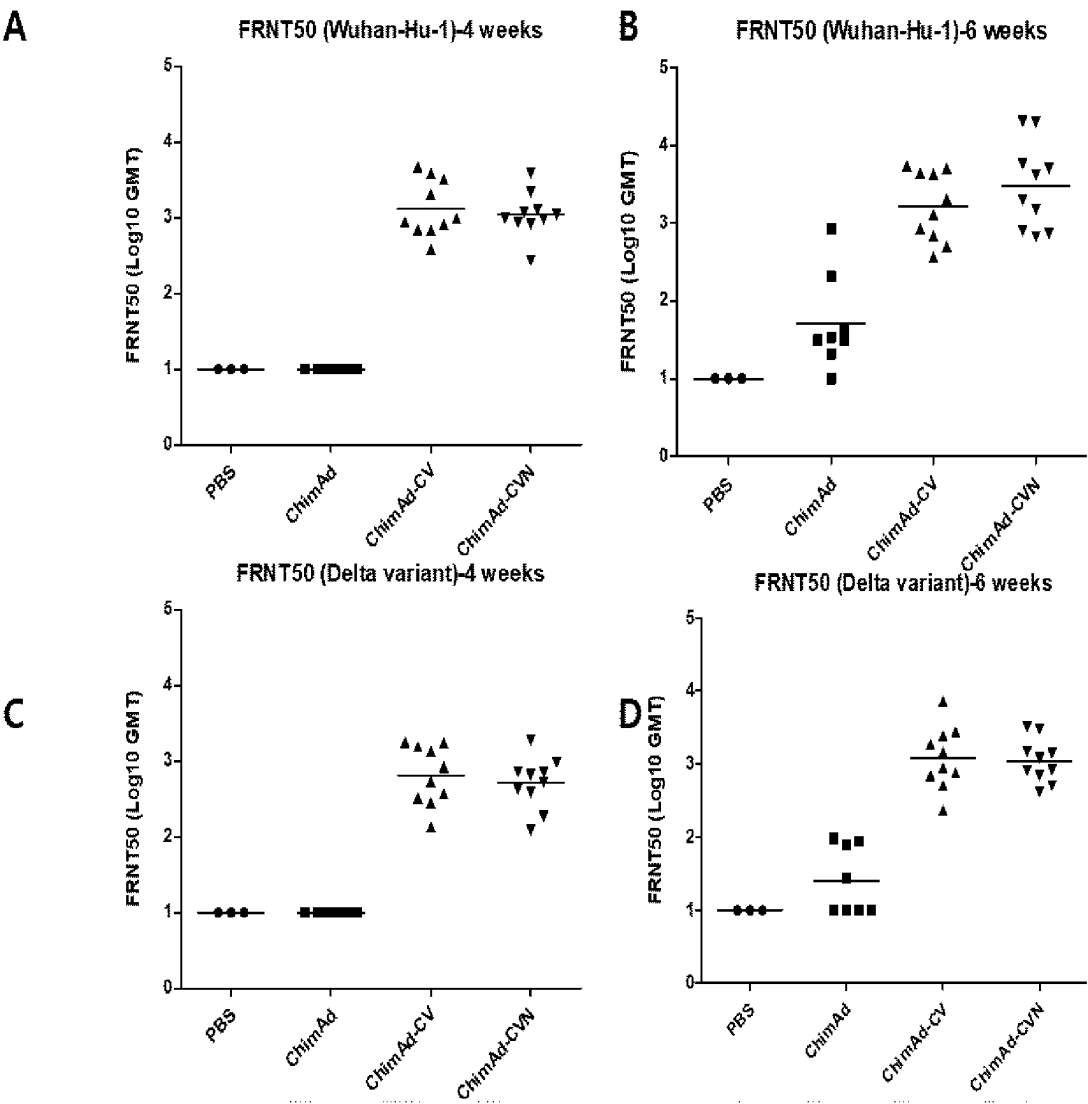
【Fig. 18】



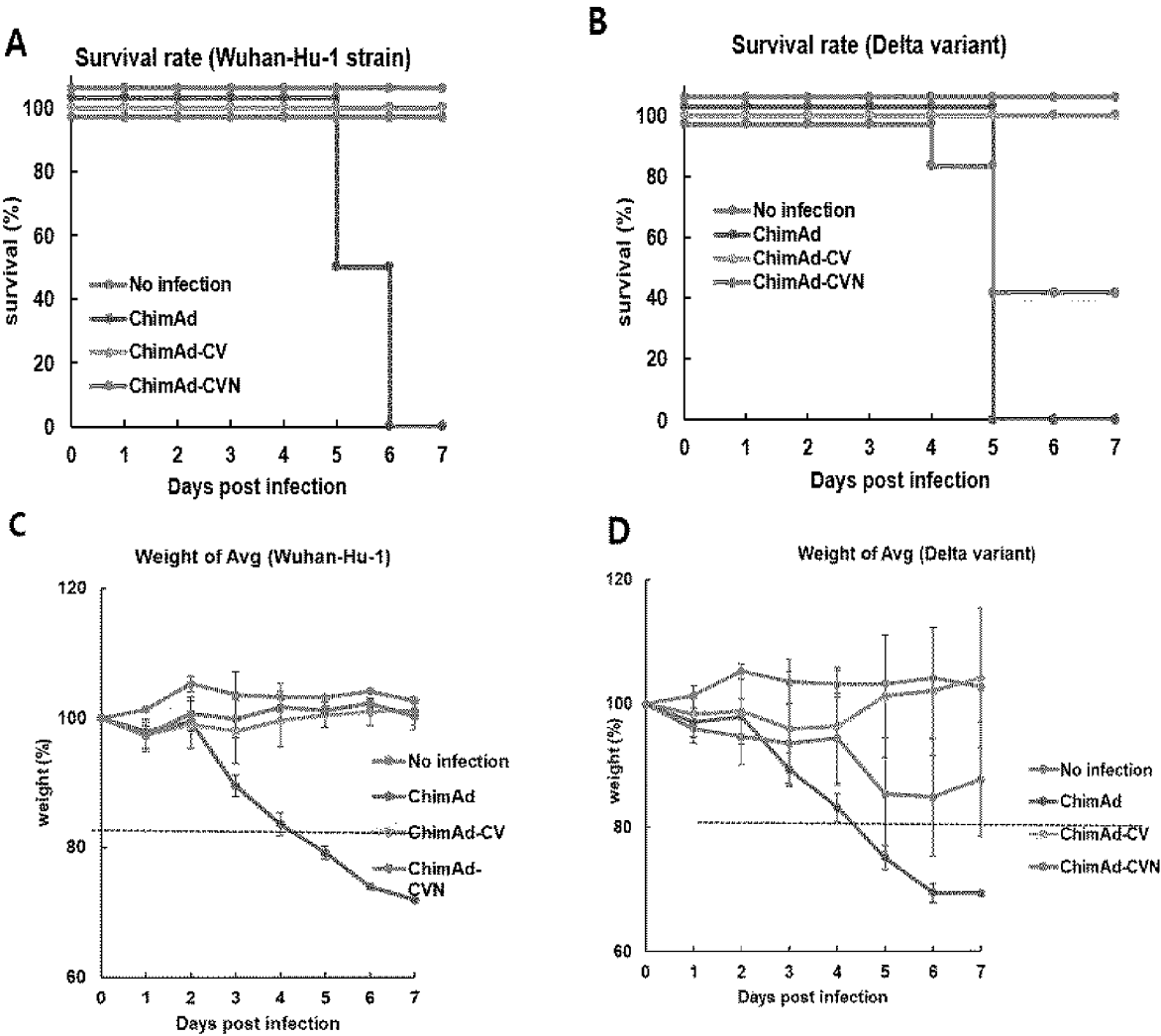
【Fig. 19】



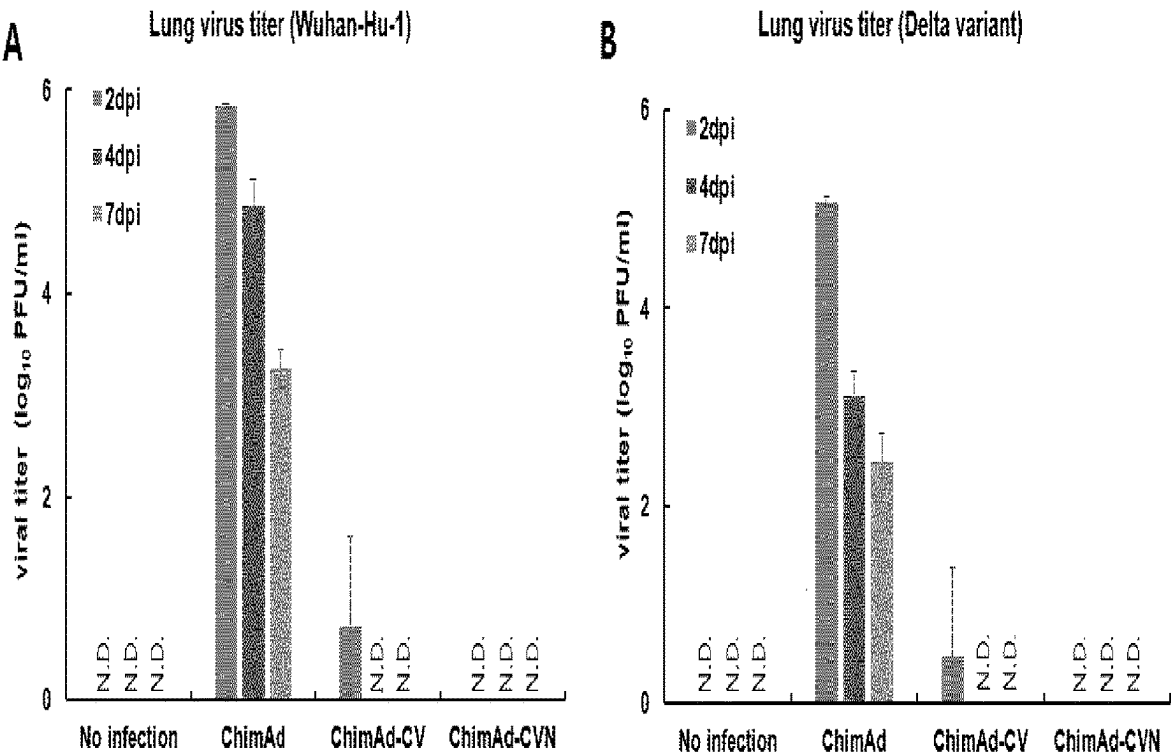
【Fig. 20】



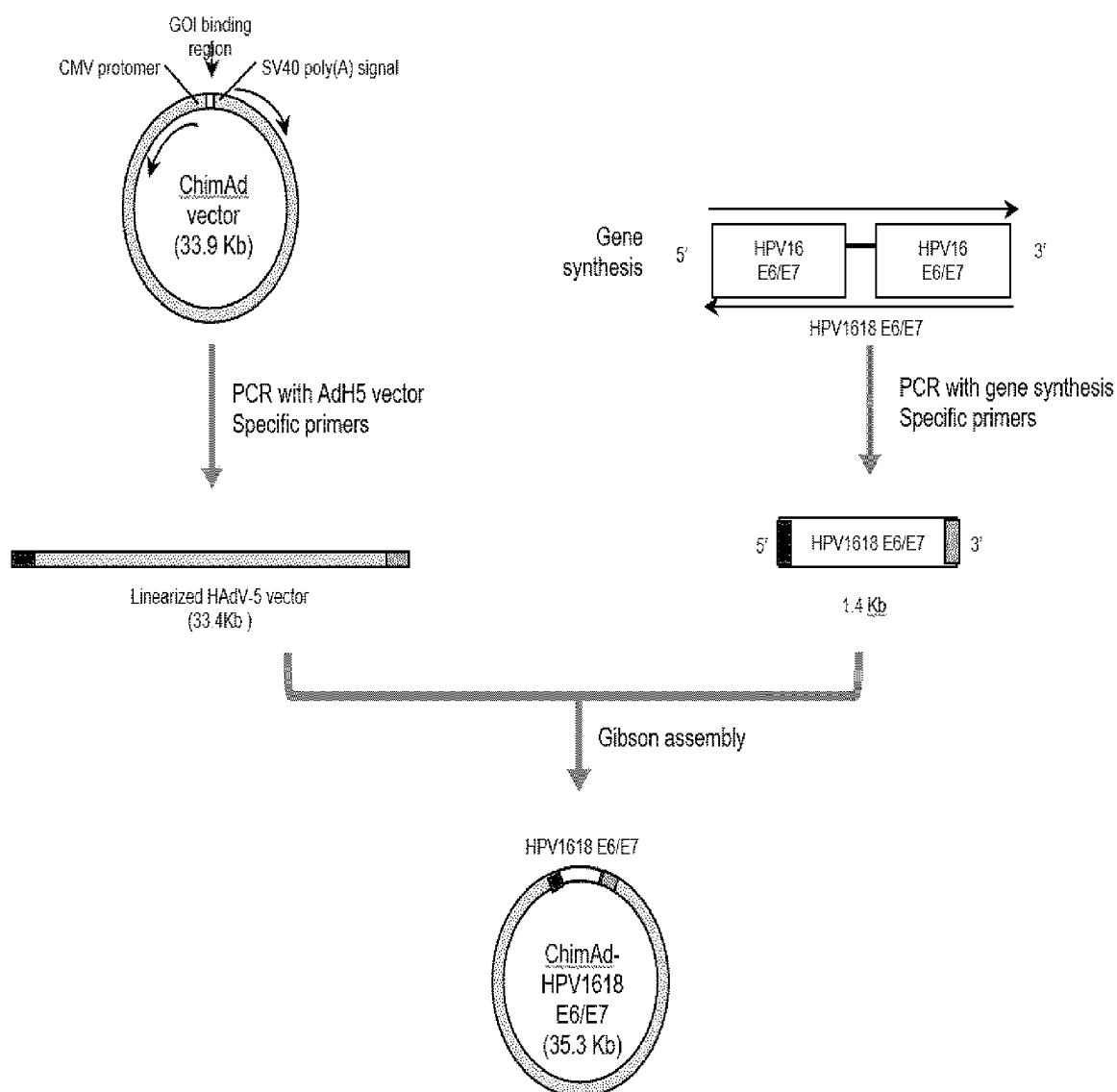
【Fig. 21】



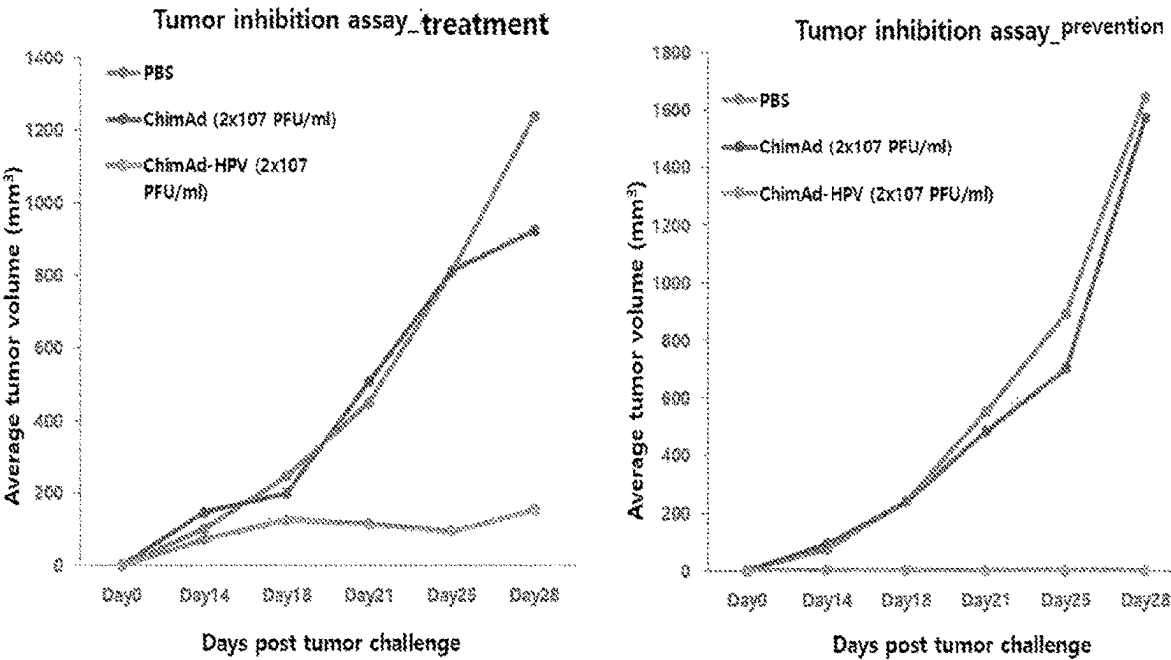
【Fig. 22】



【Fig. 23】



【Fig. 25】



**RECOMBINANT CHIMERIC ADENOVIRAL
VECTOR SUBSTITUTED BY KNOB GENE
OF CHIMPANZEE ADENOVIRUS SEROTYPE
6, AND APPLICATION THEREOF**

TECHNICAL FIELD

[0001] The present invention relates to a chimeric adenovirus vector having optimal conditions for avoiding pre-existing immunity against human adenovirus serotype 5 (HAdV-5) in a host. Specifically, it relates to a chimeric adenovirus vector expressing the knob protein and/or hexon protein of an adenovirus serotype with low seroprevalence based on the HAdV-5 gene sequence, and prophylactic and therapeutic vaccine applications against SARS-COV-2 and human papillomavirus based on the chimeric adenovirus vectors.

**REFERENCE TO AN ELECTRONIC SEQUENCE
LISTING**

[0002] The content of the electronic sequence listing (OP22-0008HSUS_SEQLIST.TXT; Size: 50,320 bytes; and Date of Creation: Jun. 14, 2024), filed via Patent Center on Jun. 17, 2024, is herein incorporated by reference in its entirety.

BACKGROUND ART

[0003] Adenovirus is a medium-sized nonenveloped virus with a diameter of about 90-100 nm containing linear double stranded DNA in a nucleocapsid. It is found in a variety of vertebrates (amphibians, birds, and mammals), most of which cause respiratory infections, conjunctivitis and gastroenteritis. In contrast to retroviruses, since it is not inserted into the genome of host cells, transformation of host cells or tumorigenesis by adenovirus DNA does not occur, transduction into non-dividing cells is possible, and virus productivity is high. Because of their characteristics, adenoviral vectors have been widely studied for gene therapy and vaccine applications.

[0004] First-generation recombinant adenovirus vectors delete E1 and E3 genes, which are early transcription genes involved in viral replication, and from a clinical point of view, they are replication-incompetent in the host. Side effects were reduced by making it, and there were fewer restrictions on inserting a foreign gene (up to about 8 Kb). So far, human adenovirus serotypes 2 and 5 (AdH2 and HAdV-5) belonging to subgroup C have been studied most as vectors, and recombinant HAdV-5 vector-based vaccines have been developed in preclinical trials. Excellent vaccine efficacy was demonstrated by showing highly immunogenic results through various animal models.

[0005] However, a clinical trial of an HIV vaccine developed based on a recombinant HAdV-5 vector showed insufficient efficacy different from non-clinical results. HAdV-5 is one of the most infected adenovirus serotypes, and it has been reported that about 80% or more of the population has anti-HAdV-5 neutralizing antibodies due to pre-existing immunity. As shown in the results of HIV vaccine clinical trials, anti-HAdV-5 neutralizing antibodies formed by pre-existing immunity bind to recombinant HAdV-5 vector-based vaccines loaded with foreign genes, reducing vaccine efficacy.

[0006] The icosahedral envelope protein of adenovirus is the main capsid protein and is composed of hexon, fiber and

penton proteins. Hexon is a protein that occupies the largest portion of adenovirus envelope proteins and includes a hypervariable region (HVR) with low genetic homology between adenovirus serotypes. Penton is a protein responsible for the structural stability and flexibility of envelope proteins and is involved in cell entry through the Arg-Gly-Asp (RGD) motif. In addition, the fiber binds to the host cell's receptor, CAR (Coxsackie and adenovirus receptor), through a knob domain at the end, and cell entry occurs. Three coat proteins (hexon, fiber and penton) are known to act as major recognition sites for anti-Ad neutralizing antibodies. Recently, to avoid existing immunity to HAdV-5, a chimeric adenovirus vector study was reported in which the projection domain of the fiber protein was replaced with a human adenovirus serotype with low seroprevalence. However, in the case of chimeric adenoviral vectors using human adenovirus serotypes, there is a possibility of cross-reactivity with neutralizing antibodies against HAdV-5, so the ability to evade existing immunity against HAdV-5 may be reduced.

[0007] To avoid cross-reactive immunity with neutralizing antibodies to HAdV-5, adenovirus serotypes isolated from non-human animals that are genetically distant have been developed as viral vectors. Among these non-human adenovirus vectors, a viral vector using a chimpanzee adenovirus (ChAdV) serotype is most widely used. ChAdV serotypes were investigated to have lower seroprevalence rates than HAdV-5. For example, the serum retention rates of neutralizing antibodies against ChAdV-6 and ChAdV-68 were detected in only 1.5-4% of serum samples from the US and Thailand. In addition, in the phase 1 clinical trial of the ChAdV-63-based malaria vaccine, safety and antigen-specific T cell response were high in all clinical trial volunteers. Despite the low seroprevalence of ChAdV-vector-based vaccines, high cross-reactivity between serotypes is a major concern for vector-specific cellular immunity.

[0008] Coronaviruses are positive-sense single-stranded RNA viruses that have an envelope on the surface of particles with a diameter of 100-200 nm. Seven types of coronaviruses are known to cause respiratory and digestive infections in humans, and two of the seven types are alpha-genus of coronaviruses, and the remaining five species were identified as genus beta-coronaviruses. Severe acute respiratory syndrome coronavirus 2 (hereinafter referred to as 'SARS-COV-2'), which causes COVID-19 (novel coronavirus, 2019-nCoV), is a beta-coronavirus. After it was first reported in Wuhan City, China in December 2019 (Wuhan-Hu-1), confirmed cases are appearing one after another around the world, and the pandemic is intensifying through various mutations (alpha variants, beta variants, delta variants, etc.) since the outbreak.

[0009] The viral envelope of SARS-COV-2 is composed of spike glycoprotein (S protein) and hemagglutinin-esterase dimer (HE) etc., The S1 region of the S protein mainly binds to the receptor (Angiotensin-converting enzyme 2, ACE2) through the RBD (Receptor binding domain), and the S2 region is responsible for direct cell-viral fusion (initiation of infection) with dramatic changes in tertiary structure. Therefore, vaccine candidates targeting the S protein as a major antigen have been developed and commercialized (AZD-1222 (AstraZeneca), Bnt162b2 (Pfizer), mRNA-1273 (Moderna) etc.)

[0010] Unlike DNA viruses, which have DNA repair pathways (base excision repair, nucleotide excision repair, mis-

match repair) that correct errors during DNA replication, RNA viruses use RNA polymerase, and so they do not have RNA repair-related proteins to correct errors generated during genome replication, so they are characterized by frequent mutations compared to DNA viruses. In the case of the SARS-COV-2 variant virus, mutations frequently occurred in the N-terminal domain (NTD) and RBD domain of the S protein, as a result, licensed vaccines developed targeting the S protein of SARS-coronavirus-2 have reduced antibody binding ability to the S protein of the variant virus in which structural changes have occurred, resulting in reduced protective efficacy against the variant virus.

[0011] In December 2020, a fatal mutation of SARS-COV-2 occurred in South Africa and was classified as a beta mutation and designated as a variant of concern (VOC). The mutant was found to be the first mutant to have all three key mutations (E484K, K417N, N501Y) of VOCs in RBD as well as deletion in NTD, and in the case of AstraZeneca's AZD-1222 vaccine, it was confirmed that the efficacy of the vaccine in the beta variant strain (B.1.351) group was reduced by 10% compared to the existing SARS-COV-2, and a rapid decline in efficacy against licensed vaccines.

[0012] In addition, as cross-neutralization was confirmed for SARS-coronavirus-2 and other variant viruses in the serum of convalescent patients who were infected with the beta mutation, the direction is changing towards the development of a vaccine targeting the S protein against the variant virus, not the S protein against SARS-COV-2, and the need for a universal vaccine that can fight against the continuous emergence of variant viruses is emerging.

[0013] HPV has about 170 HPV genotypes, which are largely classified into two groups: high risk and low risk. HPV-6/11/40/42/43/44/54/61 and 72 types are in the low-risk group for causing genital warts, and HPV-16/18/31/35/39/45/51/Including 52/56/58/66 and 68, the high-risk type is responsible for about 99.7% of cervical cancer cases.

[0014] Particularly, HPV-16 and 18 types are the most common types associated with cervical cancer worldwide, accounting for more than 70% of cases. HPV is also responsible for many penile, vulvar, and anal-related carcinomas and accounts for more than 40% of oropharyngeal cancers.

[0015] Persistent infection with these high-risk HPVs results in the development of squamous intraepithelial lesions (SILs). This is called cervical intraepithelial neoplasia (CIN) in the cervix and vulva intraepithelial neoplasia (VIN) in the vulva, and these SILs can progress to malignant cancer.

[0016] HPV is a non-enveloped virus consisting of double-stranded DNA with a genome size of approximately 8 kb. The genome encodes six early regulatory proteins (E1, E2, E4, E5, E6 and E7) and two late structural proteins (L1 and L2). Early genes encode proteins involved in viral DNA replication, transcription and oncogenic transformation, and late genes encode viral capsid proteins. In the general high-risk HPV cancer formation process, the viral genome is integrated into the host's chromosomal DNA, and the E2 genetic sequence is disrupted during gene linearization. Since E2 protein is a transcriptional repressor of E6 and E7, disruption of E2 initiates the expression of E6 and E7 proteins. The E6 protein promotes the degradation of p53, a protein that regulates apoptosis of the host, and activates telomerase to extend cell lifespan. E7 protein also targets and degrades the tumor suppressor protein retinoblastoma

protein (pRb). As such, E6 and E7 interfere with cell cycle regulation and extend the lifespan of host cells, causing genetic instability, malignant transformation of cells, and cancer.

[0017] In HPV therapeutic vaccines, infection elimination in cellular immunity is more important than humoral immune response. HPV E6 and E7 oncoproteins are essential for the initiation and maintenance of malignant tumors and are expressed in pre-malignant and invasive lesions but absent from healthy cells.

[0018] Therefore, these oncoproteins (E6, E7) cannot avoid immune responses due to mutations, and are the most ideal targets for the development of therapeutic vaccines against HPV infections and lesions.

[0019] An ideal therapeutic vaccine can target E6 and E7 proteins to induce strong tumor-specific T cell type 1 and cytotoxic T lymphocyte (CTL) responses to kill infected and malignant cells.

PRIOR PATENT LITERATURE

[0020] Korean Patent Publication No. 1020040054796

DISCLOSURE

Technical Problem

[0021] The present invention has been made in response to the above needs, and an object of the present invention is to provide an adenoviral vector that is superior to conventional recombinant HAdV-5 vectors in terms of constructing a chimeric adenoviral vector that overcomes pre-existing immunity to HAdV-5 in the host.

[0022] Another object of the present invention is to provide a therapeutic or prophylactic vaccine vector capable of delivering foreign genes with high efficiency.

[0023] Another object of the present invention is to provide a chimeric adenovirus vector comprising the S gene sequence of the beta variant to prevent infection with SARS-coronavirus-2.

[0024] Another object of the present invention is to provide a vaccine for preventing infection of SARS-coronavirus-2.

[0025] Another object of the present invention is to provide a chimeric adenovirus vector comprising gene sequences of 16 types E6 and E7 and 18 types E6 and E7 of human papillomavirus.

[0026] Another object of the present invention is to provide a vaccine for the treatment and prevention of HPV.

Technical Solution

[0027] In order to achieve the above object, the present invention provides a chimeric adenoviral vector based on human adenovirus serotype 5 having excellent ability to evade existing immunity that the knob domain at the end of the fiber protein of human adenovirus type 5 is substituted with the knob gene of chimpanzee adenovirus serotype 6, or the hexon protein of human adenovirus type 5 is substituted with hypervariable regions 1-7 of human adenovirus serotype 28 in the substituted type 5 adenovirus.

[0028] The present invention relates to a vector capable of avoiding pre-existing immunity to human adenovirus type 5 to overcome the decrease in efficacy when exogenous gene transfer due to existing immunity in humans against human adenovirus type 5.

[0029] In the present invention, the replacement site of the HAdV-5 gene is the knob domain of the fiber protein. Fiber proteins are composed of three domains (tail, shaft, and knob), among which the knob domain is known as a major recognition site for neutralizing antibodies.

[0030] In one embodiment of the present invention, the protein expressed by the chimpanzee adenovirus serotype 6 knob gene preferably includes the amino acid sequence of SEQ ID NO: 1 but is not limited thereto.

[0031] In another embodiment of the present invention, the knob gene sequence of the chimpanzee adenovirus serotype 6 preferably comprises the nucleic acid sequence of SEQ ID NO: 2 but is not limited thereto.

[0032] The knob sequence of the present invention uses the gene sequence of a rare adenovirus serotype having a low serum retention rate in humans. For example, there are ChAdV-1, ChAdV-2, ChAdV-3, ChAdV-5, ChAdV-7, ChAdV-68, etc. In addition to chimpanzee adenovirus, adenovirus serotypes found in cattle, pigs, birds, etc. can be used.

[0033] The present invention provides a chimeric adenovirus vector comprising the tail and shaft domains of human adenovirus type 5 and the knob combined fiber protein of chimpanzee adenovirus serotype 6.

[0034] In another embodiment of the present invention, the chimeric protein in which the hexon protein of human adenovirus type 5 is substituted with hypervariable regions 1-7 of human adenovirus serotype 28 has the amino acid sequence of SEQ ID NO: 3. The hypervariable region gene sequence in the hexon protein of the human adenovirus serotype 28 preferably comprises the nucleic acid sequence of SEQ ID NO: 4 but is not limited thereto.

[0035] The hexon hypervariable region sequence of the present invention uses the gene sequence of a rare adenovirus serotype having a low serum retention rate in humans. For example, there are HAdV-14, HAdV-19, HAdV-32, HAdV-34, HAdV-36, HAdV-43, etc., and adenovirus serotypes found in cattle, pigs, birds, etc. besides humans can be used.

[0036] In the present invention, a chimeric adenoviral vector comprising a hexon protein obtained by replacing a hypervariable region in a hexon protein of human adenovirus type 5 with a gene of human adenovirus type 28 in an adenovirus vector comprising a fiber protein combined with the knob gene of chimpanzee adenovirus serotype 6 is provided.

[0037] The vector of the present invention is characterized in that it comprises Inverted terminal repeat (ITR) sequence, BR322 origin of replication, kanamycin resistance gene, ITR sequence, encapsidation signal, CMV enhancer, CMV promoter, and SV40 ploy (A) signal as vector regulators of the present invention.

[0038] The vector preferably comprises a BR322 origin of replication and an antibiotic resistance gene, and preferably uses a kanamycin resistance gene (Kan) but is not limited thereto. Antibiotic resistance genes for ampicillin, chloramphenicol, tetracycline, or neomycin may be used as antibiotics used for selection.

[0039] The vector preferably comprises an inverted terminal repeat (ITR) sequence and an encapsidation signal of adenovirus. ITR sequences and encapsidation sequences are essential elements for adenoviral DNA replication and packaging, respectively.

[0040] The vector preferably comprises, but is not limited to, a cytomegalovirus immediate early enhancer (CMV enhancer) as an enhancer. For example, an enhancer of Simian virus 40 (SV40) may be used as an enhancer.

[0041] The promoter is preferably a virus-derived or mammalian-derived promoter, and specifically refers to a cytomegalovirus (CMV) promoter. The CMV promoter is recognized as a powerful promoter that efficiently induces exogenous gene expression in all mammalian and insect cells.

[0042] The SV40 poly(A) signal sequence of the vector refers to the poly adenyl signal of Simian virus 40 (SV40) and is preferably comprised in the vector. The use of the SV40 poly(A) signal sequence increases the stability of messenger RNA (mRNA), making protein expression easier.

[0043] The vector of the present invention is preferably the chimeric adenoviral vector shown in FIG. 5 or 6 but is not limited thereto.

[0044] In one embodiment of the present invention, the vector preferably comprises a) a gene encoding the coronavirus spike protein, or b) a gene encoding the E6 and E7 proteins of human papillomavirus type 16 or 18 as an insert but is not limited to thereto.

[0045] In another embodiment of the present invention, the vector is preferably capable of infecting mammalian cells but is not limited thereto.

[0046] The present invention also provides a host cell transformed with the chimeric adenoviral vector of the present invention.

[0047] In one embodiment of the present invention, the transformed cell of the chimeric adenoviral vector that substituted with the chimpanzee adenovirus knob gene and the chimeric adenoviral vector substituted with the hexon hypervariable region of a rare human adenovirus is preferably mammalian cells including HEK293, PERC6 or 911 cells, more preferably HEK293 cells but are not limited thereto.

[0048] For example, an adenovirus-producing cell suitable for the present invention should be a cell capable of producing a virus by trans-complementation of the E1 and E3 gene deletion sites of a replication defective adenovirus, and many cells transformed with the viral early transcription genes are being developed. Preferred cells into which the E1 and E3 genes are introduced comprise A549 (carcinomic human alveolar basal epithelial cell), HeLa (Human cervical cancer cell), CHO (Chinese hamster ovary cell), MRC5 (human lung fibroblast), and PC12 (rat pheochromocytoma cell) etc but not limited thereto.

[0049] The present invention also provides an isolated polypeptide encoded by the chimeric adenoviral vector of the present invention.

[0050] In addition, the present invention provides a vaccine composition comprising the chimeric adenovirus vector of the present invention or a polypeptide encoded by the vector as an active ingredient.

[0051] In the present invention, the above elements are artificially synthesized and provide a vector through the Gibson assembly method.

[0052] In an embodiment of the present invention, the chimeric adenovirus vector-based vaccine is preferably beta variant S or human papillomavirus types 16 and 18 E6E7 proteins but is not limited thereto.

[0053] In addition, the present invention provides mouse immunity and virus neutralization efficacy by using the beta variant strain S based on the chimeric adenovirus vector.

[0054] In addition, the present invention provides the efficacy for the immune response of specific T cells according to the beta variant strain S based on the chimeric adenovirus vector. The present invention provides protective efficacy against SARS-coronavirus-2 and delta variant virus after immunization with beta variant S virus based on chimeric adenovirus vector.

[0055] The present invention provides the efficacy of mouse tumor cell therapy using HPV16 and 18 E6E7 viruses based on the chimeric adenovirus vector.

[0056] The present invention provides efficacy against the immune response of specific T cells against HPV16 and 18 E6E7 viruses based on a chimeric adenovirus vector.

[0057] The present invention provides the efficacy of mouse tumor cell prevention using HPV16 and 18 E6E7 viruses based on chimeric adenovirus vectors.

[0058] In addition, the present invention provides a vaccine composition for preventing or treating an infection of SARS-COV-2 by the chimeric adenovirus vector of the present invention or being encoded by the vector.

[0059] In addition, the present invention provides a composition characterized in that it is a vaccine for treating cervical cancer of human papillomavirus disease by the chimeric adenovirus vector of the present invention or being encoded by the vector.

[0060] The present invention will be described below.

[0061] The present inventors have made diligent efforts to develop a chimeric adenovirus vector that improves the existing immunity problems against HAdV-5 by combining the capsid protein gene sequences of adenovirus serotypes with low serum retention, and as a result, the present invention was completed by confirming that the ChimAd vector in which the knob gene sequence was replaced with the corresponding gene of ChAdV-6 based on the recombinant HAdV-5 vector, or ChimAd-H28 vector in which the hypervariable region in the hexon protein of HAdV-5 was replaced with the corresponding gene of serotype 28 can be usefully used as an optimal adenoviral vector.

[0062] Since more than 80% of the population naturally possesses neutralizing antibodies against adenovirus, vaccines based on adenovirus vectors loaded with foreign genes have a problem in that immune efficacy is reduced due to existing immunity against HAdV-5. The developed chimeric adenoviral vector presents the potential as an excellent adenoviral vector that improves these problems of the conventional recombinant HAdV-5 vector through cytopathic effect (CPE) and in vitro neutralizing antibody tests.

[0063] Specifically, the present invention provides an AdKnob vector in which the gene sequence of the HAdV-5 knob domain is replaced with the corresponding gene of chimpanzee adenovirus serotype 6, a rare serotype, based on a recombinant vector. is. based on a recombinant HAdV-5 vector composed of a BR322 origin of replication, a kanamycin resistance gene, an inverted terminal repeat (ITR) sequence, an encapsidation signal, a CMV enhancer, a CMV promoter, and a SV40 poly(A) signal sequence.

[0064] In addition, the present invention provides a ChimAd-H28 vector in which a region in the hexon protein of HAdV-5 is replaced with a corresponding gene of HAdV-28, an adenovirus serotype with low human seroprevalence, based on the ChimAd vector.

[0065] The present invention provides the existing immune avoidance effect by testing the cell infectivity and neutralizing antibody for the ChimAd and ChimAd-H28 vectors.

[0066] To analyze the cell infectivity of the chimeric adenovirus provided in the present invention, compared to the existing HAdV-5, the virus titer was measured under the same MOI virus dose (multiplicity of infection) and time conditions. As a result, it was confirmed that the number of cells infected with the chimeric adenovirus and the virus titer increased more than the HAdV-5 virus. Due to the knob gene of ChAdV-6 and the HVR gene of HAdV-28, the ability to infect host cells increased, and the virus productivity increased, compared to HAdV-5.

[0067] In addition, in the present invention, the neutralizing antibody immunity evasion ability of the ChimAd vector against HAdV-5 was evaluated through an in vitro neutralizing antibody test. In the in vitro neutralizing antibody test, anti-HAdV-5 antibody and virus were reacted, were infected to HEK293 cells, and finally cells infected with HAdV-5, ChimAd and ChimAd-H28 viruses were identified. As a result, when compared with the HAdV-5 virus, about 4 times more adenovirus-infected cells were confirmed in 10^1 - 10^2 by the ChimAd virus, indicating a higher ability to evade existing immunity than the conventional HAdV-5. In addition, about 2.8 times more infected cells were identified by the ChimAd-H28 virus.

[0068] The present invention provides a composition comprising an immunological adjuvant, the chimeric vector or a polypeptide encoded by the vector, and optionally a pharmaceutically acceptable excipient.

[0069] A composition according to the invention comprising an adjuvant can be used, for example, as a vaccine for human subjects. Immunological adjuvants, also referred to herein as "adjuvants" for short, accelerate, prolong and/or enhance the quality and/or strength of the immune response to the antigen/immunogenicity, compared to administration of the antigen alone. This reduces the amount of antigen/immunogen required for any given vaccine and/or the frequency of injections required to generate an adequate immune response to the desired antigen/immunogen.

[0070] Examples of adjuvants that may be used in connection with the composition according to the present invention are gel-like precipitates of aluminum hydroxide (allum); AlPO₄; alhydrogel; bacterial products from the outer membrane of Gram-negative bacteria, in particular monophosphoryl lipid A (MPLA), lipopolysaccharide (LPS), muramyl dipeptide and derivatives thereof; Freund's incomplete adjuvant; liposomes, particularly neutral liposomes, liposomes containing a composition and optionally a cytokine; non-ionic block copolymers; ISCOMATRIX adjuvant (Drane et al, 2007); unmethylated DNA containing CpG dinucleotides (CpG motifs), in particular CpG ODNs having a phosphorothioate (PTO) backbone (CpG PTO ODN) or phosphodiester (PO) backbones (CpG PO ODN); synthetic lipopeptide derivatives, particularly Pam3Cys; lipoarabinomannan; peptidoglycan; zymosan; heat shock proteins (HSP), especially HSP 70; dsRNA and its synthetic derivatives, particularly Poly I:poly C; polycationic peptides, especially poly-L-arginine; taxol; fibronectin; flagellin; imidazoquinoline; cytokines with adjuvant activity, in particular GM-CSF, interleukin-(IL)-2, IL-6, IL-7, IL-18, type I and type II interferons, in particular interferon-gamma, TNF-alpha; 25-dihydrophytamine D3 (calcitriol); and synthetic

oligopeptides, particularly MHCII-provided peptides. Non-ionic block polymers containing polyoxyethylene (POE) and polyoxypropylene (POP), for example POE-POP-POE block copolymers, can be used as adjuvants (Newman et al, 1998). Adjuvants of this type are particularly useful for compositions that include nucleic acids as active ingredients.

[0071] If desired, various pharmaceutically acceptable excipients may be used. Preferred pharmaceutically acceptable excipients are mentioned below when discussing the use according to the present invention.

[0072] Activation of specific receptors can stimulate an immune response. Such receptors are known to the person skilled in the art and include, for example, cytokine receptors, especially type I cytokine receptors, type II cytokine receptors, TNF receptors; and vitamin D receptors that act as transcription factors; and toll-like receptor 1 (TLR1), TLR-2, TLR 3, TLR4, TLR5, TLR-6, TLR7, and TLR9. Agonists for these receptors have adjuvant activity, ie are immunostimulatory.

[0073] According to an embodiment of the present invention, ChimAd-CV and ChimAd-CVN comprising the S sequence of the beta variant strain based on the ChimAd vector and the N partial sequence of SARS-coronavirus-2 are provided.

[0074] In another embodiment of the present invention, the modified spike protein of the beta variant strain of SARS-coronavirus-2 is preferably comprised of the amino acid sequence shown in SEQ ID NO: 7, and the modified spike protein of the beta variant strain of SARS-coronavirus-2 The gene encoding the modified spike protein of the beta variant strain of SARS-coronavirus-2 is preferably composed of the nucleotide sequence shown in SEQ ID NO: 8 but all mutant sequences that achieve the object of the present invention through mutation of one or more substitutions, deletions, inversions, etc. to the sequence are also included in the scope of the present invention.

[0075] In one embodiment of the present invention, the amino acid residues from positions 44 to 180 of the nucleocapsid protein of the SARS-coronavirus-2 is shown in the amino acid sequence of SEQ ID NO: 9, and the amino acid residues from positions 247 to 364 of the nucleocapsid protein is preferably shown in the amino acid sequence of SEQ ID NO: 10, but all mutant sequences that achieve the object of the present invention through mutation such as one or more substitutions, deletions, inversions, etc. included in the scope of the invention.

[0076] The ChimAd-CV and ChimAd-CVN vaccine compositions produced in the present invention provide excellent neutralizing antibody titers against SARS-coronavirus-2, beta variants and delta variants.

[0077] The ChimAd-CV and ChimAd-CVN vaccine compositions produced according to an embodiment of the present invention provide excellent neutralizing antibody titers against pseudo-viruses based on beta variants.

[0078] The ChimAd-CV and ChimAd-CVN vaccine compositions produced in the present invention provide efficacy against the immune response of specific T cells.

[0079] In the present invention, the ChimAd-CV and ChimAd-CVN vaccine compositions produced by the above method provide protective efficacy against SARS-coronavirus-2 and delta variants.

[0080] Body weight change and survival rate after immunization and viral infection with ChimAd-CV and ChimAd-

CVN produced according to an embodiment of the present invention were confirmed. Both experimental groups inoculated with ChimAd-CV and ChimAd-CVN against SARS-coronavirus-2 infection showed a 100% survival rate, confirming excellent defense. In addition, for delta variant infection, ChimAd-CV maintained 100% survival rate and 100% average body weight after 7 days of infection, but ChimAd-CVN showed a survival rate of 40% and average weight loss within about 20% after 7 days of infection.

[0081] According to an embodiment of the present invention, ChimAd-HPV comprising E6E7 and type 18 E6E7 sequences of human papillomavirus type 16 based on a ChimAd vector is provided.

[0082] In another embodiment of the present invention, the human papillomavirus type 16 E6E7 and type 18 E6E7 proteins are preferably comprised of the amino acid sequence shown in SEQ ID NO: 13, and The gene encoding the modified E6 and E7 of the human papillomavirus type 16 E6E7 and type 18 are preferably composed of the nucleotide sequence shown in SEQ ID NO: 14, but all mutant sequences that achieve the object of the present invention through mutation such as one or more substitutions, deletions, inversions, etc. are also included in the scope of the present invention.

[0083] In the present invention, to confirm the efficacy of human papillomavirus in treating cervical cancer, TC-1 tumor cells and the produced virus were injected. According to an embodiment of the present invention, the therapeutic efficacy was confirmed by measuring the tumor volume after injection of the recombinant antigen. As a result, the therapeutic efficacy of the ChimAd-HPV virus on tumor volume compared to the control group (ChimAd) was confirmed.

[0084] In addition, the ChimAd-HPV vaccine composition produced according to the embodiment of the present invention provides efficacy against the immune response of specific T cells.

[0085] To confirm the immune response efficacy of the specific T cells of virus produced in the present invention, ELISPOT (Enzyme-Linked ImmunoSpot, ELISPOT) experimental technique was used. In the present invention, tumor cells and antigen-injected mouse splenocytes were separated to confirm the immune response to each antigen. As a result, it is thought that the ChimAd-HPV virus induces a specific immune response against the HPV E7 protein antigen.

[0086] The present invention provides the efficacy of preventing mouse tumor cells with respect to the ChimAd-HPV vaccine composition produced by the above method. In the present invention, to confirm the preventive effect of cervical cancer by human papillomavirus, the virus and TC-1 tumor cells were injected. In the present invention, after injecting the recombinant antigen, tumor cells were injected to confirm the preventive effect on cancer. As a result, it was confirmed that the ChimAd-HPV virus did not generate tumor cells compared to the control group.

Advantageous Effects

[0087] As can be seen through the present invention, the present invention can provide an optimal adenoviral vector in the development of a therapeutic agent or vaccine for various diseases. In addition, when the chimeric adenoviral vector constructed in the present invention is infected with host cells for production, it can contribute to productivity improvement by exhibiting superior cell infectivity com-

pared to recombinant HAdV-5 vector-based vaccines. In addition, the chimeric adenovirus of the present invention can overcome existing immunity to HAdV-5 when inoculated into humans. In addition, it is considered that it will be an excellent chimeric adenovirus vector as the immune effectiveness that can more effectively transfer foreign genes is confirmed when developing a preventive or therapeutic vaccine for SARS-coronavirus-2 and human papillomavirus.

DESCRIPTION OF DRAWINGS

[0088] FIG. 1 shows the results of sequence analysis of the amino acid sequences of the fiber proteins of HAdV-5 and ChAdV-6 in the present invention through Clustal Omega and ESPript programs (A) and a schematic diagram of the fiber proteins of the chimeric adenovirus vector (B),

[0089] FIG. 2 shows the results of sequence analysis of the amino acid sequences of the hexon proteins of HAdV-5 and HAdV-28 in the present invention through Clustal Omega and ESPript programs (A) and a schematic diagram of the hexon proteins of the chimeric adenovirus vector (B),

[0090] FIG. 3 schematically shows the sequence of the vector regulator synthetic gene produced in the present invention,

[0091] FIG. 4 is a schematic diagram showing the HAdV-5 vector construction process in the present invention,

[0092] FIGS. 5 and 6 are schematic diagrams of the ChimAd vector production process (FIG. 5) and the final ChimAd vector structure (FIG. 6) produced in the present invention.

[0093] FIGS. 7 and 8 are schematic diagrams showing the production process of the ChimAd-H28 vector (FIG. 7) and the final structure of the ChimAd-H28 vector (FIG. 8) produced in the present invention,

[0094] FIG. 9 shows the result of confirming the chimeric capsid gene in the two chimeric adenovirus vectors (ChimAd, ChimAd-H28) prepared in the present invention by amplifying the knob gene (0.5 Kb) of ChAdV-6 and the hexon HVR gene of HAdV-28 (1 Kb) by PCR, separating them on a 0.8% agarose gel, and passing ultraviolet rays through them,

[0095] FIG. 10 is a photograph at 200× magnification using an optical microscope to show the formation of cytopathic effect (CPE) after induction of transformation in HEK293 cells with the HAdV-5 vector and two types of chimeric adenovirus vectors prepared in the present invention,

[0096] FIG. 11 shows the expression of hexon (about 108 kDa), fiber (about 61 kDa), and penton (about 68 kDa) proteins through Western blot after HAdV-5 and two kinds of chimeric adenovirus were produced in HEK293 cells,

[0097] FIG. 12 is a graph showing the number of positive cells infected with adenovirus taken at 200-fold magnification of an optical microscope, counting, and converting to virus titer through immunocytochemistry using an anti-hexon polyclonal antibody 48 hours after infecting HEK293 cells with 0, 5, 10, 25, or 50 MOI virus doses of HAdV-5 and chimeric adenoviruses prepared in the present invention. The X-axis represents the dose of virus infected cells, and the Y-axis represents the HAdV-5, ChimAd, and ChimAd-H28 virus titers according to each virus dose. * indicates the

significance level between the HAdV-5 virus and the chimeric adenovirus verified using a paired t-test (* $p < 0.05$, ** $p < 0.01$),

[0098] FIG. 13 shows the in vitro neutralizing antibody test in HEK293 cells by reacting HAdV-5 and chimeric adenoviruses prepared in the present invention with anti-HAdV-5 polyclonal antibodies, and then measuring the number of adenovirus-infected cells and the measured values are plotted graphically. The X axis represents the dilution factor of the anti-HAdV-5 polyclonal antibody serially diluted from 10^1 to 10^4 , and the Y axis represents the number of cells infected by HAdV-5, ChimAd and ChimAd-H28 viruses. * indicates the level of significance between the HAdV-5 virus and the chimeric adenovirus verified using a paired t-test (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$, **** $p < 0.0005$),

[0099] FIG. 14 shows the result of in vitro neutralizing antibody test in HEK293 cells by reacting HAdV-5 and chimeric adenoviruses prepared in the present invention with healthy human serum ($n=7$) and the values expressed as reciprocal numbers by measuring the number of cells infected with adenovirus shown as a dot plot. The X-axis represents the adenovirus reacted with the serum, and the Y-axis represents the adenovirus neutralizing antibody (Ad Nab) titer for 7 human serum samples. The significance level between the HAdV-5 virus, the ChimAd virus, and the ChimAd-H28 virus, verified using a paired t-test, is shown as a P value,

[0100] FIG. 15 is a schematic diagram of cloning ChimAd-CV and ChimAd-CVN using the Gibson assembly gene synthesis method in a ChimAd vector obtained through gene synthesis,

[0101] FIG. 16 shows the SARS-Coronavirus-2 S protein results (A) and beta variant S protein results (B) for virus-binding antibody titers induced by ChimAd-CV and ChimAd-CVN using BALB/c female mice,

[0102] FIG. 17 shows the neutralizing antibody titers induced by ChimAd-CV and ChimAd-CVN in BALB/c female mice as neutralizing antibody measurement results (A, B) using a beta variant (B.1.351 variant type) pseudovirus.

[0103] FIG. 18 shows the results of cellular immunization (A, B) using S peptides of ChimAd-CV and ChimAd-CVN as stimulating antigens and the results of cellular immunization (C, D) using NP peptides as stimulating antigens, using BALB/c female mice,

[0104] FIG. 19 shows the results of S protein of SARS-coronavirus-2 (A) and the S protein of delta variant strain (B) induced by ChimAd-CV and ChimAd-CVN using hACE2 transgenic female mice,

[0105] FIG. 20 shows the results of neutralizing antibody measurement using FRNT of SARS-coronavirus-2 (A, B) and of the delta variant strain (C,D) as a result of confirming the viral neutralizing antibody titer induced by ChimAd-CV and ChimAd-CVN using hACE2 transgenic female mice,

[0106] FIG. 21 shows survival rate (A, B) and the average body weight (C,D) after SARS-coronavirus-2 (Wuhan-Hu-1), delta variant (B.1.617.2) challenge test in mouse animal models immunized by ChimAd-CV and ChimAd-CVN using hACE2 transgenic female mice,

[0107] FIG. 22 shows coronavirus titers detected in the lungs after the challenge test,

[0108] FIG. 23 is a schematic diagram of cloning ChimAd-HPV by using the Gibson assembly gene synthesis

method in a ChimAd vector from a human papillomavirus gene obtained through gene synthesis,

[0109] FIG. 24 shows the results of cellular immunization (A, B) using the HPV16-type E7 peptide as a stimulating antigen, as a result of ELISPOT, showing the specific T cell immune response to the ChimAd-HPV vaccine candidate.

[0110] FIG. 25 shows the results of measuring the size of tumors, after subcutaneously injecting TC-1 cells, a tumor cell line, into C57BL/6 mice, and vaccine candidates, including a control group were injected before and after tumor cell injection to measure the size of cancer over time Results showing the lowest tumor volume in the ChimAd-HPV group (A, B) are shown.

MODE FOR INVENTION

[0111] Hereinafter, the present invention will be described in more detail by the following examples. However, the following examples are described with the intention of illustrating the present invention, and the scope of the present invention is not to be construed as being limited by the following examples.

EXAMPLE

Example 1: Gene Synthesis for Construction of Chimeric Adenoviral Vectors

1-1) Synthetizing ChAdV-6 Knob Gene

[0112] In order to construct a chimeric adenoviral vector by replacing the knob domain at the end of the fiber protein of HAdV-5 with the corresponding gene of ChAdV-6, a non-human adenovirus, genetic homology of the amino acid sequence to the fiber protein of HAdV-5 (Genebank: AY339865.1) and ChAdV-6 (Genebank: AY530877) was analyzed using the Clustal Omega program. Finally, the Clustal Omega results visualized the homology region through ESPript figure (FIG. 1A), and among the knob domains of ChAdV-6 (SEQ ID NO: 1), a base sequence corresponding to 405 to 574 amino acids (SEQ ID NO: 1), which is a region with low homology to the gene sequence of HAdV-5, was obtained by gene synthesis. (FIG. 1B, SEQ ID NO: 2).

1-2) Synthetizing HAdV-28 Hypervariable Region (HVR) Gene

[0113] To construct a chimeric adenovirus vector by replacing the hypervariable region in the hexon protein of HAdV-5 with the corresponding gene of HAdV-28, an adenovirus serotype with low seroprevalence in humans, genetic homology of the amino acid sequence to the hexon protein of HAdV-5 (Genebank: AY339865.1) and HAdV-28 (Genebank: DQ149626.1) were analyzed using the Clustal Omega program. Finally, the Clustal Omega results were visualized through the ESPript figure, and the HVR1 region to the HVR7 region with low genetic homology of amino acid sequences between HAdV-5 and HAdV-28 were set (Table 1, FIG. 2A). Subsequently, to replace the HVR region in the hexon protein of HAdV-5 with the corresponding gene of HAdV-28, the nucleotide sequence from the beginning of the HVR1 region of HAdV-28 (132 amino acids) to the end of the HVR7 region (449 amino acids) was gene synthesized. (FIG. 2B, SEQ ID NO: 4). the genetic homology of amino acid sequences between HAdV-5 and HAdV-28 was

set from the HVR1 region to the HVR7 region (Table 1, FIG. 2A). Subsequently, to replace the HVR region in the hexon protein of HAdV-5 with the corresponding gene of HAdV-28, the nucleotide sequence from the beginning of the HVR1 region of HAdV-28 to the end of the HVR7 region was gene synthesized. (FIG. 2B, SEQ ID NO: 4).

TABLE 1

Target	HVR region	Position	Sequence
HAdV-5	HVR1	132-168	PCWEDEAATALEINLEEDDDNEDEV DBQAEQKTHV (SEQ ID NO: 15)
	HVR2	188-195	VEGQTPKY (SEQ ID NO: 16)
	HVR3	209-220	SQWYETEINHAA (SEQ ID NO: 17)
	HVR4	248-265	GILVKQQNGKLESQVEMQ (SEQ ID NO: 18)
	HVR5	268-284	STTEATAGNDNLTPKV (SEQ ID NO: 19)
	HVR6	305-316	TIKEGNSRELMG (SEQ ID NO: 20)
	HVR7	418-458	GGVINTETLTQVKPKTGQENGWEKDA TEFSDKNEIRVGNNF (SEQ ID NO: 21)
HAdV-28	HVR1	132-153	SSQWDAQEKSGQSDMVTKTHT (SEQ ID NO: 22)
	HVR2	173-185	TEITADNQKKEIF (SEQ ID NO: 23)
	HVR3	199-209	ENWQENEVFG (SEQ ID NO: 24)
	HVR4	237-256	AKFKTPAEGQEPKELDIDL (SEQ ID NO: 25)
	HVR5	259-273	DTDGGTADTEYKADI (SEQ ID NO: 26)
	HVR6	294-305	GKEDDSSEINLV (SEQ ID NO: 27)
	HVR7	407-449	DGLGTNATYQGVKSTGDGATQSGWA KDDTMRQNQICRNIY (SEQ ID NO: 28)

[0114] Table 1 shows HVR regions selected through nucleotide sequence analysis between HAdV-5 and HAdV-28.

1-3) Synthesis of Vector Regulators

[0115] Vector regulators to be inserted into the chimeric adenovirus vector were synthesized by combining vector regulators known to date (e.g., promoter, origin of replication, etc.). In more detail, corresponding vector regulators were obtained by synthesizing genes in the order of ITR (inverted terminal repeat, Genebank: AY339865.1), BR322 origin of replication (Genebank: MT612434.1), kanamycin resistance gene (Genebank: MT084773.1), ITR sequence, encapsidation signal, Genebank: AY339865.1), CMV enhancer (Genebank: MT267310.1), CMV promoter (Genebank: MT267310.1), and SV40 poly(A) (Genebank: MT612432.1) (FIG. 3).

Example 2: Preparation of Chimeric Adenoviral Vectors

2-1) Preparing HAdV-5 Vector

[0116] Using DNA of HAdV-5 (ATCC, VR-1516) as a template, PCR amplification was performed using primers except for some of the E1 and E3 genes to obtain the amplified products (fragments A-G) (Table 2). The synthetic gene (4.1 Kb) comprising the 7 vector PCR amplification products and the synthesized CMV promoter sequence has the same sequence (about 20 bp) at each end, and were combined through a gibson assembly (NEB, E2611S) reaction to construct HAdV-5 vectors (FIG. 4). A ChimAd vector was constructed using this as a basic vector.

TABLE 2

Primer name	Sequence (5' to 3' direction)	PCR product
Fragment A	F: TAAGGGTGGGAAGAATATATAAGGTGGGGGTC (SEQ ID NO: 29) R: GCC TTC CAT CAA GGG CAT CCC G (SEQ ID NO: 30)	4 Kb
Fragment B	F: GTAAAGTTCCAAGAAGCGCGGGATGC (SEQ ID NO: 31) R: AGTTAATCTCCTGGTTCACCGTCTG (SEQ ID NO: 32)	4 Kb
Fragment C	F: GTAACCGCATACGAGCAGACGGTG (SEQ ID NO: 33) R: CGTGGAGCGGAAGGTCACG (SEQ ID NO: 34)	4 Kb
Fragment D	F: GCCAGACATGATGCAAGACCCCGTG (SEQ ID NO: 35) R: CGTACCACTGAGATTCTCTATTGAGGTTT (SEQ ID NO: 36)	4 Kb
Fragment E	F: CAACCTGAACCTCAAATAGGAGAATCTCAG (SEQ ID NO: 37) R: GGATAACGATCTAAAGGCGAACTTCAAAC (SEQ ID NO: 38)	4 Kb
Fragment F	F: GTCAGGCAGTAGTTTGAAGTTCGC (SEQ ID NO: 39) R: CTCTAGTTATACTAGAGGATCTTGATGTAATCCAGGGTTAG (SEQ ID NO: 40)	4.7 Kb
Fragment G	F: CCTCTAGTTATACTAGAGTACCCGGGATCTTATTCCTTTA AC (SEQ ID NO: 41) R: CGGAGTAACTTGATGTGTGGGAATTG (SEQ ID NO: 42)	5 Kb
Gene synthesis (promoter, etc.)	F: CAATTCCCAACATACAAGTTACTCCG (SEQ ID NO: 43) R: TATATTCTTTCCACCCTTACGCGTTAAGATACATTGATG (SEQ ID NO: 44)	5 Kb

[0117] Table 2 shows PCR primers used to construct HAdV-5 vector

2-2) Preparation of ChimAd Vector

[0118] A vector PCR amplification product was obtained by PCR amplification using the HAdV-5 vector prepared above as a template and using the primers shown in Table 3, excluding the knob gene sequence. The synthesized ChAdV-6 knob gene was PCR amplified to have the same sequence (about 20 bp) as the vector PCR amplification product at the 3' and 5' ends to obtain an insert gene, and was combined with the vector PCR amplification product through gibson assembly reaction to construct a ChimAd vector (FIGS. 5 and 6).

TABLE 3

Primer name	Sequence (5' to 3' direction)	PCR product
6Knob vector	F: CCTTTGCTACCAACTCATTCTCTTTTCA TACATTGCCC (SEQ ID NO: 45) R: ACAATTAGGGCTTGGGTCAGGTGTGGTCCACA AAGTTAGC (SEQ ID NO: 46)	33.4 Kb
6Knob insert	F: GCTAACTTTGTGGACCACACCTGACCCAAGCC CTAATTGT (SEQ ID NO: 47)	0.5 Kb

TABLE 3-continued

Primer name	Sequence (5' to 3' direction)	PCR product
	R: GGGCAATGTATGAAAAAGAGAATGAGTTGGTA GCAAAAGG (SEQ ID NO: 48)	

[0119] Table 3 shows the PCR primers used to amplify the vector and the knob gene for the construction of the ChimAd vector

2-3) Preparation of the ChimAd-H28 Vector

[0120] Using the ChimAd vector as a template, the vector sequence excluding the HVR region (19,879-20,858 bp) of HAdV-5 was PCR amplified using the primers in Table 4 by PCR, and the amplified product (32.9 Kb) and the PCR

amplified product of the HVR sequences of HAdV-28 (1 Kb) synthesized above were combined through Gibson assembly reaction to construct a ChimAd-H28 vector (FIGS. 7 and 8).

TABLE 4

Primer name	Sequence (5' to 3' direction)	PCR product
28HVR vector	F:	32.9 Kb
	TGCAGGGGTAACATCTACGCCATGGAAAT	
	CAATCTAAATG (SEQ ID NO: 49)	
	R:	
28HVR insert	TGCGCATCCCACTGACTGGAATTTGGGGC	1 Kb
	ACCCTTGGGAG (SEQ ID NO: 50)	
	F:	
	CTCCCAAGGGTGCCCAAATTCAGTCAG	
	TGGGATGCGCA (SEQ ID NO: 51)	
	R:	
	CATTTAGATTGATTTCCATGGCGTAGATG	
	TTACCCCTGCA (SEQ ID NO: 52)	

[0121] Table 4 shows the PCR primers used to amplify the vector and the 28-type hexon HVR gene for the construction of the ChimAd-H28 vector.

[0122] As shown in FIG. 9, only the ChAdV-6 knob gene inserted in 0.5 Kb is amplified in the ChimAd vector by the PCR method, In the ChimAd-H28 vector using this as a template, the amplification of the ChAdV-6 knob gene and the 1 Kb HAdV-28 hexon HVR gene were simultaneously confirmed. Through this, two final chimeric adenoviral vectors (ChimAd, ChimAd-H28) were obtained by confirming the presence or absence of the chimeric capsid gene in the HAdV-5 vector.

Example 3: Production of Adenovirus

[0123] To evaluate the virus formation of HAdV-5 and the two chimeric adenoviruses designed and produced in the present invention, each vector was cut and removed from the pBR322 replication start point and kanamycin resistance gene sequence unnecessary for adenovirus formation using Pac I restriction enzyme, and HEK293 cells (human embryonic kidney; ATCC, CRL-1573) expressing E1 and E3 genes subcultured the previous day at 1.6×10⁶ cells in a 100 mm cell culture dish were transformed using Lipofectamine 2000 (Invitrogen, #11668027).

[0124] Transformed cells were cultured for 10-20 days in an incubator maintained at 37° C. and 5.0% CO₂ partial pressure, and the formation of a cytopathic effect by adenovirus infection was monitored (FIG. 10). To harvest HAdV-5 and two chimeric adenoviruses from whole CPE, the cells attached to the bottom of the flask were detached using a scraper, and the supernatant was removed by centrifugation at 4° C. and 500×g together with the culture medium, and after washing with 1×PBS, adenovirus was released by repeating the freezing-thawing process three times in the resulting pellet. Cell debris was centrifuged at 14,000×g at room temperature, and the adenovirus-containing supernatant was dispensed into 1.5 mL microcentrifuge tubes and stored at -80° C.

[0125] As shown in FIG. 10, the formation of CPE by transformation in HEK293 cells using the HAdV-5, ChimAd, and ChimAd-H28 vectors prepared in the present invention was confirmed at 200× magnification through an optical microscope. From these results, it is considered that adenovirus replication was achieved in the host cell by the

three vectors constructed above, and cytotoxicity was induced by the virus propagation or the expressed adenovirus protein.

Example 4: Western Blot Analysis for Confirming Adenovirus Envelope Protein

[0126] To confirm the expression of adenoviral structural proteins for HAdV-5 and chimeric adenoviruses produced in the HEK293 cell line, Western blot analysis was performed as follows. After adding SDS-PAGE loading dye to HAdV-5, ChimAd, and ChimAd-H28 viruses and bathing at 100° C. for 15 minutes, and were separated by electrophoresis on 10% SDS-PAGE at 100 V, and then transferred to a PVDF membrane. After removing the activity of the PVDF membrane with 5% skim milk solution, the primary antibody, anti-HAdV-5 polyclonal antibody (Abcam, #ab6982) was diluted 1:1000 with TBS solution and reacted at room temperature for 1 hour. After washing three times with TBS-T buffer solution, the secondary antibody, HRP-attached goat-derived anti-mouse immunoglobulin G was diluted 1:5000 with TBS solution, reacted at room temperature for 1 hour, and then washed with TBS-T buffer solution repeatedly. Confirmation of Western results was confirmed after color development using diaminobenzidine (Sigma, #34002) (FIG. 11).

[0127] As shown in FIG. 11, in HAdV-5, ChimAd and ChimAd-H28 viruses, In the virus, hexon (about 108 kDa), fiber (about 61 kDa), and penton (about 68 kDa) proteins were detected in response to all anti-HAdV-5 polyclonal antibodies specifically. When comparing the detected fiber protein of the HAdV-5 virus and the detected fiber protein of the ChimAd virus, since fiber proteins of the same size were confirmed in ChimAd virus, it can be seen that the chimAd virus knob protein replaced with the ChAdV-6 knob gene was normally expressed. In addition, hexon protein of the same size was confirmed in HAdV-5 and ChimAd-H28, indicating that the hexon protein of the ChimAd-H28 virus replaced with the HVR gene of HAdV-28 was normally expressed. In addition, it can be confirmed that the main coat proteins (hexon, fiber, and penton) of adenovirus are stably expressed by the two chimeric adenovirus vectors.

Example 5: Analysis of Adenovirus Infectivity in HEK293 Cells

[0128] To analyze the cell infectivity of two chimeric adenoviruses in which the knob protein sequence that binds to the CAR receptor of the host cell and acts on cell entry is replaced with the gene sequence of ChAdV-6, cells infected with HAdV-5, ChimAd and ChimAd-H28 viruses were observed under the same MOI virus dose condition.

[0129] Adenovirus infection titer was measured using the QuickTiter™ titer immunoassay kit (Cell biolabs, #VPK-109) of immunocytochemistry using an anti-hexon polyclonal antibody that binds to adenovirus hexon protein. The adenovirus to be used for titer measurement was diluted from 10¹ to 10⁶ by 10-fold serial dilution using 1×PBS, and infection was induced by dispensing 100 μl of adenovirus dilution into HEK293 cells in a 24-well plate (SPL, #30024) prepared with a cell count of 2.5×10⁵ by subculture 1 hour before. Immunostaining for detection of adenovirus was performed 48 hours after infection in an incubator maintained at a temperature of 37° C. and a partial pressure of CO₂ of 5.0%. Cells infected for 48 hours were fixed by

removing the supernatant, dispensing cold methanol, and leaving the cells at -20°C . for 20 minutes. After sufficiently removing methanol from the fixed cells with 1×PBS solution, the activity was inhibited with 1% skim milk (Difco, #232100) solution, and the primary antibody, anti-hexon antibody, was diluted with 1×PBS and reacted at room temperature for 1 hour. After washing three times with 1×PBS, the HRP-conjugated secondary antibody was diluted to 1× and reacted. Then, using 3,3'-diaminobenzidine stain kit (DAB) (Thermo Fisher, #34002), color was developed for 10 minutes, and then, using an optical microscope, after select the virus dilution interval in which brown spots, which are positive cells infected with adenovirus, are observed between 50 and 100, the infection titer (pfu/mL) of HAdV-5, ChimAd and ChimAd-H28 viruses was measured by applying the adenovirus titer conversion formula provided by Cell Biolabs.

[0130] For the infectivity analysis, HEK293 cells subcultured in a 24-well plate at 2.5×10^5 cells were incubated for 1 hour in an incubator at a temperature of 37°C . and a CO_2 partial pressure of 5.0%, and then based on the pfu/mL value obtained from the virus titer measurement, HAdV-5 and chimeric adenoviruses were diluted with 1×PBS at 0, 5, 10, 25, and 50 MOI virus doses, respectively, to induce infection in the cells. 48 hours after infection, the number of adenovirus-infected cells at each MOI virus dose was counted in at least 4 fields at 200× magnification under an optical microscope through immunocytochemical staining using an anti-hexon polyclonal antibody (Table 5), and the number of positive cells (brown spots) was converted into pfu/mL values, and the infectivity of HAdV-5 and chimeric adenoviruses for each MOI virus dose was compared (FIG. 12).

TABLE 5

Ad infected cell spots	Virus dose (MOI)			
	5	10	25	50
HAdV-5	4.0	8.1	16.6	22.2
ChimAd	10.1	17.9	21.6	34.6
ChimAd-H28	9.5	17.7	19.8	32.3

[0131] Table 5 shows the number of cells infected by HAdV-5 and chimeric adenovirus at the same MOI. As shown in Table 5, when HEK293 cells were infected with the same time and MOI virus dose, the number of cells infected with ChimAd and ChimAd-H28 viruses was higher than that of HAdV-5 virus.

[0132] In addition, when comparing the virus titers for each MOI virus dose in FIG. 12, the ChimAd virus and ChimAd-H28 virus had 1.3-3 times and 1-2 times more virus titers than HAdV-5 virus, respectively. showed a scientifically significant difference. Through these results, it was confirmed that not only cell infectivity but also virus productivity was increased due to replacement of the ChAdV-6 knob gene and the HAdV-28 hexon HVR gene.

Example 6: In Vitro Neutralizing Antibody Test Using Anti-HAdV-5 Polyclonal Antibody

[0133] To evaluate the existing immune evasion ability of the HAdV-5 vector due to replacement of the knob gene and HVR region of hexon, which is the main neutralizing antibody recognition site, in the two chimeric adenoviruses produced, an in vitro neutralizing antibody test was per-

formed using an anti-HAdV-5 polyclonal antibody (Abcam, #ab6982) as a neutralizing antibody against HAdV-5.

[0134] In the in vitro neutralizing antibody test, the anti-HAdV-5 antibody was diluted using 1×PBS solution to a 10^{-4} dilution factor through serial dilution, and after neutralizing with 1.25×10^7 VP of HAdV-5, ChimAd, and ChimAd-H28 viruses corresponding to a 50 MOI virus dose for 1 hour at room temperature, 2.5×10^5 cells were dispensed into HEK293 cells in a prepared 24-well cell culture plate, respectively and cultured in an incubator maintained at a temperature of 37°C . and a CO_2 partial pressure of 5.0%, and viral infection was induced for 48 hours. 48 hours after infection, immunocytochemical staining using an anti-hexon polyclonal antibody was performed, and by counting the number of positive cells infected with HAdV-5, ChimAd and ChimAd-H28 viruses in at least 4 fields at 200× magnification using an optical microscope for each anti-HAdV-5 polyclonal antibody dilution factor (10^1 - 10^4), the ability to avoid neutralization against anti-HAdV-5 antibodies was analyzed (Table 6).

TABLE 6

Ad infected cell spots	Dilution of anti-HAdV-5 antibody			
	10^1	10^2	10^3	10^4
HAdV-5	1.0	12.5	44.1	57.8
ChimAd	4.4	40.1	97.0	100.8
ChimAd-H28	2.6	13.5	69.5	100.1

[0135] Table 6 shows the number of cells infected with HAdV-5 and chimeric adenovirus after reaction with anti-HAdV-5 polyclonal antibody. As shown in FIG. 13, compared to the HAdV-5 virus, by ChimAd virus about 3.1 to about 4.5 times at low antibody dilution factor (10^1 - 10^2) and about 1.8 to about 2.2 times more adenovirus-infected cells at high antibody dilution factor (10^3 to 10^4). ChimAd-H28 virus also confirmed about 2.8 times more adenovirus-infected cells than HAdV-5 virus at a low antibody dilution factor and about 1.5 to about 1.7 times at a high antibody dilution factor, indicating a statistically significant difference. Through this, it was obtained that the neutralizing antibody reaction can be avoided up to 4.5 times only by replacing the knob gene in the adenovirus capsid protein. In addition, the above results are 2 times higher avoidance activity than the result of in vitro neutralizing antibody test of a chimeric adenoviral vector (2 fold avoidance compared to HAdV-5 vector) in which the recently developed HAdV-5 structure was replaced with the HAdV-3 knob gene.

[0136] As a result, the ChimAd vector constructed in the present invention is considered to be an adenoviral vector with superior immune evasion ability against HAdV-5 than conventional chimeric adenoviral vectors.

Example 7: In Vitro Neutralizing Antibody Test Using Human Serum

[0137] To evaluate the neutralizing antibody titer in human serum capable of recognizing ChimAd and ChimAd-H28 viruses, which are the chimeric adenoviral vectors constructed in the present invention, an in vitro neutralizing antibody test was conducted using the serum of 7 healthy adults. The complement cascade was inactivated in human serum by heat-inactivation at 56°C ., for 30 minutes, and then serially diluted two-fold with 1×PBS. Thereafter, the

inactivated sera were neutralized with adenovirus under the same 50 MOI virus dose condition at room temperature for 1 hour, and then through the in vitro neutralizing antibody test method described in Example 6, the number of positive cells (brown spots) was measured in at least four fields at 200 $\times$ magnification using an optical microscope and expressed as an average to determine whether HAdV-5 and chimeric adenoviruses were infected in HEK293 cells on day 2 of infection (Table 7). The number of stained cells was converted into a reciprocal number and then multiplied by 100 to derive neutralizing antibody titers for HAdV-5, ChimAd, and ChimAd-H28 viruses in human serum, and expressed as a dot plot (FIG. 14).

TABLE 7

Ad infected	Human serum (N = 7)						
	#1	#2	#3	#4	#5	#6	#7
cell spots							
HAdV-5	18.5	14.4	1.4	8.6	19.25	11.4	8.6
ChimAd	45.2	50.8	23.2	22	39	31.2	22
ChimAd-H28	44	44	6	27.2	43.6	27.6	23.8

[0138] Table 7 shows the number of cells infected by HAdV-5 and the chimeric adenovirus following reaction with human sera. As can be seen in Table 7, in the seven human serogroups, we identified cells infected on average 4.6 times more by ChimAd virus than by HAdV-5 virus and by average 2.9 times more cells infected by ChimAd-H28 virus than by HAdV-5 virus, which was a statistically significant difference. In addition, as can be seen in FIG. 13, which shows the number of stained cells as the reciprocal number, it can be seen that the average 2.6-2.7 times lower Ad neutralizing antibody (NAb) titer in the ChimAd virus than in the HAdV-5 virus. These results indicate that vector immunity is lower when administering ChimAd vector-based or ChimAd-H28 vector-based vaccines to humans than HAdV-5 vector-based vaccines, so it will be an optimal adenoviral vector that can deliver foreign genes more efficiently.

Example 8: Application of ChimAd Vector-Based Corona Vaccine

8-1) Genetic Modification Encoding SARS-COV-2 S Protein and N Protein

[0139] To develop a vaccine using the coronavirus structural protein, two S polypeptide synthetic genes comprising NTD and CTD of Spike protein and N protein including modifications such as amino acid substitution were obtained.

[0140] Specifically, in the case of the CV gene, SARS-Coronavirus-2 of the beta-variant S protein was introduced with glycine at 614, acidic amino acid substitution of residues 682 to 685, and consecutive proline substitution mutations at 986 and 987. [Pallesen, Jesper, et al. Proceedings of the National Academy of Sciences 114.35 (2017): E7348-E7357. Bangaru, Sandhya, et al. Science 370.6520 (2020): 1089-1094. Plante, Jessica A., et al. Nature 592.7852 (2021): 116-121.].

[0141] In the case of the CVN gene, positions 44 to 180 and 247 to 364 of the N protein were linked to the N-terminus and C-terminus of the Spike protein through a (GGGGS) 3 linker, respectively, and at this time, additional structure prediction was performed through prediction tools

such as trRosetta and Alphafold. [Du, Zongyang, et al. Nature Protocols (2021): 1-18. Jumper, John, et al. Nature 596.7873 (2021): 583-589.]

8-2) Gene Synthesis to be Used as SARS-COV-2 Vaccine Material

[0142] To develop a vaccine using coronavirus structural proteins, synthetic genes encoding Spike and Nucleocapsid proteins among structural proteins were obtained. As shown in Table 8, for each vaccine candidate, S protein derived from coronavirus B.1.351 and N protein derived from Wuhan-Hu-1 were selected and codon optimized for easy expression in adenovirus. The region comprising the S protein was named CV, and the chimera form in which a portion of the N protein and the S protein were fused was named CVN.

TABLE 8

	Region	Accession no.
CV	Spike surface glycoprotein	QUT64557(B.1.351)
CVN	NP(NTD)-Spike-NP(CTD)	QUT64557(B.1.351), MN908947(Wuhan-Hu-1)

8-3) Cloning Chimeric Adenovirus Comprising SARS-COV-2 Gene

[0143] In order to clone the two SARS-COV-2 vaccine antigens synthesized above into the ChimAd vector, which is a chimeric adenovirus vector, using the plasmid DNA of the ChimAd vector as a template, PCR amplification was performed except for a part of the GOI binding region gene, and the amplified product (about 33.4 Kb) was purified through gel extraction. The insertion gene for cloning was PCR amplified to comprise the same sequence (homologous region, about 20 bp) as the 5' and 3' ends of the GOI binding region in the ChimAd vector, and the amplification product CV (about 3.8 Kb) and CVN (about 4.7 Kb) were obtained, and then combined with the previously purified DNA PCR amplification products through a gibbon assembly reaction, respectively, to finally obtain two types of ChimAd-CV and ChimAd-CVN plasmids (FIG. 15).

TABLE 9

Primer name	Sequence (5' to 3' direction)	PCR product
ChimAd vector	F: CTAAGCTTCTAGATAAGATATCCGATCCAC (SEQ ID NO: 53) R: CGCGTCGACGGTACCAGATCTCTAGCGGATC (SEQ ID NO: 54)	33.4 Kb
CV region	F: GATCTGGTACCGTCGACGCGATGTTTGTGTT (SEQ ID NO: 55) R: TATCTTATCTAGAAGCTTAGTCAGGTATAGTGCA GTTTCAC (SEQ ID NO: 56)	3.8 Kb
CVN region	F: GATCTGGTACCGTCGACGCGATGGGCCTGCC (SEQ ID NO: 57) R: TATCTTATCTAGAAGCTTAGTCAGGGGAAGGTCT TGTAGGC (SEQ ID NO: 58)	4.7 Kb

8-4) Production and Purification of Chimeric Adenovirus-Based Corona Vaccine Candidate

[0144] To produce a chimeric adenovirus-based coronavirus vaccine candidate using the ChimAd-CV and ChimAd-CVN plasmids produced in the present invention, each plasmid was treated with Pac I restriction enzyme (NEB, R0547L) to cut and remove the pBR322 replication start point and kanamycin resistance gene sequence that are unnecessary for adenovirus formation, and HEK293 cells were transformed with Lipofectamine 2000 (Invitrogen, 11668027) and cultured for 10-20 days in an incubator at 37° C. and CO₂ partial pressure of 5.0% to obtain a primary virus stock.

[0145] ChimAd-CV and ChimAd-CVN virus production was performed by infecting HEK293 cells at a cell density of about 1×10⁶ cells/ml with a titer of 2 MOI using the primary virus stock obtained above, and after 48 hours, 0.45 µm CFF microfiltration filter (Cytiva, CFP-4-E-6A) was used to enrich the cells. The recovered cells were treated with 0.5% Tween20 (Sigma-aldrich, P2287-500ML) and 20 U/ml benzonase (Miliopore, E1014-25KU) for 4 hours to induce cell lysis and host cell DNA degradation. Thereafter, cell debris was removed using 2 µm and 0.6 µm ULTA GF filters (Cytiva, KGF-A-0210TT, KGF-A-9606TT), and virus concentration and buffer exchange were performed through tangential flow filtration (TFF). ChimAd-CV and ChimAd-CVN virus were isolated from virus samples in PBS buffer through anion exchange chromatography using a HiTrap Capto Q ImpRes column (Cytiva, 17547051) and fast protein liquid chromatography (FPLC) using a Capto core 700 column (Cytiva, 17548115), and formulated into a vaccine buffer, and finally filtered with a 0.2 µm ULTA GF filter (Cytiva, KGF-A-9610TT) to obtain two vaccine candidates.

8-5) Immunogenicity Test Results Through Binding Antibody Titer/Neutralizing Antibody Titer/Cellular Immunity Analysis

(1) Immune Induction Using BALB/c Female

[0146] Mice were immunized using the ChimAd-CV and ChimAd-CVN vaccine candidates obtained by the above manufacturing method. The test group includes a total of 4 groups: buffer, ChimAd used as a mock-up, ChimAd-CV, and ChimAd-CVN. For the immune induction test, 5-week-old BALB/c female mice were used, and each adenovirus test group was injected at a concentration of 2.5×10⁹ ifu/dose. At 4-week intervals, the first and second vaccinations were administered by intramuscular injection twice (days 0 and 28), and blood was collected 4 and 6 weeks after the first vaccination (days 27 and 42), serum was separated and the spleen was removed.

(2) ELISA; Binding Antibody Titer of Mouse Anti-Serum by ChimAd-CV and ChimAd-CVN

[0147] A binding antibody titer test was performed to confirm the presence of antibodies in the serum through a mouse immunity induction test. To perform the test, coronavirus SARS-coronavirus-2 and beta variant S protein were dispensed into microplates at a concentration of 1 µg/ml, 100 ng/well each, and coated at 4° C. for more than 16 hours. After reaction, the plate was washed at least 4 times with 1×PBS comprising Tween 20 (PBS-T) and 100 µl of 0.5% casein was dispensed and reacted at 37° C. for 1 hour. After

washing with PBS-T, each serum was diluted 1:100 and then serially diluted 3-fold to approximately 1×10⁶. After dispensing and reacting at 37° C. for 1 hour, the mixture was washed with PBS-T, and HRP-conjugated Goat anti-mouse IgG HRP (Thermo) was diluted 1:5000 and reacted for 1 hour. After the reaction was completed, to proceed with color development, 50 µl of TMB solution (Thermo) was dispensed and color development was performed for 15-20 minutes. Afterwards, the reaction was stopped by adding an equal amount of Stop solution (Thermo), and the absorbance was measured at a wavelength of 450 nm.

[0148] As a result of the binding antibody titer, it was confirmed that all experimental groups inoculated with ChimAd-CV and ChimAd-CVN showed high binding antibody titers of 10⁴ or more to SARS-coronavirus-2 and beta variant strains compared to the control group. In the experimental group inoculated with the control group, binding antibody titers to both SARS-coronavirus-2 and beta variant strains were measured, but the values were confirmed to be invalid (FIG. 16).

(3) pVNT (Pseudovirus Neutralization Test);

Measurement of Neutralizing Antibody Titers Against Pseudoviruses in Mouse Anti-Serum by ChimAd-CV and ChimAd-CVN

[0149] Neutralizing antibodies in serum were tested through a mouse immune induction test. The pseudovirus system is a virus that uses the backbone of Lentivirus to express the luciferase gene and the Spike protein of SARS-coronavirus-2 or beta variants. In this experiment, Takara's Lenti-X™ Packaging Mix product was used to produce pseudoviruses comprising the Spike protein of SARS-coronavirus-2 and beta variant strains. The produced pseudovirus was transformed into 293T cells overexpressing the ACE2 protein, luciferase expression was quantified and measured, and a neutralizing antibody test was performed using a certain RLU (Relative Light Units) value. For the neutralizing antibody test, each mouse serum was first inactivated at 56° C. for 30 minutes, and DMEM medium comprising 10% FBS was used. The serum was diluted by 1/8 and then serially diluted 2-fold to about 10³. Pseudoviruses with a certain RLU value were mixed with the same amount of diluted serum and neutralization reaction was performed at 37° C. and 5% CO₂ incubator for 1 time. hACE2-293T cells cultured a day earlier were treated with 100 µl of mixture of the virus and serum and cultured in an incubator at 37° C. and 5% CO₂ for about 72 hours. After completion of incubation, about 100 µl of the culture medium was removed, 30 µl of One-Glo luciferase reagent (Promega) was added, and reacted for 3-5 minutes. Then, 60 µl of the mixture was transferred to a white plate and the luminescence value was measured using a luminometer. The measured RLU value was used to calculate the IC₅₀ value according to the serum dilution factor using a statistical program.

[0150] As shown in Table 10, the results of neutralizing antibodies against the beta variant virus showed IC₅₀=1253 for the ChimAd-CV inoculated group and IC₅₀=1131 for the ChimAd-CVN inoculated group, which were all higher than 10³, confirming a higher neutralizing antibody titer than the negative control group (FIG. 17).

TABLE 10

No	Group	pVNT(IC50)	
		4 wk	6 wk
1	PBS	7	42
2	ChimAd	9	0
3	ChimAd-CV	244	1253
4	ChimAd-CVN	409	1131

(4) ELISpot; Specific T Cell Immune Response Caused by ChimAd-CV, ChimAd-CVN Adenovirus

[0151] In the mouse immunity induction test, spleens were removed from each group 4 weeks after the first vaccination and 2 weeks after the second vaccination (6 weeks after the first vaccination). Spleen cells were recovered using Cell strainer (Falcon) and washed with PBS buffer solution. After washing, red blood cells were removed using red blood cell lysis buffer (Sigma), and the number of spleen cells was counted and diluted to the cell concentration used in the test. This experiment used R&D system's mouse IFN-gamma ELISpot kit, and the test was performed according to the manufacturer's instructions. For each test, 1×10^5 cells, 2×10^4 cells of spleen cells and Genscript's SARS-coronavirus-2 Spike protein peptide pool or Nucleoprotein peptide pool were used as stimulating antigens and incubated together for 16 hours in a 37°C , 5% CO_2 incubator. After incubation, the cells were washed with a washing solution, and 100 μl of biotin-conjugated IFN-gamma detection antibody was added and reacted by stirring at room temperature for 2 hours. After the reaction, the cells were washed four times using a washing solution, and 100 μl of streptavidin-AKP (alkaline phosphatase) was added and incubated at room temperature for 2 hours. After the final washing, 100 μl of BCIP/NBT substrate was added and reacted at room temperature for about 45 minutes. When color development occurred due to the reaction, the spot was washed with sterilized water, the number of colored spots was counted, and the number of cells in the Spot Forming Unit (SFU) was calculated.

[0152] As a result of measuring the induction of specific T-cell immunity to each stimulating antigen of S protein and N protein through the ELISpot test, T-cell immunity was higher after the second vaccination (week 6) than after the first vaccination (week 4). Particularly it was confirmed that a specific immune response to the Spike protein peptide was induced in ChimAd-CVN, which was about two times higher than that in ChimAd-CV. In addition, it was confirmed that ChimAd-CVN induces a specific immune response to Nucleocapsid peptide (FIG. 18).

TABLE 11

No	Group	SPF/ 10^6 splenocytes (Spike peptide pool)	
		4 wk	6 wk
1	PBS	34	17
2	ChimAd	61	42
3	ChimAd-CV	2367	3824
4	ChimAd-CVN	3020	6949

8-6) Antibody Titer, Neutralizing Antibody Titer and Challenge Test after ChimAd-CV, ChimAd-CVN Animal Immunization

A. Immune Induction and Challenge Test Using hACE2 Transgenic Female Mouse

[0153] The immune induction and challenge tests conducted at IVI used 6-week-old female human ACE2 transgenic mice (Tg(K18-ACE2)2PrImn) purchased from Jackson Laboratory (ME, USA). All the above tests were conducted in a BSL-3 level facility within the International Vaccine Research Institute. The immune induction test group includes a total of 4 groups: buffer, ChimAd used as a mock-up, ChimAd-CV, and ChimAd-CVN. Mice were inoculated intramuscularly twice at 4-week intervals, and blood was collected 4 and 6 weeks after the first priming inoculation.

B. ELISA; Measurement of Antibody Titers of Mouse Anti-Serum by ChimAd-CV, ChimAd-CVN

[0154] After the mouse immune induction test, it was performed to measure the antibody titer in serum. In a 96-well plate, 100 ng/well of SARS-coronavirus-2 S protein and delta variant S protein were dispensed at a concentration of 2 $\mu\text{g}/\text{ml}$ and coated at 4°C . for more than 16 hours. After the reaction, 100 μl of PBS containing 1% BSA was dispensed and allowed to react. Each serum was diluted 1:100, then serially diluted 5-fold and reacted at 4°C . for more than 16 hours. HRP-conjugated goat anti-mouse IgG HRP (Southern Biotech) was diluted 1:3000 and reacted at 37°C . for 1 hour. After the reaction was completed, the TMB solution (Millipore) was dispensed to proceed with the color development. Afterwards, the reaction was stopped by adding an equal amount of 0.5 N HCl solution (Merck), and the absorbance was measured at a wavelength of 450 nm.

[0155] It was confirmed that both the experimental group inoculated with ChimAd-CV and the experimental group inoculated with ChimAd-CVN showed high binding antibody titers of 10^5 or more against SARS-coronavirus-2 and delta variant strains compared to the control group (FIG. 19).

C. FRNT (Focus Reduction Neutralization Test); Measurement of Neutralizing Antibody Titers Against Coronavirus in ChimAd-CV and ChimAd-CVN Mouse Anti-Serum

[0156] After the mouse immune induction test, a test was performed to measure neutralizing antibodies in the serum. For the neutralizing antibody test, each mouse serum was first inactivated at 56°C . for 30 minutes and serially diluted 2-fold using DMEM medium containing 2% FBS. Diluted serum was mixed with 4.5×10^2 PFU/25 μl of SARS-coronavirus-2 coronavirus (NCCP #43326 (BetaCoV/Korea/KCDC03/2020) and 4.0×10^2 PFU/25 μl of delta variant coronavirus (NCCP #43390 (hCoV-19/Korea119861)/KDCA/2021) and reacted for 30 minutes at 37°C . After reaction, 50 μl of virus and serum mixture was treated with Vero cells (1.5×10^4 cells/well) cultured one day before and incubated at 37°C . in a CO_2 incubator for 4 hours. After 4 hours of reaction, the supernatant was removed, washed with a PBS buffer solution, and 300 μl of 4% formaldehyde solution (Sigma) was added and fixed in a place without light for more than 16 hours. After the reaction, it was washed with PBS. After washing, 100% cold methanol was

added and reacted for 10 minutes, followed by reaction at room temperature for 1 hour using blocking solution. Afterwards, it was treated with anti-SARS-COV-2 NP rabbit monoclonal antibody (Sino Biological) diluted 1:3000 and incubated at 37° C. for 1 hour. After incubation, it was washed with a PBS solution containing 0.1% Tween20 (PBS-T), and it was treated with HRP-conjugated goat anti-rabbit IgG (Bio-Rad) diluted 1:2000 and incubated at 37° C. for 1 hour. After washing with PBS-T, the TMB solution was reacted at room temperature for 30 minutes in the absence of light, the reaction was stopped with water, and then dried. For FRNT50 analysis, a Spot reader (Cytation 7 Cell Imaging Multi-Mode Reader) was used, and the value was calculated according to the dilution factor of the serum. The FRNT method is a method of measuring neutralizing antibodies by identifying virus-infected cells by attaching a virus-derived monoclonal antibody to a virus-infected cell and then attaching a secondary antibody labeled with HRP to develop a color as a substrate.

[0157] As a result of the neutralizing antibody against the SARS-coronavirus-2 virus, the neutralizing antibody titer can be confirmed as shown in Table, which is more than 100 times higher than the negative control group in both the ChimAd-CV and ChimAd-CVN groups after the first vaccination. After the second vaccination, neutralizing antibody titers were confirmed to be more than 30 times higher in the ChimAd-CV vaccinated group and more than 50 times higher in the ChimAd-CVN vaccinated group.

[0158] In addition, as a result of neutralizing antibodies against the delta variant virus, it was confirmed that both the ChimAd-CV vaccinated group and the ChimAd-CVN vaccinated group showed neutralizing antibody titers more than 40 times higher than the negative control group after the second vaccination (FIG. 20).

TABLE 12

No	Group	SARS-coronavirus-2(FRNT50)		Delta variant virus (FRNT50)	
		4 wk	6 wk	4 wk	6 wk
1	PBS	10	10	10	10
2	ChimAd	10	50	10	26
3	ChimAd-CV	1288	1660	646	1202
4	ChimAd-CVN	1096	3020	513	1096

D. Virus Challenge Test after ChimAd-CV, ChimAd-CVN Immunization

[0159] The challenge test was intranasal infection with 5×10^5 PFU SARS-COV-2 virus (SARS-coronavirus-2, delta variant strain) 4 weeks after boosting vaccination. After vaccination, survival and body weight were monitored daily, and on the 2nd, 4th, and 7th days after infection, blood, lungs, liver, kidneys, and spleen were collected to measure organ weights, and virus titers in the lungs were measured.

[0160] As a result of confirming the vaccine protective ability of ChimAd-CV and ChimAd-CVN against SARS-Coronavirus-2 (Wuhan-Hu-1) attack, both survival rates were found to be 100%, and it was confirmed that the ChimAd-CVN vaccinated group did not lose weight and the body weight of the ChimAd-CV vaccinated group was maintained without loss in all but one animal, and that there were individuals whose body weight decreased by more than 10% and then recovered.

[0161] In addition, as a result of confirming the vaccine protective ability of ChimAd-CV and ChimAd-CVN against Delta variant attack, ChimAd-CVN showed a survival rate of 40% and average body weight reduction of about 20% after 7 days of infection. Confirmed what was visible. The ChimAd-CV results showed that the survival rate was 100% and the average body weight was maintained at 100% after 7 days of infection (FIG. 21).

E. Lung Virus Titer for ChimAd-CV, ChimAd-CVN Immune Induction and Challenge Test (Plaque Assay)

[0162] After the mouse immune induction and challenge test, a plaque assay test was performed to measure the remaining virus titer in the lungs. To measure titer, Vero cells (ATCC, Cat #CCL-81) were used and cultured in an incubator at 37° C. and 5% CO₂ for more than 16 hours. The virus isolated from the lungs collected from all groups was stepwise diluted 4-fold up to 10⁶, and 200 μ l of the diluted virus was added to the cultured Vero cells and reacted at room temperature for 30 minutes. After reaction, the virus mixture was removed, covered with DMEM (Overlay media) containing 0.8% Agarose and 2% FBS, reacted at room temperature for 15 minutes, and then cultured in a 37° C., 5% CO₂ incubator for 72 hours. After fixation with 10% formalin solution for 1 hour, overlay media was removed, and plaques were stained for 5 minutes using crystal violet (Sigma). To confirm the virus titer, the number of plaques was counted, each dilution factor and the total amount (ml) were calculated to calculate the final titer.

[0163] As a result of measuring the residual virus titer in the lungs after the challenge test, in the ChimAd-CV vaccinated group against SARS-coronavirus-2 and delta variant attack, the lung virus titer decreased by more than 20,000 times 2 days after the attack compared to the control group. It was confirmed that the virus was cleared and not measured 4 and 7 days after the attack. It was confirmed that the ChimAd-CVN vaccinated group achieved virus clearance and that the lung virus titer was not measured after 2 days, 4 days, and 7 days (FIG. 22).

Example 9: Application of Chimeric Adenovirus-Based HPV Treatment and Prevention Vaccine

9-1) Genetic Modification Encoding the Proteins of HPV16 Type E6 and E7 and HPV Type 18 E6 and E7 Proteins.

[0164] For genetic modification of the anti-carcinogenic protein of HPV type 16, in the case of the E6 protein, a deletion mutation of residues 113 to 122 was induced to inhibit the interaction with carcinogenic proteins caused by the binding domain of the anti-cancer protein p53. In the case of the E7 protein, a deletion mutation of residues 21 to 26 was induced to strongly inhibit the interaction with pRb, the main target, and to suppress the transforming activity ability of E7 itself. In addition, for genetic modification of the anti-cancer protein of HPV type 18, in the case of the E6 protein, a deletion mutation of residues 113 to 122 was induced to inhibit the interaction with carcinogenic proteins caused by the binding domain of the anti-cancer protein p53. In the case of the E7 protein, a deletion mutation of residues 24 to 29 was induced to strongly inhibit the interaction with pRb, the main target, and to suppress the transforming activity ability of E7 itself.

[0165] Linear polypeptide (E6E7) in which the amino acids of E6 and E7, tumor proteins derived from human papillomavirus type 16, have been deleted, and linear polypeptide (Linear; E6E7) in which the amino acids of E6 and E7, tumor proteins derived from human papillomavirus type 18, have been deleted. were combined with Linker.

9-2) Chimeric Adenovirus-Based HPV Type 16, Type 18 E6, E7 Gene Synthesis and Chimeric Adenovirus Cloning

[0166] The E6 and E7 genes, which are anticancer proteins of human papillomavirus types 16 and 18, were modified and synthetic genes encoding these proteins were obtained. For each protein, the strain was selected as shown in Table 13 and the codon was optimized to facilitate expression in HEK cells.

TABLE 13	
Accession no.	
HPV 16E6E7	E6; AAV91660.1, E7; BAO18680.1
HPV 18 E6E7	E6; 1ADC35716.1, E7; 1ADC35717.1

[0167] Cloning of the HPV vaccine antigen cloned into the chimeric adenovirus vector was performed in the same manner as the SARS-coronavirus vaccine production (Example 8-3), and finally the ChimAd-HPV plasmid was obtained (FIG. 23).

TABLE 14		
Primer name	Sequence (5' to 3' direction)	PCR product
ChimAd vector	F: CTAAGCTTCTAGATAAGATATCCGATCCAC (SEQ ID NO: 59)	33.4 Kb
	R: CGCGTCGACGGTACCAGATCTCTAGCGGATC (SEQ ID NO: 60)	
HPV16/18 E6E7	F: GATCTGGTACCGTCGACGCGATGCACCAGAA GAGAAC (SEQ ID NO: 61)	1.4Kb
	R: TATCTTATCTAGAAGCTTAGTTATTGTTGACTAG CGCACCATGG (SEQ ID NO: 62)	

9-3) Immunization Induction Using C57BL/6 Female

[0168] Mouse immunization was performed using the ChimAd-HPV vaccine candidate obtained by the above preparing method. The test group includes three groups: buffer, ChimAd used as a mock-up, and ChimAd-HPV. For the immune induction test, 5-week-old C57BL/6 female mice were used, and each adenovirus test group was injected at a concentration of 2*10⁷ PFU/ml. At 4-week intervals, the first and second vaccinations were administered by intramuscular injection twice (days 0 and 14), and blood was collected and serum was separated and the spleen was removed at 2 and 4 weeks after the first vaccination (days 14 and 28). In addition, to confirm the effect of treating cervical cancer, 1×10⁵ TC-1 tumor cells were subcutaneously injected into the flank of 6-week-old C57BL/6 female mice 7 days before the first vaccination, and 6 mice were used for each group. In addition, to confirm the effect of preventing cervical cancer, 1×10⁵ TC-1 tumor cells were subcutaneously injected into the flanks of 6-week-old C57BL/6 female mice 7 days after the second vaccination, and 6 mice were used in each group.

9-4) ELISPOT; Specific T Cell Immune Response by ChimAd-HPV Adenovirus

[0169] On the 35th day after the second vaccination, spleens were removed from each group. The spleen was crushed into small pieces using a Cell strainer (Falcon) and the spleen cells were recovered and washed with a PBS buffer solution. Afterwards, red blood cells were removed using red blood cell lysis buffer (Sigma), and the number of spleen cells was counted and diluted to the number of cells to be used in the test. This experiment used R&D system's mouse IFN-gamma ELISpot kit, and the test was performed according to the manufacturer's instructions.

[0170] Then, 2×10⁵ cells and 4×10⁴ cells of splenocytes recovered from each mouse and the 49-57 amino acid peptide of Genscript's HPV E7 protein were used as stimulating antigens on a plate coated with mouse IFN-gamma monoclonal antibody and incubated for 16 hours at 37° C. in a 5% CO₂ incubator. After incubation, the cells were washed four times using 1× Wash buffer, 100 μl of streptavidin-AKP (alkaline phosphatase) was added and incubated at room temperature for 2 hours. After the final washing, 100 μl of BCIP/NBT substrate was added and incubated at room temperature for 45 minutes. When color development due to the reaction appeared, it was washed with sterilized water and the number of colored dots was counted.

[0171] As a result of measuring the specific T cell immune response to the HPV E7 protein antigen using ELISPOT, the ChimAd-HPV group showed higher results than the control group, confirming that it induced a specific immune response to the antigen. (FIG. 24)

9-5) Tumor Inhibition Assay; Confirmation of Tumor Treatment and Prevention Effects by ChimAd-HPV Adenovirus

[0172] To confirm the treatment effect of ChimAd-HPV adenovirus on cervical cancer, 1×10⁵ TC-1 tumor cells were subcutaneously injected into the flank of 6-week-old C57BL/6 female mice, and 6 mice were used per group. To confirm treatment efficacy, 7 days after injection of TC-1 cells, intramuscular injection of control group and ChimAd-HPV adenovirus was performed, and a second injection was performed on the 21st day after injection of tumor cells. In addition, to confirm the preventive efficacy, control and ChimAd-HPV adenoviruses were injected intramuscularly twice at two-week intervals, and 14 days after the second injection, TC-1 tumor cells (1×10⁵ cells) were injected subcutaneously into the flank. Tumor cells were measured every 4-5 days by measuring their longest (length) and shortest dimensions (width) using digital calipers to estimate tumor size in mice. Tumor volume was calculated by the following equation: tumor volume=(length×width²)/2.

[0173] As shown in FIG. 24, as a result of measuring the tumor volume using ChimAd-HPV adenovirus, the tumor volume was found to be lower in the ChimAd-HPV adenovirus group compared to the negative control group and positive control group in both therapeutic and preventive efficacy. It was confirmed that no tumor cells occurred in the ChimAd-HPV adenovirus group. (FIG. 25)

[Sequence list pre text]

SEQ ID NO: 1

Chimpanzee adenovirus serotype 6 Fiber knob domain a.a
sequence(synthesized ChAdV-6 knob a.a sequences are underlined)
PDPSPNCQLLSDRDAKFLLCLTKCGSQILGTVAVAAVTVGSAIIVFLR
FDSDGVLMNSSMVGDYWNFRGQTTQSVAYTNAVGFMPNIGAYPKTQSKTPKNSI
VSQVYLTGETTMPMTLITITFNGTDEKDTTPVSTYSMTFTWQWTDGYDKDKNITFATN
SFS

SEQ ID NO: 2

Chimpanzee adenovirus serotype 6/Pan6/Sad23 Fiber knob domain
nucleotide sequence
CCTGACCCCAAGCCCTAATTGTCAATTACTTTTCAGACAGAGATGCCAAATTTACTC
TCTGTCTTACAAAATGCGGTAGTCAAATACTAGGCACTGTGGCAGTGGCGGCTGT
TACTGTAGGATCAGCACTAAATCCAATTAATGACACAGTCAAAGCGCCATAGT
TTTCCTTAGATTGATTCGGATGGTGTACTCATGTCAAACCTCATCAATGGTAGGTG
ATTACTGGAACTTTAGGGAGGGACAGACCACTCAAAGTGTAGCCTATACAAATG
CTGTGGGATTCTATGCCAAATATAGGTGCATATCCAAAACCCCAAAGTAAACAC
CTAAAAATAGCATAGTCAGTCAGGTATATTTAACTGGAGAACTACTATGCCAAT
GACACTAACCATAAATTTCATGGCACTGATGAAAAAGACACAACCCAGTTAG
CACCTACTCTATGACTTTACATGGCAGTGGACTGGAGACTATAAGGACAAAAAT
ATTACCTTTGCTACCAACTCATTCTCT

SEQ ID NO: 3

Chimeric hexon HVR 1 to 7 region a.a sequence (HAdV-5 hexon + HAdV-
28 HVR 1-7 region)
SSQWDAQEKSGQGSMDVTKHTFTGVAAMGGENITKNGLQIGTEITADNQKKEIFAN
KTYQPEPQVQGEENWQENEVFYGGRAKKETKMKPCYGSFARPTNENGGAQKFKTP
AEQQEPKELDIDLAFDFTDGGTADTEYKADIVMYAENVNLETPDTHVYKPGKEDD
SSEINLVQSQMPNRPNIYIGFRDNFVGLMYNSTGNMGVLAGQASQLNAVVDLQDR
NTELSYQLLLDSLGRTRYFSMWNSAVDSYDPDVRIENHGVDELPNYCFPLDGLG
TNATYQGVKYSTGDGATQSGWAKDDTMARQNQICRGNLY

SEQ ID NO: 4

Chimeric hexon HVR 1 to 7 region nucleotide sequence (HAdV-5 hexon +
HAdV-28 HVR 1-7 region)
TCCAGTCAGTGGGATGCGCAAGAAAAAGTGGACAAGGAAGTGACATGG
TTACTAAAACTCACACATTGGGCGGCGCCTTATTCTGGTATAAATATTACAAA
GGAGGGTATTCAAATAGGTACTGAAATTACCGCCGACAATCAAAAGAAAGAAAT
TTTTGCCGATAAAACATTTCAACCTGAACCTCAAATAGGAGAAGGAAGTGGCA
AGAAAATGAAGTCTTCTATGGCGGGAGAGTCTAAAAAAGACTACCCCAATGAA
ACCATGTTACGGTTTATATGCAAAACCCACAATGAAATGGAGGGCAAGCAAA
ATTCAAAACACCTGCTGAAGGACAGGAACCTAAAGAAGTTCACATTCACCTTGC
TTTTTTTCGACACTGACGGCGGAACAGCTGACACTGAATATAAGCAGATATTGTA
TTGTACAGTGAAGATGTAGATATAGAAACCCAGACACTCATATTTCTTACATGC
CCGGCAAGGAAGATGACAGTTTCAAAATCAATCTAGTTCAACAATCTATGCCCCA
ACAGGCCTAATTACATTGCTTTTAGGGACAATTTATTGGTCTAATGTATTACAAC
AGCACGGGTAATATGGGTGTTCTGGCGGGCAAGCATCGAGTTGAATGCTGTT
GTAGATTTGCAAGACAGAAACACAGAGCTTTTATACAGCTTTTGCTTGATTCCA
TTGGTGATAGAACCGGTACTTTTCTATGGGAATCAGGCTGTTGACAGCTATGA
TCCAGATGTTAGAAATATTGAAAATCATGGAAGTGAAGTGAAGTCCAAATTAC
TGCTTTCCACTGGACGGATGGGAAGTAACTGCTACCTATCAAGGTGTAAGAGTAT
CAACTGGAGATGGTCTACGCAAGCGGATGGGCAAAAGACGACACAATGGCC
AGGCAAAACCAATATGACAGGGGTACATCTAC

SEQ ID NO: 5

Chimeric fiber gene sequence (HAdV-5 tail, shift domain + ChAdV-
6 knob domain)

ATGAAGCGCGCAAGACCGTCTGAAGATACCTTCAACCCCGTGATCCATATGAC
ACGGAACCCGCTCTCCAACTGTGCCTTTCTTACTCCTCCCTTTGTATCCCCCAA
TGGGTTTCAAGAGAGTCCCCCTGGGGTACTCTCTTTGCGCCTATCCGAACCTCTA
GTTACCTCCAATGGCATGCTTGCCTCAAATGGGCAACGGCTCTCTCTGGACG
AGGCCGGCAACCTTACCTCCCAAAATGTAACCACTGTGAGCCCACTCTCAAAA
AAACCAAGTCAAAACATAAACCTGGAAATATCTGCACCCCTCACAGTTACCTCAG
AAGCCCTAACTGTGGCTGCCGCCGACCTCTAATGGTCGCGGGCAACACACTCAC
CATGCAATCACAGGCCCGCTAACCGTGCACGACTCCAACTTAGCATTGCCACC
CAAGGACCCCTCACAGTGTCAGAAGGAAAGCTAGCCCTGCAAAATCAGGCCCC
CTCACCCACCGATAGCAGTACCTTACTATCACTGCCTCACCCTCTCAACTA
CTGCCACTGGTAGCTTGGGCATTGACTTGAAGAGGCCATTATACACAAAATGG
AAAACCTAGGACTAAAGTACGGGGCTCCTTTGCATGTAACAGACGACCTAAACAC
TTTGACCGTAGCAACTGGTCCAGGTGTGACTATTAATAATACTTCTTGCAAACT
AAAGTTACTGGAGCCTTGGGTTTGATTACAAAGCAATATGCAACTTAATGTAG
CAGGAGGACTAAGGATTGATTCTCAAAACAGACGCTTATACTTGATGTTAGTTA
TCCGTTTGATGCTCAAAACCAACTAAATCTAAGACTAGGACAGGGCCCTTTTT
ATAAACTCAGCCCAACTTGATATTAACACAAAGGCCCTTACTTGTTTA
CAGCTTCAACAATTCAAAAGCTTGAAGTTAACTAAGCACTGCCAAGGGGT
TGATGTTGAGCTACAGCCATAGCCATTAAATGCAGGAGATGGGCTGAATTTGG
TTCACCTAATGCACCAACACAAATCCCCTCAAAACAAAATTGCCATGCGCT

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AGAATTGTGATTCAAACAAGGCTATGGTTCCTAAACTAGGAAGCTGGCCTTAGTTTT
 GACAGCACAGGTGCCATTACAGTAGGAACAAAAATAATGATAAGCTAACTTTG
 TGGACCACACCTGACCCAGCCCTAATTGTCAATTACTTTGACAGAGATGCCA
 AATTTACTCTGTCTTACAAAATGCGGTAGTCAAATACTAGGCAGCTGTGGCAGT
 GGCGGCTGTACTGTAGGATCAGCACTAAATCCAATTAATGACACAGTCAAAAG
 CGCCATAGTTTTCTTAGATTGATTCGGATGGTGTACTCATGTCAAACCTCATCAA
 TGGTAGGTGATTACTGGAACTTTAGGGAGGGACAGACCACCTCAAAGTGTAGCCT
 ATACAAATGCTGTGGGATTATGCCAAATATAGGTGCATATCCAAAACCCAAA
 GTAAACACCTAAATAAGCATAGTCAGTCAGGTATATTTAACTGGAGAACTA
 CTATGCCAATGACACTAACCATAACTTTCAATGGCACTGATGAAAAAGACACAA
 CCCCAGTTAGCACCTACTCTATGACTTTTACATGGCAGTGGACTGGAGACTATAA
 GGACAAAAATATTACCTTTGCTACCACTCATCTCT

SEQ ID NO: 6

Chimeric hexon gene sequence (HAdV-5 hexon + HAdV-28 HVR 1-
 7region)
 ATGGCTACCCCTTCGATGATGCCGAGTGGTCTTACATGCACATCTCGGGCCAGG
 ACGCTCGGAGTACCTGAGCCCGGGCTGGTGCAGTTTGCCCGCGCCACCGAGA
 CGTACTTCAGCCTGAATAACAAGTTTAGAAACCCACGGTGGCGCCTACGCACG
 ACGTGACCACAGACCGGTCCCGAGCTTTGACGCTGCGGTTTATCCCTGTGGACCG
 TGAGGATACCTGCGTACTCGTACAAGGCGCGGTTACCCCTAGCTGTGGGTGATAAC
 CGTGTGCTGGACATGGCTTCCACGTACTTTGACATCCGCGCGGTGCTGGACAGGG
 GCCCTACTTTTAAGCCCTACTCTGGCACTGCCTACAACGCCCTGGCTCCCAAGGG
 TGCCCCAAATTCAGTCAGTGGGATGCGCAAGAAAAAGTGGACAAGGAAGTGA
 CATGGTTACTAAAACCTCACACATTTGGGCGAGCGCCTTATCTGGTATAAATATT
 ACAAGGAGGGTATTCAAAATAGGTACTGAAATTAACGCGCAATCAAAAGAAA
 GAAATTTTGGCCGATAAAACATTTCAACCTGAACCTCAAATAGGAGAAGAGAAC
 TGGCAAGAAAATGAAGTCTTCTATGGCGGAGAGTCTAAAAAGACTACCCCA
 ATGAAACCATGTTACGGTTCATATGCAAAACCCCAAAATGAAAATGGAGGGCAA
 GCAAAATTCAAAACACCTGCTGAAGGACAGGAACCTAAAGAACTTGACATTGAC
 CTTGCTTTTTTCGACACTGACGGCGGAACAGCTGACACTGAATATAAAGCAGATA
 TTGTATTGTACAGTGAAGATGTAGATATAGAAACCCAGACACTCATATTTCTTA
 CATGCCCGGCAAGGAAGATGACAGTTTCAAGAAATCAATCTAGTTCAACATCTAT
 GCCCAACAGGCTAATTACATTGCTTTTAGGGACAATTTTATGGTCTAATGTATT
 ACAACAGCACGGGTAAATATGGGTGTTCTGGCGGGCCAAGCATCGCAGTTGAATG
 CTGTTGTAGATTGCAAGACAGAAACACAGAGCTTTCATACACAGCTTTTGCTTGA
 TTCCATTGGTGATAGAACCAGGTACTTTTCTATGTGGAATCAGGCTGTTGACAGC
 TATGATCCAGATGTTAGAAATTTTGAATAATCATGGAACCTGAAGATGAACCTCCAA
 ATTACTGCTTTCCACTGGACGGATTGGGAACCTAATGCTACCTATCAAGGTGTAAA
 AGTATCAACTGGAGATGGTGTCTACGCAAGCGGATGGGCAAAAGACGACACAAT
 GGCCAGGCAAAACCAAAATATGCAAGGGTAACTCTACGCCATGGAAATCAATCT
 AATGCAACCTGTGGAGAAATTTCTGTACTCCAACATAGCGCTGTATTTGCC
 GACAAGCTAAAGTACAGTCCTTCCAACGTAAAAATTTCTGATAACCCAAACACCT
 ACGACTACATGAACAAGCGAGTGGTGGCTCCCGGGCTAGTGGAGTGTACATTA
 ACCTTGGAGCACGCTGGTCCCTTGAATATATGGACAACGTCAACCCATTTAACCA
 CCACCGCAATGCTGGCTGCGCTACCGCTCAATGTTGCTGGGCAATGGTCGCTAT
 GTGCCCTTCCACATCCAGGTGCTCAGAAATTTCTTGCCATTAATAAACCCTCTCT
 CTTGCGGGCTCATACACCTACGAGTGGAACTTACGGAAGGATGTTAATGTTG
 CTGACAGCTCCCTAGGAAATGACCTAAGGGTTGACGGAGCCAGCATTAAGTTT
 GATAGCATTTGCTTTACGCCACCTTCTTCCCATGGGCCCAACACCGCCTCCA
 CGCTTGAGGCCATGCTTAGAAACGACCAACGACCGAGTCTTTAACGACTATCT
 CTCCGCCGCCAACATGCTCTACCTTATACCCGCCAACGCTACCAACGTGCCCAT
 TCCATCCCTCCCGCAACTGGCGGCTTCCGCGGCTGGGCTTACCGCGCTTAC
 AGACTAAGGAAACCCATCACTGGGCTCGGGCTACGACCTTATTACACCTACTC
 TGGCTCTATACCTTACCTAGATGGAACCTTTTACCTCAACACACCTTTAAGAAG
 GTGGCCATTACCTTTGACTCTTCTGTGAGCTGGCCTGGCAATGACCGCTGCTTAC
 CCCCACGAGTTTGAATTAAGCGCTCAGTTGACGGGGAGGGTTACAACGTTGC
 CCAGTGTAACATGACCAAGACTGGTTCTGGTACAAATGCTAGCTAACTATAAC
 ATTGGCTACAGGGCTTCTATATCCAGAGAGCTACAAGGACCGCATGTACTCT
 TCTTTAGAAACTTCCAGCCCATGAGCCGTGAGGTGGTGGATGATACTAAATACAA
 GGACTACCAACAGGTGGGCATCTTACCAACACACAACTCTGGATTGTTGG
 CTACCTTGCCCCCACCATGCGGGAAGGACAGGCTTACCCTGCTAACTTCCCCTAT
 CCGCTTATAGGCAAGACCGAGTTGACAGCATTACCAGAAAAAGTTTCTTTGCG
 ATCGCACCTTTTGGCGCATCCCATTTCTCCAGTAACCTTATGTCATGGGCGCACTC
 ACAGACCTGGGCCAAAACCTTCTCTACGCCAACTCCGCCACGCGCTAGACATG
 ACTTTTGAAGTGGATCCCATGACGAGCCACCTTCTTTATGTTTGTGTTGAAGT
 CTTTGACGTGGTCCGTGTGACACGCGCACCGCGGCTCATGAAACCGGTGAC
 CTGCGCACGCCCTTCTCGGCCGCAACGCCACAACATAA

[ChimAd-based coronavirus vaccine sequence]

SEQ ID NO: 7

SARS-COV-2_B.1.351(beta) spike modified amino acid sequence
 MFVFLVLLPLVSSQCVNFTRTQLPPAYTNSFTRGVYYPDKVFRSSVLHSTQDLFLPF
 FSNVTFWFAIHVSGTNGTKRFANPVLFPNDGVYFASTEKSNIIIRGWIPTGLDSTQS
 LLIVNNATNVVIKVCFQFCNDPFLGVYHKNNKSWMESEFRVYSANNCTFEYVS
 QPFLMDLEKQGNFKNLREFVFKNIDGYFKIYSKHPTINLVRGLPQGFSALEPLVDLP
 IGINITRFQTLHISYLTPODSSSGWTAGAAAYVGYLQPRFTLLKYNENGTITDAVDC
 ALDPLSETKCTLKSFTVEKGIYQTSNFRVQPTESIVRFPNITNLCPFGEVFNATRFASV

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YAWNKRISNCVADYSVLNSASFSTFKCYGVSPTKLNDLCFTNVYADSFVIRGDEV
 RQIAPGQGTGNIADYNYKLPPDDFTGCVIAWNSNNLDSKVGNNYLYRLFRKSNLKP
 FERDISTEIYQAGSTPCNGVKGFCNYFPLQSYGFQPTYGVGYQPYRVVVLSELLHAP
 ATVCGPKKSTNLVKNKCNVFNENGLTGTGVLTESNKKFLPPQQFGRDIADTTDAVR
 DPQTLEILDITPCSPGGVSVITPGTNTSNQVAVLYQGVNCTEVPVAIHADQLTPTWRV
 YSTGSNVFQTRAGCLIGAHEVNNSYECDIPIGAGICASYQTQNTSPQQAQSVASQSIIA
 YTMSLGVENSVAYSNNSIAIPTNFTISVTEILPVSMTKTSVDCTMYICGDSTECSNLL
 LQYGSFCTQLNRALTGIAVEQDKNTQEVFAQVKQIYKTPPIKDFGGFNFSQILPDPSPK
 PSKRSFIEDLLFNKVTLADAGFIKQYGDCLGDI AARDLICAQKFNGLTVLPPLLDEMI
 AQYTSALLAGTITSGWTFGAGAAALQIPFAMQMAYRENGIGVTONVLYENQKLIANQ
 FNSAIGKIQDLSSTASALGKLQDVVNQNAQALNTLVKQLSSNFGAISSVLNDILSRL
 DPPEAEVQIDRLITGRLQSLQTYVTQQLIRAAEIRASANLAATKMSCEVLGQSKRVDF
 CGKGHYLMSFPQSAHPGVVFLHVTVYPAQEKNTTAPAI CHDGKAHFPREGVFEVSN
 GTHWFTVQTNFYPEPQIIITDNTFVSGNCDVIGIVNNTVYDPLQPELDSFKEELDKYF
 KNHTSPDVLGDISGINASVVNIQKEIDRLNEVAKNLNESLIDLQELGKYEYQYIKWPW
 YIWLGFIAGLIAIVMVTIMLCMTSCCCLKGCCSCGSCCKFDEDDSEPVLGKVKLH
 YTSEQ

SEQ ID NO: 8

SARS-COV-2_B.1.351(beta) spike modified nucleic acid sequence

ATG TTT GTG TTT CTG GTG CTG CTG CCT CTG GTG AGC AGC CAG TGC GTG
 AAC TTC ACC ACC AGA ACC CAG CTG CCT CCT GCC TAC ACC AAC AGC TTT
 ACC AGA GGC GTG TAT TAC CCC GAC AAA GTG TTT AGA TCC TCC GTG CTG CAC
 AGC ACC CAG GAC CTG TTT CTG CCC TTT TTC AGC AAC GTG ACC TGG TTC
 CAT GCC ATC CAC GTG AGC GGC ACC AAC GGC ACC AAG AGA TTC GCC AAT
 CCC GTG CTG CCT TTT AAT GAC GGC GTG TAC TTT GCC AGC ACC GAG AAG
 AGC AAT ATC ATC AGA GGC TGG ATC TTC GGC ACC ACC CTG GAT AGC AAG ACC
 CAG AGC CTG CTG ATC GTG AAT AAT GCC ACC AAT GTG GTG ATC AAA GTG
 TGC GAG TTC CAG TTT TGT AAC GAC CCC TTT CTG GGC GTG TAT TAC CAC
 AAG AAC AAC AAA AGC TGG ATG GAG AGC GAA TTC AGA GTG TAC AGC AGC
 GCC AAT AAC TGC ACC TTT GAG TAC GTG AGC CAG CCC TTT CTG ATG GAC CTG
 GAG GGC AAG CAG GGC AAC TTC AAG AAC CTG AGA GAG TTC GTG TTT AAA
 AAT ATC GAC GGC TAC TTC AAG ATC TAC AGC AAG CAT ACA CCC ATC AAC
 CTG GTG AGA GGC CTG CCC CAG GGC TTC AGC GCC CTG GAG CCC CTG GTG
 GAC CTG CCC ATC GGC ATC AAC ATT ACC CGG TTT CAG ACC CTG CAC AGA
 AGC TAT CTG ACC CCC GGC GAC AGC AGC AGC GGC TGG ACC GCC GGA GCC
 GCC GCC TAT TAT GTG GGC TAC CTG CAG CCC AGA ACC TTC CTG CTG AAA TAC
 AAC GAG AAC GGC ACC ATC ACC GAC GCC GTG GTC TGC GCC CTG GAC CCC
 CTG AGC GAG ACC AAG TGT ACC CTG AAG AGC TTT ACC GTG GAG AAG GGC
 ATC TAC CAG ACC AGC AAC TTT AGA GTG CAG CCC ACC GAG AGC ATC GTG
 AGA TTT CCC AAC ATC ACC AAC CTG TGC CCC TTC GGG GAG GTG TTT AAT
 GCC ACC AGA TTC GCC AGC GTG TAC GCC TGG AAT AGA AAA AGA ATC AGC
 AAC TGC GTG GCC GAC TAC AGC GTG CTG TAC AAC TCT GCC AGC TTT TCC
 ACC TTT AAG TGC TAC GGC GTG AGC CCA ACC AAA CTG AAC GAC CTG TGC TTC
 ACC AAC GTG TAC GCC GAC AGC TTC GTG ATC AGA GGC GAC GAG GTG AGA
 CAG ATC GCC CCC GGC CAG ACC GGC AAT ATC GCC GAT TAC AAC TAC AAG
 CTG CCC GAC GAC TTT ACC GGC TGC GTG ATC GCC TGG AAC AGC AAC AAC
 CTG GAC AGC AAG GTG GGC GGC AAC TAC AAC TAC CTG TAC AGA CTG TTT
 AGA AAG AGC AAC CTG AAG CCC TTT GAG AGA GAC ATC AGC ACC GAG ATC
 TAC CAG GCC GGC AGC ACC CCC TGC AAC GGC GTG AAA GGC TTT AAT TGT
 TAC TTC CCA CTG CAG TCC TAC GGC TTC CAG CCT ACC TAC GGC GTG GGA TAT
 CAG CCC TAT AGA GTG GTG GTG CTG TCC TTC GAG CTG CTG CAC GCC CCC
 GCC ACC GTG TGC GGC CCA AAG AAG AGC ACC AAC CTG GTG AAG AAC AAG
 TGC GTG AAC TTC AAC TTC AAC GGC CTG ACC GGC ACC GGC GTG CTG ACC
 GAG TCC AAC AAG AAA TTT CTG CCC TTC CAG CAG TTC GGA AGA GAC ATT GCC
 GAT ACC ACC GAC GCC GTG AGA GAC CCC CAG ACC CTG GAG ATC CTG GAC
 ATC ACC CCC TGT AGC TTC GGC GGC GTG TCC GTG ATC ACC CCA GGC ACC
 AAC ACC TCC AAC CAG GTG GCC GTG CTG TAC CAG GGC GTG AAC TGT ACC
 GAG GTG CCT GTG GCC ATC CAC GCC GAT CAG CTG ACC CCA ACC TGG AGA
 GTG TAC TCC ACC GGC TCC AAC GTG TTC CAG ACC AGA GCC GGC TGT CTG
 ATC GGC GCC GAG CAC GTG AAC AAC TCC TAC GAG TGC GAC ATC CCT ATC GGC
 GCC GGC ATC TGC GCC AGC TAC CAG ACC CAG ACC AAT GAC CCC CAG
 CAG GCC CAG
 TCC GTG GCC AGC CAG TCC ATC ATT GCC TAC ACC ATG TCC CTG GGC GTG
 GAG AAC AGC GTG GCC TAC TCC AAC AAC AGC ATC GCC ATC CCC ACC AAT TTC
 ACC ATC AGC GTG ACC ACC GAA ATC CTG CCA GTG AGC ATG ACC AAA ACC
 TCC GTG GAT TGC ACC ATG TAC ATC TGC GGA GAC TCC ACC GAG TGC TCC
 AAC CTG CTG TGC CAG TAC GGC TCC TTT TGC ACC CAG CTG AAC AGA GCC
 CTG ACC GGA ATC GCC GTG GAG CAG GAT AAA AAC ACC CAG GAA GTG TTC
 GCC CAG GTG AAA CAG ATC TAC AAG ACC CCC CCC ATC AAA GAT TTT GGA
 GGC TTT AAT TTT AGC CAG ATC CTG CCT GAT CCC TCC AAA CCC AGC AAG AGA
 TCC TTT ATC GAG GAT CTG CTG TTC AAC AAA GTG ACA CTG GCC GAT GCC
 GGA TTT ATC AAA CAG TAC GGA GAC TGC CTG GGC GAC ATC GCC GCC AGA
 GAC CTG ATC TGC GCC CAG AAG TTT AAC GGC CTG ACC GTG CTG CCA CCC
 CTG CTG ACC GAC GAG ATG ATC GCC CAG TAT ACC TCC GCC CTG CTG GCC GGC
 ACC ATC ACC TCC GGC TGG ACC TTT GGC GCC GGA GCC GCC CTG CAG ATC
 CCA TTT GCC ATC CAG ATG GCC TAC AGA TTT AAT GGA ATC GGA GTG ACC
 CAG AAC GTG CTG TAT GAG AAC CAG AAA CTG ATC GCC AAT CAG TTC AAC

(SEQ ID NO: 9)

MGLPNNTASWFALTQHGKEDLKFPGRGQVPINTNSSPDDQIGYYRRATR
RIRGGDGMKMDLSPRWYFYLLGTGPEAGLPYGANKDGIWVATEGALNTP
KDHIGTRNPANNAATVLOLPGTTLPKGFYAEGSRGGS

(SEQ ID NO: 10)

GGGSGGGSGGGGS|GGGSGGGSGGGSGT KKSAAEASKPRQKRTAT
KAYNVTQAFRRGPEQTQGNFGDQELIRGTDYKHWPQIAQFAPSASAFF
GMSRIGMEVTPSGTWLTYTGAIKLLDDKDPNFKDOVILLNKHIDAYKTEP*

(SEQ ID NO: 11)

ATGGGCCTGCCCAACAACCCGCCAGCTGGTTTACCGCCTGACCCAGCA
CGGCAGGAGGAGCTGAAGTCTCCAGAGCCAGGCGTCCCATTAACA
CCCAACAGCCGCCCGCAGCAGCAGTCCGTACTACGAGAAGAGCCACAGA
AGAATCAGAGCGCGCGCAGCGCAAGATGAAGGACCTGAGCCCATGGTATGGTA
CTTCTACTATCTGGGAACCGGCCGGAAGCGCGCTGCCCTATGGCGCCA
ACAAAGATGGCATCATCTGGGTGGCCACCGAAGCGCGCTGAACACCCCC
AAGGATCACATCGGAACAGAAATCCGCCAACATGCCGCCATCGTGCT
GCAGCTGCCCGCCAGGAACCCCTGCCATAAGGCTTTTATGCCGAAGGAA
CGAGAGGAGGAGG

(SEQ ID NO: 12)

GGCGGCGGAGGAGGACGGCGGAGGAGCGAGCGCGGAGGCGGGAAGC|GGCG
GAGGAGGAGGACGGCGGAGGCGGAGGCGGCGGAGGAGGACGCCAAAAAG
AGGCGCGCGAGGCGCAGCAAAAAGCCAGACGAGAAAGAACGCCACCAAG
GGCCTCAATTGTGACCCAGGCCCTTTGTGACGAAGAGGACCCGACGACACCC
AGGGCAAAATTTGGCAGCCAGGAGTGTATCAGACAGGGCACCGCATCAAA
CACTGGCCCGAGATCGCCAGTTTCGCCCCAGCGCCAGCGCCTTCTTCGG
CATGAGCAGAATCGGCATGGAGGTGACCCCCAGCGGCACCTGGCTGACCT
ACACCGCGCGCCATCAAGCTGGACGACAGGACGCCAACTTCAAGGACCAAG
GTGATCCCTGTGAACAAGACACATCGACGCGCTACAAGACCTTCCCGCTGA

[ChimAD vector-based HPV vaccine sequence]

SEQ ID NO: 13

Human papillomavirus type 16, E6, E7 & Human
pappillomavirus type 18, E6, E7 modified amino
acid sequence (Linkder underlined)

MHQKRTAMFQDPQPRKPLQLCLETQLTTIHDIIECVYCKQQLLREVEY
DFAERDLICVIRVDGNPYAVCDKCLKFYSGIRSHYCYSLYGLTLEQQYN
KPLCDLLLCIRQKRLDKKQRFNDSRWGTCSCCSCCRSSRTTRETQLMH
GDPTTLEHYMLDQLPQTTQFNSSSEEDIEDGAGQAEPDRAHYNIIVTFC
CKCDSTLRCLVCQSTHVDIRLTEDLLMGTGLGIVCIPICSQKPRGKRKRSMA
FEDPTRRPYKLPDLCELNTLSLQDIEITCVSKQTLVLETFVEFAPFKDLF
VYVRDLSIPHAACHCKIDFYSRIRELHYRHSVYGDTEFLKILINTGLYLLI
RCLKLRLHLENEKRPFHNIAGHYRGQCHSCCNRAQRERLQRRRETQVMHGPK
ATLQDILVLEHPQENIIPVQLSSSEENDIEDGVNQHQLPARRAEQRPHMT
LCMCKCEARIELLVESASDDLRAPOLFNLTLFSVCPWCASAO

SEQ ID NO: 14

Human papillomavirus type 16, E6, E7 & Human papillomavirus type 18, E6, E7 modified nucleic acid sequence (linker underlined)

ATGCAGCCAGAAAGAACAGCCATGTTCAGGACCCCTCAGGAAGCGGCCAGCGAAGCTCCCACAACCTTTGTACAGAGCTGCACAAACACCCATTATGCATCACTCTTGATGCGTTTACTGCAGAACGCAGCACTCTCGCGCCGGGAAGCTTACGATTTTCGCTTTTCAGAGATCTGTGCATCGTGTATAGAGACGGGAACCCCTACCGCGCTGTGTGACAAATGCCCTTAAATTTCTACAGTAAGATTTTCAGAGTACCCGGCATATTGTCTACAGCGCTTTATGTAGCTACATTGGAGCAGCAGTACAACAAACCTTTTGTGCATCTTCTTATCCGGTGCATTAAAGCAACCGCACTTGGACAAAACCAACCGCTTCATATACATCAGGGGCGAGGTGGACTGGACGCTGCAATGAGTTGCTGTCTCAAGCCGCATCCGGAGAGAGATCAACTCATGTGCATGGTGATACCTCAACTTTGTCACGAATATATGTTGGACCTCAACCGGAAAC TACCCAACCTGATGACAGCTCTTGAGGAAGAGGACGAATTCGACCGCCCTGCAGGGCAAGCGCAACCTGCAGGGGCCCTTATAAACCATTGTGCATCTTTTGC TGCAAATGCGATAGTACTTTGAGACCTCTGTGTGCAGTCAACTCATGTGTTGACATTAGAACAACCTTGAAGACCTCCTCATGGGGACACTGGGGATTGTTTGCCCAATCTGTGTCACAGAAACCCAAAGAGGCCGCAAAAGGCGCGTCTATGTGGCTCGCTTAGGACCCCTACAAGGAGGCCCTACAAGCTCCCCGATTCTGTGTACACACTGAATCCAGCTCTTCAGGATTTGAATACTCTTGGCTTACTGTGCAAAA ACTGTGCTCGAAGCTTACCGAAGTCTTTGAATTTGGCTTCAAGGCACTTTTGTTGTGTATCGCGGATTCATCCCAACGCAGCGCTGTCAACAAATGTATTGATCTTACTCTAGGATCAGAGAATTGAGGCATATTCAGACTCTGTGTATGGCGACACCTTGGAAAAATTGACCAACACTGGATTGTATAATCTGCTTATTCGGCTGCTTTAAGCTCGGGCACTTGAAGCAAAAACCGAGATTTCCACAACATCGGCGGCAATTACGCGGCAATGCCATCTTTGTGTGCAATCGCGCCGCGCAGGAGCGGCTGCAAAAGGCGCAGAGAGACTCAGGTGATGCATGGACCCAA GGCCACTCTCCAGGACATCGTCTTCATCTTGAACCCCAAGACGAATTC CCGTTCAACTTAGCGACTCAGAAGAGAAAATGATGAATCGACGGAGTG AACCATCAGACTCTCCCGACAAGAGAGCCGCGACCAAGAAGACACAAT GCTCTGTATGTGCTGCAAGTGTGAAGCAGGATCGAAGCTCGTCTGTGCAATCTAGTCAGATGATCTCAGAGGCTTTCAACAGCTCTTCTTGAATACACTG AGCTTTGTCTGTGCTCATGGTGGCTAGTCAACAA

 SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 62

<210> SEQ ID NO 1

<211> LENGTH: 173

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Chimpanzee adenovirus serotype 6 Fiber knob domain a.a sequence(synthesized ChAdV-6 knob a.a sequences are underlined)

<400> SEQUENCE: 1

Pro Asp Pro Ser Pro Asn Cys Gln Leu Leu Ser Asp Arg Asp Ala Lys
1 5 10 15

Phe Thr Leu Cys Leu Thr Lys Cys Gly Ser Gln Ile Leu Gly Thr Val
20 25 30

Ala Val Ala Ala Val Thr Val Gly Ser Ala Leu Asn Pro Ile Asn Asp
35 40 45

Thr Val Lys Ser Ala Ile Val Phe Leu Arg Phe Asp Ser Asp Gly Val
50 55 60

Leu Met Ser Asn Ser Ser Met Val Gly Asp Tyr Trp Asn Phe Arg Glu
65 70 75 80

Gly Gln Thr Thr Gln Ser Val Ala Tyr Thr Asn Ala Val Gly Phe Met
85 90 95

Pro Asn Ile Gly Ala Tyr Pro Lys Thr Gln Ser Lys Thr Pro Lys Asn
100 105 110

Ser Ile Val Ser Gln Val Tyr Leu Thr Gly Glu Thr Thr Met Pro Met
115 120 125

Thr Leu Thr Ile Thr Phe Asn Gly Thr Asp Glu Lys Asp Thr Thr Pro
130 135 140

Val Ser Thr Tyr Ser Met Thr Phe Thr Trp Gln Trp Thr Gly Asp Tyr
145 150 155 160

Lys Asp Lys Asn Ile Thr Phe Ala Thr Asn Ser Phe Ser
165 170

<210> SEQ ID NO 2

<211> LENGTH: 519

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Chimpanzee adenovirus serotype 6/Pan6/Sad23 Fiber knob domain nucleotide sequence

<400> SEQUENCE: 2

cctgacccaa gccctaattg tcaattactt tcagacagag atgccaaatt tactctctgt 60

cttacaaaat gcggtagtagc aatactaggc actgtggcag tggcggctgt tactgttagga 120

tcagcactaa atccaattaa tgacacagtc aaaagcgcca tagttttcct tagatttgat 180

tccgatgggt tactcatgtc aaactcatca atggttaggtg attactggaa cttaggggag 240

ggacagacca ctcaaagtgt agcctataca aatgctgtgg gattcatgcc aaatataggt 300

gcataatccaa aaacccaaag taaaacacct aaaaatagca tagtcagtca ggtatatatta 360

actggagaaa ctactatgcc aatgacacta accataactt tcaatggcac tgatgaaaaa 420

gacacaaccc cagtttagcac ctactctatg acttttacat ggcagtggac tggagactat 480

aaggacaaaa atattacctt tgctaccaac tcattctct 519

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<210> SEQ ID NO 3
<211> LENGTH: 318
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Chimeric hexon HVR 1 to 7 region a.a sequence
(HAdV-5 hexon + HAdV-28 HVR 1-7 region)

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<400> SEQUENCE: 3

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Ser Ser Gln Trp Asp Ala Gln Glu Lys Ser Gly Gln Gly Ser Asp Met
1      5      10      15
Val Thr Lys Thr His Thr Phe Gly Gln Ala Pro Tyr Ser Gly Ile Asn
20     25     30
Ile Thr Lys Glu Gly Ile Gln Ile Gly Thr Glu Ile Thr Ala Asp Asn
35     40     45
Gln Lys Lys Glu Ile Phe Ala Asp Lys Thr Phe Gln Pro Glu Pro Gln
50     55     60
Ile Gly Glu Glu Asn Trp Gln Glu Asn Glu Val Phe Tyr Gly Gly Arg
65     70     75     80
Val Leu Lys Lys Thr Thr Pro Met Lys Pro Cys Tyr Gly Ser Tyr Ala
85     90     95
Lys Pro Thr Asn Glu Asn Gly Gly Gln Ala Lys Phe Lys Thr Pro Ala
100    105    110
Glu Gly Gln Glu Pro Lys Glu Leu Asp Ile Asp Leu Ala Phe Phe Asp
115    120    125
Thr Asp Gly Gly Thr Ala Asp Thr Glu Tyr Lys Ala Asp Ile Val Leu
130    135    140
Tyr Ser Glu Asp Val Asp Ile Glu Thr Pro Asp Thr His Ile Ser Tyr
145    150    155    160
Met Pro Gly Lys Glu Asp Asp Ser Ser Glu Ile Asn Leu Val Gln Gln
165    170    175
Ser Met Pro Asn Arg Pro Asn Tyr Ile Ala Phe Arg Asp Asn Phe Ile
180    185    190
Gly Leu Met Tyr Tyr Asn Ser Thr Gly Asn Met Gly Val Leu Ala Gly
195    200    205
Gln Ala Ser Gln Leu Asn Ala Val Val Asp Leu Gln Asp Arg Asn Thr
210    215    220
Glu Leu Ser Tyr Gln Leu Leu Leu Asp Ser Ile Gly Asp Arg Thr Arg
225    230    235    240
Tyr Phe Ser Met Trp Asn Gln Ala Val Asp Ser Tyr Asp Pro Asp Val
245    250    255
Arg Ile Ile Glu Asn His Gly Thr Glu Asp Glu Leu Pro Asn Tyr Cys
260    265    270
Phe Pro Leu Asp Gly Leu Gly Thr Asn Ala Thr Tyr Gln Gly Val Lys
275    280    285
Val Ser Thr Gly Asp Gly Ala Thr Gln Ser Gly Trp Ala Lys Asp Asp
290    295    300
Thr Met Ala Arg Gln Asn Gln Ile Cys Arg Gly Asn Ile Tyr
305    310    315

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<210> SEQ ID NO 4
<211> LENGTH: 954
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Chimeric hexon HVR 1 to 7 region nucleotide sequence (HAdV-5 hexon + HAdV-28 HVR 1-7 region)

<400> SEQUENCE: 4

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tccagtcagt gggatgcgca agaaaaagt ggacaaggaa gtagcatggt tactaaaact    60
cacacatttg ggcaggcgcc ttattctggt ataaatatta caaaggaggg tattcaaata    120
ggtactgaaa ttacggcgga caatcaaaag aaagaaattt ttgccgataa aacatttcaa    180
cctgaacctc aaataggaga agagaactgg caagaaaatg aagtcttcta tggcgggaga    240
gtcctaaaaa agactacccc aatgaaacca tgttacggtt catatgcaa acccacaat    300
gaaaatggag ggcaagcaaa attcaaaaca cctgctgaag gacaggaacc taaagaactt    360
gacattgacc ttgctttttt cgacactgac ggcggaacag ctgacactga atataaagca    420
gatattgtat tgtacagtga agatgtagat atagaaaccc cagacactca tatttcttac    480
atgcccgga aggaagatga cagttcagaa atcaatctag ttcaacaatc tatgcccaac    540
aggcctaatt acattgcttt tagggacaat tttattggtc taatgtatta caacagcacg    600
ggtaatatgg gtgttctggc gggccaagca tcgcagttga atgctgtgt agatttgcaa    660
gacagaaaca cagagctttc ataccagctt ttgcttgatt ccattggtga tagaaccagg    720
tacttttcta tgtggaatca ggtgttgac agctatgac cagatgttag aattattgaa    780
aatcatggaa ctgaagatga acttccaaat tactgcttcc cactggacgg attgggaact    840
aatgtacct atcaagggtg aaaagtatca actggagatg gtgctacgca aagcggatgg    900
gcaaaagacg acacaatggc caggcaaac caaatatgca ggggtaacat ctac    954

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<210> SEQ ID NO 5

<211> LENGTH: 1731

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Chimeric fiber gene sequence (HAdV-5 tail, shift domain + ChAdV-6 knob domain)

<400> SEQUENCE: 5

```

atgaagcgcg caagaccgtc tgaagatacc ttcaaccccg tgtatccata tgacacggaa    60
accggtcctc caactgtgcc ttttcttact cctccctttg tatccccc aa tgggtttcaa    120
gagagtcccc ctgggggtact ctctttgcgc ctatccgaac ctctagttac ctccaatggc    180
atgcttgccg tcaaaatggg caacggcctc tctctggacg aggcgggcaa ccttacctcc    240
caaaatgtaa ccaactgtgag ccacactctc aaaaaaacca agtcaaacat aaacctggaa    300
atatctgcac ccctcacagt tacctcagaa gccctaactg tggctgccc cgcacctcta    360
atggtcgccg gcaacacact caccatgcaa tcacaggccc cgctaaccgt gcacgactcc    420
aaacttagca ttgccaccca aggacccctc acagtgtcag aaggaaagct agccctgcaa    480
acatcaggcc ccctcaccac caccgatagc agtaccctta ctatcactgc ctcacccctc    540
ctaactactg ccaactggtag cttgggcatt gacttgaaag agccattta tacacaaaat    600
ggaaaactag gactaaagta cggggctcct ttgcatgtaa cagacgacct aaacactttg    660
accgtagcaa ctgggtccagg tgtgactatt aataaactt ccttgcaaac taaagttact    720
ggagccttgg gttttgatc acaaggcaat atgcaactta atgtagcagg aggactaagg    780
attgattctc aaaacagacg cttatactt gatgttagt atccgtttga tgctcaaac    840

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caactaaatc taagactagg acagggccct ctttttataa actcagccca caacttggat    900
attaactaca acaaaggcct ttacttgttt acagcttcaa acaattccaa aaagcttgag    960
gttaacctaa gcactgccaa ggggttgatg ttgacgcta cagccatagc cattaatgca   1020
ggagatgggc ttgaatttgg ttcaccta atgcacaaaca caaatccct caaaacaaaa   1080
attggccatg gcctagaatt tgattcaaac aaggctatgg ttcctaaact aggaactggc   1140
cttagttttg acagcacagg tgccattaca gtaggaaaca aaaataatga taagctaact   1200
ttgtggacca cacctgacce aagcccta atgtcaattac tttcagacag agatgccaaa   1260
tttactctct gtcttacaaa atgcggtagt caaatactag gcactgtggc agtggcggct   1320
gttactgtag gatcagcact aaatccaatt aatgacacag tcaaaagcgc catagttttc   1380
cttagatttg attcogatgg tgtactcatg tcaaactcat caatggtagg tgattactgg   1440
aactttaggg agggacagac cactcaaagt gtagcctata caaatgctgt gggattcatg   1500
ccaaatatag gtgcataatc aaaaacccaa agtaaaacac ctaaaaatag catagtcagt   1560
caggatatatt taactggaga aactactatg ccaatgacac taaccataac tttcaatggc   1620
actgatgaaa aagacacaac cccagtttagc acctactcta tgacttttac atggcagtgg   1680
actggagact ataaggacaa aaatattacc ttgctacca actcattctc t          1731

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<210> SEQ ID NO 6

<211> LENGTH: 2832

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Chimeric hexon gene sequence (HAdV-5 hexon + HAdV-28 HVR 1-7region)

<400> SEQUENCE: 6

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atggctaccc cttcgatgat gccgcagtgg tottacatgc acatctcggg ccaggacgcc    60
tcggagtacc tgagccccgg gctgggtgcag ttgccccg ccaccgagac gtacttcagc   120
ctgaataaca agtttagaaa ccccacggtg gcgcctacgc acgacgtgac cacagaccgg   180
tcccagcgtt tgacgctgcg gttcatccct gtggaccgtg aggatactgc gtactcgtac   240
aaggcgcggt tcaccctagc tgtgggtgat aaccgtgtgc tggacatggc ttccacgtac   300
tttgacatcc gcggcgtgct ggacaggggc cctactttta agccctactc tggcactgcc   360
tacaacgccc tggctcccaa gggtgcccca aattccagtc agtgggatgc gcaagaaaaa   420
agtggacaag gaagtgcacat ggttactaaa actcacacat ttgggcaggc gccttattct   480
ggtataaata ttacaaagga gggatttcaa ataggtactg aaattaccgc cgacaatcaa   540
aagaaagaaa tttttgccga taaaacattt caacctgaac ctcaaatagg agaagagaac   600
tggcaagaaa atgaagtctt ctatggcggg agagtcctaa aaaagactac cccaatgaaa   660
ccatgttacg gttcatatgc aaaaccacaa aatgaaaatg gagggcaagc aaaattcaaa   720
acacctgctg aaggacagga acctaaagaa cttgacattg accttgcttt tttcgacact   780
gacggcggaa cagctgacac tgaatataaa gcagatattg tattgtacag tgaagatgta   840
gatatagaaa cccagacac tcatatctct tacatgcccg gcaaggaaga tgacagttca   900
gaaatcaatc tagttcaaca atctatgccc aacaggccta attacattgc ttttagggac   960
aattttattg gtctaattga ttacaacagc acgggtaata tgggtgttct ggcgggccaa  1020

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gcacgcagc tgaatgctgt ttagatttg caagacagaa acacagagct ttcataccag 1080
cttttgcttg attccattgg tgatagaacc aggtactttt ctatgtggaa tcaggctgtt 1140
gacagctatg atccagatgt tagaattatt gaaaatcatg gaactgaaga tgaacttcca 1200
aattactgct ttccactgga cggattggga actaatgcta cctatcaagg tgtaaaagta 1260
tcaactggag atgggtgtac gcaaagcgga tgggcaaaag acgacacaat ggccaggcaa 1320
aaccaaatat gcaggggtaa catctacgcc atggaaatca atctaaatgc caacctgtgg 1380
agaaatttcc tgtactccaa catagcgctg tatttgcccg acaagctaaa gtacagtcct 1440
tccaacgtaa aaatttctga taacccaaac acctacgact acatgaacaa gcgagtgggtg 1500
gctcccgggc tagtgactg ctacattaac ctgggagcac gctggtcctt tgactatatg 1560
gacaacgtca acccatttaa ccaccaccgc aatgctggcc tgcgtaccg ctcaatgttg 1620
ctgggcaatg gtcgtatgt gcccttcac atccagggtc ctcagaagtt ctttgccatt 1680
aaaaacctcc ttctctgcc gggctcatac acctacgagt ggaacttcag gaaggatgtt 1740
aacatgggtc tgcagagctc cctaggaaat gacctagggt ttgacggagc cagcattaag 1800
tttgatagca tttgccttta cgccaccttc ttcccatggg cccacaacac cgctccacg 1860
cttgaggcca tgetttagaa cgacaccaac gaccagtcct ttaacgacta tctctccgcc 1920
gccaacatgc tctacctat acccgccaac gctaccaacg tgcccatatc catccctcc 1980
cgcaactggg cggttttcg cggttgggcc ttcacgcgcc ttaagactaa ggaacccca 2040
tactggggt cggtgtacga ccttattac acctactctg gctctatacc ctacctagat 2100
ggaacctttt acctcaacca cacctttaag aaggtggcca ttaccttga ctcttctgtc 2160
agctggcctg gcaatgaccg cctgcttacc cccaacgagt ttgaaattaa gcgctcagtt 2220
gacggggagg gttacaacgt tgcccagtg aacatgacca aagactgggt cctggtacaa 2280
atgctagcta actataacat tggtaccag ggcttctata tccagagag ctacaaggac 2340
cgcatgtact cctctttag aaacttcag cccatgagcc gtcagggtgt ggatgatact 2400
aaatacaagg actaccaaca ggtgggcac ctacaccaac acaacaactc tggatttgtt 2460
ggctaccttg ccccccacat gcgcgaagga caggcctacc ctgctaactt cccctatccg 2520
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ctttggcgca tccattctc cagtaacttt atgtccatgg gcgcactcac agacctgggc 2640
caaaaccttc tctacgcaa ctcgcccac gcgctagaca tgacttttga ggtggatccc 2700
atggacgagc ccaccttct ttatgtttt tttgaagtct ttgacgtggt ccgtgtgcac 2760
cagccgcacc gcggcgctcat cgaaccgtg tacctgcgca cgccctctc ggccggcaac 2820
gccacaacat aa 2832

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<210> SEQ ID NO 7

<211> LENGTH: 1270

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: SARS-CoV-2_B.1.351(beta) spike modified amino acid sequence

<400> SEQUENCE: 7

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val

1

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10

15

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Asn	Phe	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
			20					25					30		
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
	50					55					60				
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Ala
65					70					75					80
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85					90						95	
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
		100						105					110		
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
		115					120					125			
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
	130					135					140				
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145					150					155					160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165						170					175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180						185					190		
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
		195					200					205			
Pro	Ile	Asn	Leu	Val	Arg	Gly	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215					220				
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
225					230					235					240
Leu	His	Ile	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	Gly	Trp	Thr
			245						250					255	
Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	Arg	Thr	Phe
		260						265					270		
Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	Val	Asp	Cys
		275					280					285			
Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	Ser	Phe	Thr
	290					295					300				
Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	Gln	Pro	Thr
305					310					315					320
Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	Pro	Phe	Gly
			325						330					335	
Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	Trp	Asn	Arg
		340						345					350		
Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	Tyr	Asn	Ser
	355						360					365			
Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	Thr	Lys	Leu
	370					375					380				
Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	Val	Ile	Arg
385					390					395					400
Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	Asn	Ile	Ala
			405						410					415	
Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	Val	Ile	Ala

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420							425					430				
Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	Tyr	Asn	Tyr	
435							440					445				
Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	Glu	Arg	Asp	
450							455					460				
Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	Asn	Gly	Val	
465							470					475				
Lys	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	Phe	Gln	Pro	
485							490					495				
Thr	Tyr	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	Leu	Ser	Phe	
500							505					510				
Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	Lys	Ser	Thr	
515							520					525				
Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	Gly	Leu	Thr	
530							535					540				
Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	Pro	Phe	Gln	
545							550					555				
Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	Arg	Asp	Pro	
565							570					575				
Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	Gly	Gly	Val	
580							585					590				
Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	Ala	Val	Leu	
595							600					605				
Tyr	Gln	Gly	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	His	Ala	Asp	
610							615					620				
Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	Asn	Val	Phe	
625							630					635				
Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	Asn	Asn	Ser	
645							650					655				
Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	Ser	Tyr	Gln	
660							665					670				
Thr	Gln	Thr	Asn	Ser	Pro	Gln	Gln	Ala	Gln	Ser	Val	Ala	Ser	Gln	Ser	
675							680					685				
Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Val	Glu	Asn	Ser	Val	Ala	Tyr	
690							695					700				
Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	Ser	Val	Thr	
705							710					715				
Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	Asp	Cys	Thr	
725							730					735				
Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	Leu	Leu	Gln	
740							745					750				
Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	Gly	Ile	Ala	
755							760					765				
Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	Val	Lys	Gln	
770							775					780				
Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	Asn	Phe	Ser	
785							790					795				
Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	Phe	Ile	Glu	
805							810					815				
Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	Phe	Ile	Lys	
820							825					830				

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Gln Tyr Gly Asp Cys Leu Gly Asp Ile Ala Ala Arg Asp Leu Ile Cys
 835 840 845
 Ala Gln Lys Phe Asn Gly Leu Thr Val Leu Pro Pro Leu Leu Thr Asp
 850 855 860
 Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly Thr Ile Thr
 865 870 875 880
 Ser Gly Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile Pro Phe Ala
 885 890 895
 Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr Gln Asn Val
 900 905 910
 Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn Ser Ala Ile
 915 920 925
 Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala Leu Gly Lys
 930 935 940
 Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn Thr Leu Val
 945 950 955 960
 Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val Leu Asn Asp
 965 970 975
 Ile Leu Ser Arg Leu Asp Pro Pro Glu Ala Glu Val Gln Ile Asp Arg
 980 985 990
 Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val Thr Gln Gln
 995 1000 1005
 Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn Leu Ala Ala
 1010 1015 1020
 Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys Arg Val Asp
 1025 1030 1035
 Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro Gln Ser Ala
 1040 1045 1050
 Pro His Gly Val Val Phe Leu His Val Thr Tyr Val Pro Ala Gln
 1055 1060 1065
 Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His Asp Gly Lys
 1070 1075 1080
 Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn Gly Thr His
 1085 1090 1095
 Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln Ile Ile Thr
 1100 1105 1110
 Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val Val Ile Gly
 1115 1120 1125
 Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro Glu Leu Asp
 1130 1135 1140
 Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn His Thr Ser
 1145 1150 1155
 Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn Ala Ser Val
 1160 1165 1170
 Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu Val Ala Lys
 1175 1180 1185
 Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu Gly Lys Tyr
 1190 1195 1200
 Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu Gly Phe Ile
 1205 1210 1215

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Ala Gly	Leu Ile	Ala Ile	Val	Met Val	Thr Ile	Met	Leu Cys Cys
1220			1225			1230	
Met Thr	Ser Cys	Cys Ser	Cys	Leu Lys	Gly Cys	Cys	Ser Cys Gly
1235			1240			1245	
Ser Cys	Cys Lys	Phe Asp	Glu	Asp Asp	Ser Glu	Pro	Val Leu Lys
1250			1255			1260	
Gly Val	Lys Leu	His Tyr	Thr				
1265			1270				

<210> SEQ ID NO 8
 <211> LENGTH: 3813
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: SARS-CoV-2_B.1.351(beta) spike modified
 nucleic acid sequence

<400> SEQUENCE: 8

atgtttgtgt ttctgggtgct gctgcctctg gtgagcagcc agtgcgtgaa cttcaccacc	60
agaacccagc tgccctctgc ctacaccaac agctttacca gaggcgtgta ttaccccgac	120
aaagtgttta gatcctccgt gctgcacagc acccaggacc tgtttctgcc ctttttcagc	180
aacgtgacct ggttccatgc catccacgtg agcggcacca acggcaccaa gagattcgcc	240
aatcccgctg tgccctttaa tgacggcgtg tactttgcc gcaccgagaa gagcaatac	300
atcagaggct ggatcttcgg caccaccctg gatagcaaga cccagagcct gctgatcgtg	360
aataatgcc ccaatgtggt gatcaaagtg tgcgagtcc agttttgtaa cgacccttt	420
ctgggcgtgt attaccacaa gaacaacaaa agctggatgg agagcgaatt cagagtgtac	480
agcagcgcca ataactgcac ctttgagtac gtgagccagc cctttctgat ggacctggag	540
ggcaagcagg gcaactcaa gaacctgaga gagttcgtgt ttaaaaaat cgacggctac	600
ttcaagatct acagcaagca tacaccctac aacctggtga gaggcctgcc ccagggttc	660
agcgccctgg agccctgggt ggacctgccc atcggcacaa acattaccg gtttcagacc	720
ctgcacagaa gctatctgac ccccgcgac agcagcagcg gctggaccgc cggagccgcc	780
gcctattatg tgggctacct gcagcccaga accttctgc tgaaatacaa cgagaacggc	840
acctcaccg acgcctgga ctgcgacctg gacccctga gcgagaccaa gtgtacctg	900
aagagcttta ccgtggagaa gggcatctac cagaccagca actttagagt gcagcccacc	960
gagagcatcg tgagatttcc caacatcacc aacctgtgcc ccttcgggga ggtgtttaat	1020
gccaccagat tcgccagcgt gtacgcctgg aatagaaaa gaacagcaa ctgcgtggcc	1080
gactacagcg tgctgtacaa ctctgccagc ttttccacct ttaagtgcta cggcgtgagc	1140
ccaaccaaac tgaacgacct gtgcttcacc aacgtgtacg ccgacagctt cgtgatcaga	1200
ggcgacgagg tgagacagat cgcgccggc cagaccggca atatcgccga ttacaactac	1260
aagctgcccc acgaacttac cgctgcgtg atcgctgga acagcaaaa cctggacagc	1320
aagggtggcg gcaactacaa ctacctgtac agactgttta gaaagagcaa cctgaagccc	1380
tttgagagag acatcagcac cgagatctac caggccggca gcacccctg caacggcgtg	1440
aaaggcttta attgttactt cccactgcag tctacggct tccagcctac ctacggcgtg	1500
ggatatcagc cctatagagt ggtggtgctg tccttcgagc tgctgcacgc ccccgccacc	1560
gtgtgcggcc caaagaagag caccaacctg gtgaagaaca agtgcgtgaa cttcaacttc	1620

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aacggcctga cgggcacgg cgtgctgacc gagtccaaca agaaatttct gcccttcag	1680
cagttcggaa gagacattgc cgataccacc gacgcctga gagacccca gacctggag	1740
atcctggaca tcacccctg tagcttcggc ggctgtgccc tgatcacccc aggcaccaac	1800
acctccaacc aggtggcgt gctgtaccag ggctgaact gtaccgaggt gctgtggcc	1860
atccacggc atcagctgac cccaacctgg agagtgtact ccacggctc caacgtgttc	1920
cagaccagag cggctgtct gatcgccgc gagcacgtga acaactccta cgagtgcgac	1980
atccctatcg gcgcggcgt ctgcgccagc taccagaccc agaccaatag ccccagcag	2040
gcccagtcgg tggccagcca gtccatcatt gcctacacca tgtccctggg cgtggagAAC	2100
agcgtggcct actccaaca cagcatcgcc atccccacca atttcacat cagcgtgacc	2160
accgaaatcc tgccagtga catgacaaa acctcgtgg attgcacat gtacatctgc	2220
ggagactcca ccgagtgtc caacctgtg ctgcagtacg gctcctttg caccagctg	2280
aacagagccc tgaccggaat cgcctgggag caggataaaa acaccagga agtggtcgcc	2340
caggtgaaac agatctaca gaccccccc atcaagatt ttggaggctt taattttagc	2400
cagatcctgc ctgatccctc caaaccagc aagagatcct ttatcgagga tctgtgttc	2460
aacaaagtga cactggcga tgccggattt atcaaacagt acggagactg cctgggcgac	2520
atgcgcgcca gagacctgat ctgcgccag aagttaacg gcctgacgt gctgccacc	2580
ctgtgaccg acgagatgat cgcacagtat acctccgccc tgcctggcgg caccatcacc	2640
tcggctgga cctttggcg cggagccgc ctgcagatcc catttgccat gcagatggcc	2700
tacagattta atggaatcg agtgaccag aacgtgtgt atgagaacca gaaactgatc	2760
gccaatcagt tcaacagcg catcggaag atccaggatt ctctgagctc caccgcctct	2820
gccctgggca agctgcagga cgtggtgaat cagaacgccc aggcctgaa caccctggtg	2880
aaacagctga gcagcaactt tggcgccatc agcagcgtgc tgaatgatat cctgagcaga	2940
ctggatcctc cagaggccga ggtgcagatc gacagactga tcacaggag actgcagagc	3000
ctgcagacct atgtgacaca gcagctgac agagccgccc aaattagagc cagcgccaac	3060
ctggccgcca ccaaatgag cgagtgtgtg ctgggacaga gcaaaagagt ggactttctg	3120
ggcaaaggct accatctgat gagcttccc cagagcgccc cccacggcgt ggtgttctg	3180
catgtgacct acgtgccgc ccaggaaaa aattttacca ccgcccctgc catctgtcac	3240
gacggcaaag ccattttcc cagagagggc gtgtttgtga gcaacggcac cactgggtt	3300
gtgaccaga gaaactttta cgaacccag atcatcaca cagataatac ctttgtgagc	3360
ggaaattgtg atgtggtgat cggcatcgtg aataataccg tgtatgatcc tctgcagcca	3420
gagctggaca gctttaagga ggagctggac aagtatttta aaatcatac cagccccgac	3480
gtggatctgg gagatatcag cggaatcaac gcctcgtgg tgaatatcca gaaggagatc	3540
gacagactga acgaggtggc caaaaatctg aatgagtccc tgatcgacct gcaggagctg	3600
ggcaaatatg aacagtacat caagtggcca tggtagatct ggctggggtt tatcgccgga	3660
ctgatcgcca tctgtatggt gaccatcatg ctgtgttgca tgaccagctg ttgtagctgt	3720
ctgaaaggct gttgtagctg cggcagctgc tgcaaatctg acgaggatga ttccgagcca	3780
gtgctgaagg gagtgaact gcactatacc tga	3813

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<210> SEQ ID NO 9
 <211> LENGTH: 138
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Sequence bound to the N-terminus of
 SARS-CoV-2-B.1.1.351 Spike protein

<400> SEQUENCE: 9

Met Gly Leu Pro Asn Asn Thr Ala Ser Trp Phe Thr Ala Leu Thr Gln
 1 5 10 15
 His Gly Lys Glu Asp Leu Lys Phe Pro Arg Gly Gln Gly Val Pro Ile
 20 25 30
 Asn Thr Asn Ser Ser Pro Asp Asp Gln Ile Gly Tyr Tyr Arg Arg Ala
 35 40 45
 Thr Arg Arg Ile Arg Gly Gly Asp Gly Lys Met Lys Asp Leu Ser Pro
 50 55 60
 Arg Trp Tyr Phe Tyr Tyr Leu Gly Thr Gly Pro Glu Ala Gly Leu Pro
 65 70 75 80
 Tyr Gly Ala Asn Lys Asp Gly Ile Ile Trp Val Ala Thr Glu Gly Ala
 85 90 95
 Leu Asn Thr Pro Lys Asp His Ile Gly Thr Arg Asn Pro Ala Asn Asn
 100 105 110
 Ala Ala Ile Val Leu Gln Leu Pro Gln Gly Thr Thr Leu Pro Lys Gly
 115 120 125
 Phe Tyr Ala Glu Gly Ser Arg Gly Gly Ser
 130 135

<210> SEQ ID NO 10
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Sequence bound to the C-terminus of
 SARS-CoV-2-B.1.1.351 Spike protein

<400> SEQUENCE: 10

Thr Lys Lys Ser Ala Ala Glu Ala Ser Lys Lys Pro Arg Gln Lys Arg
 1 5 10 15
 Thr Ala Thr Lys Ala Tyr Asn Val Thr Gln Ala Phe Gly Arg Arg Gly
 20 25 30
 Pro Glu Gln Thr Gln Gly Asn Phe Gly Asp Gln Glu Leu Ile Arg Gln
 35 40 45
 Gly Thr Asp Tyr Lys His Trp Pro Gln Ile Ala Gln Phe Ala Pro Ser
 50 55 60
 Ala Ser Ala Phe Phe Gly Met Ser Arg Ile Gly Met Glu Val Thr Pro
 65 70 75 80
 Ser Gly Thr Trp Leu Thr Tyr Thr Gly Ala Ile Lys Leu Asp Asp Lys
 85 90 95
 Asp Pro Asn Phe Lys Asp Gln Val Ile Leu Leu Asn Lys His Ile Asp
 100 105 110
 Ala Tyr Lys Thr Phe Pro
 115

<210> SEQ ID NO 11
 <211> LENGTH: 414
 <212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Gene coding polypeptide bound to the N-terminus
 of SARS-CoV-2-B.1.1.351 Spike protein

<400> SEQUENCE: 11

```
atgggacctgc ccaacaacac cgccagctgg ttaccgccc tgaccagca cggcaaggag      60
gacctgaagt tccccagagg ccagggcggtg cccattaaca ccaacagcag ccccgacgac      120
cagatcggtt actacagaag agccaccaga agaatcagag gcggcgacgg caagatgaag      180
gacctgagcc ccagatggta cttctactat ctgggaaccg gcccgaagc cggctgccc      240
tatggcgcca acaaatgatg catcatctgg gtggccaccg aaggcgccct gaacaccccc      300
aaggatcaca tcggaaccag aaatcccgcc aacaatgccg ccatcgtgct gcagctgccc      360
cagggaaacca cctgcctaa aggcctttat gccgaaggaa gcagaggagg aagc          414
```

<210> SEQ ID NO 12

<211> LENGTH: 357

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Gene coding polypeptide bound to the C-terminus
 of SARS-CoV-2-B.1.1.351 Spike protein

<400> SEQUENCE: 12

```
acaaaaaaga gcgcccgcga ggccagcaaa aagcccagac agaaaagaac cgccaccaag      60
gcctacaatg tgaccaggc ctttggcaga agaggaccgg agcagacca gggcaatttt      120
ggcgaccagg agctgatcag acagggcacc gactacaaac actggcccca gatcgcccag      180
ttcgccccca gcgcagcgcg cttcttcggc atgagcagaa tcggcatgga ggtgaccccc      240
agcggcacct ggctgaccta caccggcgcc atcaagctgg acgacaagga ccccaacttc      300
aaggaccagg tgatcctgct gaacaagcac atcgacgcct acaagacctt cccctga          357
```

<210> SEQ ID NO 13

<211> LENGTH: 494

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Human papillomavirus type 16, E6, E7 & Human
 papillomavirus type 18, E6, E7 modified amino acid sequence

<400> SEQUENCE: 13

```
Met His Gln Lys Arg Thr Ala Met Phe Gln Asp Pro Gln Glu Arg Pro
1           5           10          15
Arg Lys Leu Pro Gln Leu Cys Thr Glu Leu Gln Thr Thr Ile His Asp
20          25          30
Ile Ile Leu Glu Cys Val Tyr Cys Lys Gln Gln Leu Leu Arg Arg Glu
35          40          45
Val Tyr Asp Phe Ala Phe Arg Asp Leu Cys Ile Val Tyr Arg Asp Gly
50          55          60
Asn Pro Tyr Ala Val Cys Asp Lys Cys Leu Lys Phe Tyr Ser Lys Ile
65          70          75          80
Ser Glu Tyr Arg His Tyr Cys Tyr Ser Leu Tyr Gly Thr Thr Leu Glu
85          90          95
Gln Gln Tyr Asn Lys Pro Leu Cys Asp Leu Leu Ile Arg Cys Ile Lys
100         105         110
```

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Gln	Arg	His	Leu	Asp	Lys	Lys	Gln	Arg	Phe	His	Asn	Ile	Arg	Gly	Arg	115	120	125
Trp	Thr	Gly	Arg	Cys	Met	Ser	Cys	Cys	Arg	Ser	Ser	Arg	Thr	Arg	Arg	130	135	140
Glu	Thr	Gln	Leu	Met	His	Gly	Asp	Thr	Pro	Thr	Leu	His	Glu	Tyr	Met	145	150	155
Leu	Asp	Leu	Gln	Pro	Glu	Thr	Thr	Gln	Leu	Asn	Asp	Ser	Ser	Glu	Glu	165	170	175
Glu	Asp	Glu	Ile	Asp	Gly	Pro	Ala	Gly	Gln	Ala	Glu	Pro	Asp	Arg	Ala	180	185	190
His	Tyr	Asn	Ile	Val	Thr	Phe	Cys	Cys	Lys	Cys	Asp	Ser	Thr	Leu	Arg	195	200	205
Leu	Cys	Val	Gln	Ser	Thr	His	Val	Asp	Ile	Arg	Thr	Leu	Glu	Asp	Leu	210	215	220
Leu	Met	Gly	Thr	Leu	Gly	Ile	Val	Cys	Pro	Ile	Cys	Ser	Gln	Lys	Pro	225	230	235
Arg	Gly	Arg	Lys	Arg	Arg	Ser	Met	Ala	Arg	Phe	Glu	Asp	Pro	Thr	Arg	245	250	255
Arg	Pro	Tyr	Lys	Leu	Pro	Asp	Leu	Cys	Thr	Glu	Leu	Asn	Thr	Ser	Leu	260	265	270
Gln	Asp	Ile	Glu	Ile	Thr	Cys	Val	Tyr	Cys	Lys	Thr	Val	Leu	Glu	Leu	275	280	285
Thr	Glu	Val	Phe	Glu	Phe	Ala	Phe	Lys	Asp	Leu	Phe	Val	Val	Tyr	Arg	290	295	300
Asp	Ser	Ile	Pro	His	Ala	Ala	Cys	His	Lys	Cys	Ile	Asp	Phe	Tyr	Ser	305	310	315
Arg	Ile	Arg	Glu	Leu	Arg	His	Tyr	Ser	Asp	Ser	Val	Tyr	Gly	Asp	Thr	325	330	335
Leu	Glu	Lys	Leu	Thr	Asn	Thr	Gly	Leu	Tyr	Asn	Leu	Leu	Ile	Arg	Cys	340	345	350
Leu	Lys	Leu	Arg	His	Leu	Asn	Glu	Lys	Arg	Arg	Phe	His	Asn	Ile	Ala	355	360	365
Gly	His	Tyr	Arg	Gly	Gln	Cys	His	Ser	Cys	Cys	Asn	Arg	Ala	Arg	Gln	370	375	380
Glu	Arg	Leu	Gln	Arg	Arg	Arg	Glu	Thr	Gln	Val	Met	His	Gly	Pro	Lys	385	390	395
Ala	Thr	Leu	Gln	Asp	Ile	Val	Leu	His	Leu	Glu	Pro	Gln	Asn	Glu	Ile	405	410	415
Pro	Val	Gln	Leu	Ser	Asp	Ser	Glu	Glu	Glu	Asn	Asp	Glu	Ile	Asp	Gly	420	425	430
Val	Asn	His	Gln	His	Leu	Pro	Ala	Arg	Arg	Ala	Glu	Pro	Gln	Arg	His	435	440	445
Thr	Met	Leu	Cys	Met	Cys	Cys	Lys	Cys	Glu	Ala	Arg	Ile	Glu	Leu	Val	450	455	460
Val	Glu	Ser	Ser	Ala	Asp	Asp	Leu	Arg	Ala	Phe	Gln	Gln	Leu	Phe	Leu	465	470	475
Asn	Thr	Leu	Ser	Phe	Val	Cys	Pro	Trp	Cys	Ala	Ser	Gln	Gln			485	490	

<210> SEQ ID NO 14

<211> LENGTH: 1482

<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Human papillomavirus type 16, E6, E7 & Human
        papillomavirus type 18, E6, E7 modified nucleic acid sequence

<400> SEQUENCE: 14

atgcaccaga agagaacagc catgttccag gacctcagg aacggccacg gaagctccca    60
caactttgta cagagctgca aacaaccatt catgacatca tccttgaatg cgtttactgc    120
aagcagcaac ttctgcgcgc ggaagtctac gatttcgctt tcagagatct gtgcatcggt    180
tatagagacg ggaaccoccta cgccgtgtgt gacaaatgcc ttaaattcta cagtaagatt    240
tcagagtacc ggcattattg ctacagcctt tatggaacta cattggagca gcagtacaac    300
aaacctttgt gcgatcttct tatccggtgc attaagcaac ggcacttgga caaaaaacia    360
cggttccata acatcagggg caggtggact ggacgctgca tgagttgctg tcgctcaagc    420
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ttggacctcc aaccggaac tacccaactg aatgacagct ctgaggaaga ggacgaaatc    540
gacggccctg cagggcaagc cgaacctgac agggcccatc ataacattgt gacattttgc    600
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cgcgcccgcg aggagcggct gcaaaggcgc agagagactc aggtgatgca tggacccaag   1200
gccactctcc aggacatcgt ccttcactct gaaccccaga acgaaattcc cgttcaactt   1260
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agaagagccg agccacaaag acacacaatg ctctgtatgt gctgcaagtg tgaagcacgg   1380
atcgaaactg tcgtcgaatc tagtgagat gatctcagag cctttcaaca gctcttcttg   1440
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<223> OTHER INFORMATION: HVR1

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<400> SEQUENCE: 15

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Glu Glu Asp Asp Asp Asn Glu Asp Glu Val Asp Glu Gln Ala Glu Gln
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Gln Lys Thr His Val
                35

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<212> TYPE: PRT
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<220> FEATURE:
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<400> SEQUENCE: 16

Val Glu Gly Gln Thr Pro Lys Tyr
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<210> SEQ ID NO 17
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: HVR3

<400> SEQUENCE: 17

Ser Gln Trp Tyr Glu Thr Glu Ile Asn His Ala Ala
1 5 10

<210> SEQ ID NO 18
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: HVR4

<400> SEQUENCE: 18

Gly Ile Leu Val Lys Gln Gln Asn Gly Lys Leu Glu Ser Gln Val Glu
1 5 10 15

Met Gln

<210> SEQ ID NO 19
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: HVR5

<400> SEQUENCE: 19

Ser Thr Thr Glu Ala Thr Ala Gly Asn Gly Asp Asn Leu Thr Pro Lys
1 5 10 15

Val

<210> SEQ ID NO 20
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 20

Thr Ile Lys Glu Gly Asn Ser Arg Glu Leu Met Gly
1 5 10

<210> SEQ ID NO 21
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<212> TYPE: PRT
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<220> FEATURE:
<223> OTHER INFORMATION: HVR7

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Gly Gly Val Ile Asn Thr Glu Thr Leu Thr Lys Val Lys Pro Lys Thr
1 5 10 15

Gly Gln Glu Asn Gly Trp Glu Lys Asp Ala Thr Glu Phe Ser Asp Lys
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Asn Glu Ile Arg Val Gly Asn Asn Phe
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<210> SEQ ID NO 22

<211> LENGTH: 22

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

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<223> OTHER INFORMATION: HVR1

<400> SEQUENCE: 22

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1 5 10 15

Val Thr Lys Thr His Thr
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<210> SEQ ID NO 23

<211> LENGTH: 13

<212> TYPE: PRT

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<220> FEATURE:

<223> OTHER INFORMATION: HVR2

<400> SEQUENCE: 23

Thr Glu Ile Thr Ala Asp Asn Gln Lys Lys Glu Ile Phe
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<210> SEQ ID NO 24

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<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: HVR3

<400> SEQUENCE: 24

Glu Asn Trp Gln Glu Asn Glu Val Phe Tyr Gly
1 5 10

<210> SEQ ID NO 25

<211> LENGTH: 24

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 25

Gly Thr Ala Ala Cys Cys Gly Cys Ala Thr Ala Cys Gly Ala Gly Cys
1 5 10 15

Ala Gly Ala Cys Gly Gly Thr Gly
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<210> SEQ ID NO 26

<211> LENGTH: 15

<212> TYPE: PRT

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<220> FEATURE:

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<223> OTHER INFORMATION: HVR5

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Asp Thr Asp Gly Gly Thr Ala Asp Thr Glu Tyr Lys Ala Asp Ile
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<210> SEQ ID NO 27

<211> LENGTH: 12

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: HVR6

<400> SEQUENCE: 27

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<210> SEQ ID NO 28

<211> LENGTH: 43

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: HVR7

<400> SEQUENCE: 28

Asp Gly Leu Gly Thr Asn Ala Thr Tyr Gln Gly Val Lys Val Ser Thr
1 5 10 15

Gly Asp Gly Ala Thr Gln Ser Gly Trp Ala Lys Asp Asp Thr Met Ala
20 25 30

Arg Gln Asn Gln Ile Cys Arg Gly Asn Ile Tyr
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<210> SEQ ID NO 29

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<212> TYPE: DNA

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<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 29

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<212> TYPE: DNA

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<220> FEATURE:

<223> OTHER INFORMATION: Primer

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<210> SEQ ID NO 31

<211> LENGTH: 26

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

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<223> OTHER INFORMATION: Primer

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<400> SEQUENCE: 33

gtaaccgcat acgagcagac ggtg 24

<210> SEQ ID NO 34
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<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 34

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<210> SEQ ID NO 35
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 35

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Primer

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<210> SEQ ID NO 37
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<223> OTHER INFORMATION: Primer

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caacctgaac ctcaaataagg agaatctcag 30

<210> SEQ ID NO 38
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 38

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<210> SEQ ID NO 39
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 39

gtcaggcagt agtttgaagt tcgc 24

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

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<210> SEQ ID NO 41
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 41

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<210> SEQ ID NO 42
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 42

cggagtaact tgtatgtgtt ggaattg 28

<210> SEQ ID NO 43
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 43

caattcccaa cacatacaag ttactccg 28

<210> SEQ ID NO 44
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<212> TYPE: DNA
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<400> SEQUENCE: 44

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<400> SEQUENCE: 45

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 46

acaattaggg cttgggtcag gtgtgggtcca caaagtttag 40

<210> SEQ ID NO 47
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<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 47

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<210> SEQ ID NO 48
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 48

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<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 49

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<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 50

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<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 51

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<210> SEQ ID NO 52
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 52

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<212> TYPE: DNA
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<400> SEQUENCE: 53

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<223> OTHER INFORMATION: Primer

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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 55

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 56

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<210> SEQ ID NO 57
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 57

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<210> SEQ ID NO 58
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 58

tatcttatct agaagcttag tcagggaag gtctgtagg c 41

<210> SEQ ID NO 59
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Primer

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 60

cgcgctgacg gtaccagatc tctagcgat c 31

<210> SEQ ID NO 61
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 61

gatctggtac cgctcgacgcg atgcaccaga agagaaca 38

<210> SEQ ID NO 62
<211> LENGTH: 44
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 62

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1. A chimeric adenovirus vector in which the knob domain of the end of the fiber protein of human adenovirus type 5 has been replaced with the knob gene of chimpanzee adenovirus serotype 6.

2. The chimeric adenovirus vector according to claim 1, wherein the hexon protein of human adenovirus type 5 is further replaced with hypervariable regions 1-7 of human adenovirus serotype 28.

3. The chimeric adenovirus vector according to claim 1, wherein the protein expressed by the knob gene of chimpanzee adenovirus serotype 6 comprises the amino acid sequence of SEQ ID NO: 1.

4. The chimeric adenovirus vector according to claim 1, wherein the knob gene sequence of chimpanzee adenovirus serotype 6 comprises the nucleic acid sequence of SEQ ID NO: 2.

5. The chimeric adenovirus vector according to claim 1, wherein the vector further comprises a fiber protein combined with the tail and shaft domains of human adenovirus type 5 and the knob domain of chimpanzee adenovirus serotype 6.

6. The chimeric adenovirus vector according to claim 2, wherein the chimeric protein in which the hexon protein of

human adenovirus type 5 is replaced with the hypervariable region 1-7 of human adenovirus serotype 28 comprises the amino acid sequence of SEQ ID NO: 3.

7. The chimeric adenovirus vector according to claim 2, wherein the gene encoding a chimeric protein in which the hexon protein of human adenovirus type 5 is replaced with hypervariable regions 1-7 of human adenovirus serotype 28 comprises the nucleic acid sequence of SEQ ID NO: 4.

8. The chimeric adenovirus vector according to claim 1, wherein the vector is shown in FIG. 6 or FIG. 8.

9. The chimeric adenovirus vector of claim 1, wherein the vector comprises as an insert a) a gene encoding the spike protein of the coronavirus, or b) a gene encoding the E6 and E7 proteins of human papillomavirus type 16 or 18.

10. A host cell transformed with the chimeric adenovirus vector of claim 1.

11. The host cell according to claim 10, wherein the cell is HEK293, PERC6 or 911 cell.

12. An isolated polypeptide encoded by the chimeric adenovirus vector of claim 1.

13-15. (canceled)

* * * * *